

# Highlights

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- *First HSI survey from the perspective of computer vision* Even though HSI has been a well-explored technique since it was developed, the related research mainly focuses on remote sensing or some specific tasks. No literature analyzes HSI from a computer vision perspective. Recent advances in artificial intelligence, especially foundation model, further sped up the development of RGB-based computer vision, leaving HSI research lagging behind. To this end, we present the first systematic review of HSI research from a computer vision perspective. Beyond surveying applications, we build a coherent theoretical pipeline from the physical optical imaging process to computer vision tasks, summarize state-of-the-art artificial intelligence techniques applied to HSI, and curate HSI datasets for computer vision.,
- *First discussion of differences between RS and CV* Existing HSI research usually designed sophisticated algorithms targeting on remote sensing applications, some also evaluated their developed algorithms on several computer vision HSI datasets, while few of them discussed the differences within two kinds of datasets. In this view, we provide the first in-depth discussion of the differences between HSI in remote sensing and HSI in computer vision, highlighting distinctions in illumination conditions, materials, sensors, network architectures, and datasets.
- *First evaluation of spectral foundation models on CV tasks* Thanks to the powerful representation capability and satisfactory performance of visual foundation models, several spectral foundation models in remote sensing are proposed. However, their performance on computer vision tasks is still unclear. We therefore conduct the first evaluation of existing remote sensing spectral foundation models on computer vision tasks, demonstrating their potential, visualizing their performance, and analyzing their strengths and limitations.
- *First comparison between SOTA RGB networks and current HSI networks* Reviewing the state-of-the-art networks employed in RGB-based computer vision, we initially analyze the networks' generation gap between state-of-the-art RGB foundation models and current spectral foundation models, and identify key directions for future research.

# A Comprehensive Survey on Hyperspectral Imaging in Computer Vision

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## ABSTRACT

Hyperspectral imaging (HSI) captures rich spectral information beyond conventional RGB images, enabling detailed analysis of material properties and offering significant potential for computer vision. However, existing research predominantly focuses on remote sensing or isolated tasks, lacking a unified perspective tailored to computer vision. Moreover, the fundamental differences between remote sensing HSI and computer vision HSI remain underexplored, limiting the transferability of existing methods. In this survey, we present the first systematic review of HSI from a computer vision perspective. We establish a unified pipeline that connects physical imaging theory, dataset characteristics, and downstream vision tasks. Specifically, we begin by analyzing the electromagnetic spectrum and light-matter interactions, and provide a step-by-step comparison across spectroscopy, remote sensing, and computer vision domains in terms of illumination, sensor noise, and data patterns. Building on this foundation, we organize existing literature into low-level, high-level vision tasks and visual AI, and summarize representative methods within a coherent framework. We further review commonly used hyperspectral datasets in computer vision, detailing their imaging configurations and evaluating algorithm performance to bridge data and methodology. Furthermore, we benchmark state-of-the-art spectral foundation models developed for remote sensing on representative computer vision datasets. Results demonstrate strong generalization limitations and notable domain gaps. Finally, we discuss key challenges and future directions from the perspectives of theory, data, and applications. This survey aims to provide a comprehensive foundation and promote the development of HSI in computer vision.

## 1. Introduction

Light forms the foundation of the color world and human visual perception. Beyond the visible range, electromagnetic radiation spans a much broader spectral region containing abundant physical and chemical information. Hyperspectral imaging (HSI) captures this rich spectral information by acquiring images across numerous contiguous wavelength bands, enabling detailed analysis of material composition and properties. HSI has been applied to many research fields, including agriculture [212, 155, 107], medical imaging [213], pathology diagnosis [261, 314], document analysis [264], food quality inspection [269], airborne and spaceborne sensing [254, 142], environmental monitoring [280], and remote sensing [153, 25]. Despite this, the application of HSI in computer vision (CV) remains relatively underexplored, necessitating further investigation.

The theoretical value of employing HSI in computer vision is two-fold. On the one hand, compared with traditional RGB images, HSI offers a distinct advantage by recording richer, often continuous spectral bands of light reflectance at each pixel, namely, spectral signature. This characteristic enables deeper perception of the intrinsic physical and chemical properties of materials [220]. Such spectral richness allows differentiation between objects that exhibit visually similar textures but possess different spectral characteristics [68]. Moreover, metamerism—where two surfaces appear identical under one illumination condition but different under another [136]—can be effectively mitigated using HSI rather than RGB imagery. On the other hand, although HSI technique has been extensively studied in remote

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<sup>1</sup>Qi Bi started and contributed this work when he was affiliated with the Computer Vision Research Group, University of Amsterdam

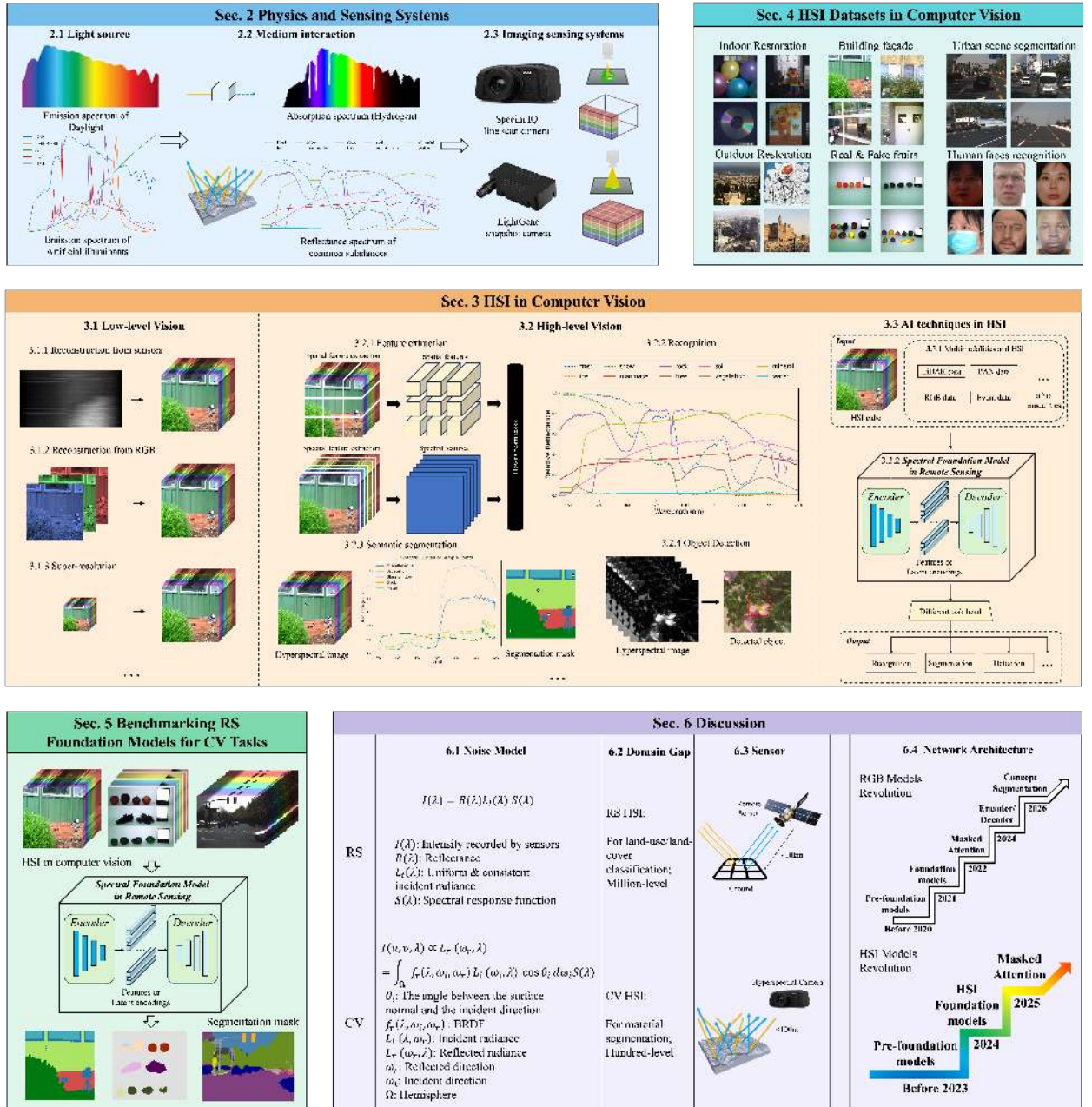


Figure 1: Overview of the survey.

sensing (RS), noise models developed for remote sensing scenarios cannot be directly transferred to computer vision settings. Remote sensing HSIs are typically captured from high-altitude platforms such as aircraft or satellites, where the large distance between the sensor and the scene results in approximately uniform global illumination, and the geometric structure of the scene can often be neglected. In contrast, computer vision applications operate in close-range environments where scene geometry plays a crucial role, introducing significant local illumination variations. Complex interactions between light and matter—including absorption, scattering, reflection, and refraction—must therefore be explicitly considered. Furthermore, natural images contain abundant self-repeating patterns, meaning neighboring regions usually exhibit strong spatial similarity. Effectively exploiting this property requires modeling

spatial correlations over larger contextual regions [93]. Consequently, it is necessary to investigate hyperspectral imaging from a computer vision-specific perspective.

Building upon these advantages, HSI has been increasingly adopted across a wide range of visual tasks, even though sometimes in a naive or suboptimal manner. From a computer vision perspective, these applications can be broadly categorized into low-level and high-level tasks. Low-level vision tasks focus on restoring or enhancing raw, often degraded HSI data to produce more accurate three-dimensional data cubes. These tasks are typically unsupervised and rely on the intrinsic properties of the data. Representative examples include spectral reconstruction [36, 352, 361], spectral unmixing [28], super-resolution [311], denoising [364], demosaicing [81], and illumination spectrum estimation [150]. In contrast, high-level vision tasks incorporate human knowledge as supervision to achieve specific semantic goals. Typical applications include classification [176, 12, 179, 80], semantic segmentation [104], face recognition [370], object detection [191, 254, 209, 132], and object tracking [259]. Furthermore, with the rapid advancement of artificial intelligence (AI), there has been growing interest in applying AI techniques to HSI data. In particular, spectral foundation models [53, 125, 297] and spectral generative models [249, 248, 229, 173, 172, 356, 247, 350, 89] have recently emerged as promising research directions.

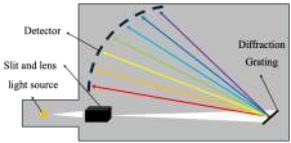
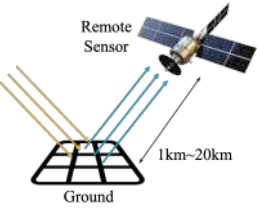
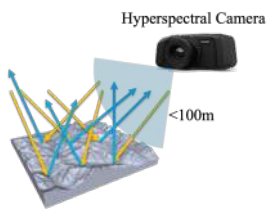
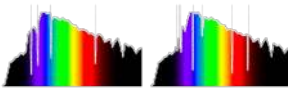
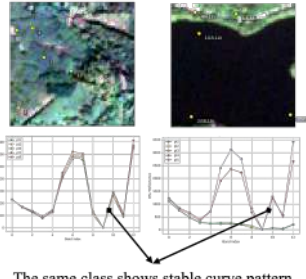
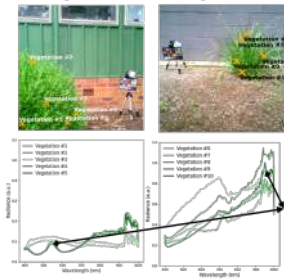
Despite significant progress, hyperspectral imaging research in computer vision does not evolve as smoothly as expected, primarily due to substantial differences between idealized HSI assumptions and real-world scenarios, both in theory and in practice. From a theoretical perspective, although HSI has been extensively studied in the remote sensing community, existing literature rarely investigates hyperspectral imaging explicitly from a computer vision standpoint. While some studies propose task-specific algorithms under particular conditions [314, 261], and others evaluate remote sensing-oriented methods on selected computer vision datasets to demonstrate their effectiveness [37, 320], a systematic theoretical analysis tailored to computer vision remains largely absent.

As illustrated in Fig. 2, key distinctions among spectroscopy, remote sensing and computer vision HSIs—including imaging pipeline models, data domain distributions, sensor-induced noise characteristics, algorithm design principles, and scene complexity—have received limited attention. These differences are theoretically critical and may constitute fundamental bottlenecks that restrict the performance and generalization of existing HSI algorithms. From a practical perspective, notable discrepancies also arise when handling remote sensing versus computer vision HSI data. Variations in spectral stability, scene characteristics, sensor configurations, network architectures, and dataset properties can significantly influence experimental outcomes and hinder the direct transfer of methodologies across domains.

Where current challenges persist, promising opportunities for advancement also emerge. On the theoretical side, existing gaps can be addressed by developing more accurate and interpretable imaging pipeline models that explicitly account for noise sources and acquisition mechanisms, thereby improving algorithm robustness. From a data perspective, constructing large-scale, computer vision-oriented hyperspectral datasets or leveraging transfer learning techniques offers a viable path to mitigating domain gaps. Methodologically, drawing inspiration from state-of-the-art RGB-based computer vision research and designing advanced network architectures that are intrinsically compatible with the spectral-spatial characteristics of HSIs are expected to further propel progress in this field.

In a nutshell, our contributions can be summarized as follows.

- *First HSI survey from the perspective of computer vision* Even though HSI has been a well-explored technique since it was developed, the related research mainly focuses on remote sensing or other specific tasks. No literature analyzes HSI from a computer vision perspective. Recent advances in artificial intelligence, especially foundation model, further sped up the development of RGB-based computer vision, leaving HSI research lagging behind. To this end, we present the first systematic review of HSI research from a computer vision perspective. Beyond surveying applications, we build a coherent theoretical pipeline from the physical optical imaging process to computer vision tasks, summarize state-of-the-art artificial intelligence techniques applied to HSI, and curate HSI datasets for computer vision.
- *First evaluation of spectral foundation models on CV tasks* Thanks to the powerful representation capability and satisfactory performance of visual foundation models, several spectral foundation models in remote sensing are proposed. However, their performances on computer vision tasks are still unknown. We therefore conduct the first evaluation of existing remote sensing spectral foundation models on computer vision tasks, demonstrating their potential, visualizing their performance, and analyzing their strengths and limitations.
- *First discussion of differences between RS and CV* Existing hyperspectral imaging (HSI) literature primarily focuses on designing sophisticated algorithms for remote sensing applications. Some studies further evaluate

| Domain                           |            | Spectroscopy                                                                                                                                                                                                                                                          | Remote sensing                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Computer vision                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|----------------------------------|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Type of Material                 |            | Chemical and physical (molecular-level)                                                                                                                                                                                                                               | Semantic (forest, pasture, etc.)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Semantic (metal, vegetation, etc.)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Light Source                     |            | Single source under strict control;<br>Consistent & Parallel                                                                                                                                                                                                          | Sunlight and other uniform sources;<br>Consistent & Parallel                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Random multi-source;<br>Inconsistent & Unparallel                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Sensor                           |            | Spectrometer                                                                                                                                                                                                                                                          | Airborne/Spaceborne imaging system                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Close-range hyperspectral camera                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Imaging Model                    |            |                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Noise Model                      | Setting    | $I(\lambda) = E_a(\lambda) S(\lambda)$ <ul style="list-style-type: none"> <li><math>I(\lambda)</math>: Intensity recorded by sensors</li> <li><math>E_a(\lambda)</math>: Absorption intensity</li> <li><math>S(\lambda)</math>: Spectral response function</li> </ul> | $I(\lambda) = \int_{\Omega} f_r(\lambda) \cos \theta_l d\omega_l L_l(\lambda) S(\lambda)$ $= R(\lambda) L_l(\lambda) S(\lambda)$ <p>where <math>R(\lambda) = \int_{\Omega} f_r(\lambda) \cos \theta_l d\omega_l = \pi f_r(\lambda)</math></p> <ul style="list-style-type: none"> <li><math>\Omega</math>: hemisphere</li> <li><math>f_r(\lambda)</math>: Simplified BRDF</li> <li><math>\theta_l</math>: The angle between the surface normal and the incident direction</li> <li><math>\omega_l</math>: Incident direction</li> <li><math>R(\lambda)</math>: Reflectance</li> <li><math>L_l(\lambda)</math>: Uniform &amp; consistent incident radiance</li> </ul> | $I(u, v, \lambda) \propto L_r(\omega_r, \lambda)$ $= \int_{\Omega} f_r(\lambda, \omega_l, \omega_r) L_l(\omega_l, \lambda) \cos \theta_l d\omega_l S(\lambda)$ <ul style="list-style-type: none"> <li><math>u, v</math>: Pixel coordination of the camera</li> <li><math>L_l(\omega_l, \lambda)</math>: Incident radiance</li> <li><math>\omega_r</math>: Reflected direction</li> <li><math>L_r(\omega_r, \lambda)</math>: Reflected radiance</li> <li><math>f_r(\lambda, \omega_l, \omega_r)</math>: BRDF</li> </ul> |
|                                  | Assumption | Well controlled light sources;<br>pure simple substances;<br>no complex geometric structure                                                                                                                                                                           | $R(\lambda)$ does not depend on incident direction;<br>$L_l(\lambda)$ maintain globally consistent while may change slightly in local region                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Highly variable                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|                                  | Objective  | Given $I(\lambda)$ and $S(\lambda)$ ,<br>predict $E_a(\lambda)$                                                                                                                                                                                                       | Given $I(\lambda)$ and $S(\lambda)$ ,<br>predict $R(\lambda)$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Given $I(u, v, \lambda)$ and $S(\lambda)$ ,<br>predict $f_r(\lambda, \omega_l, \omega_r)$                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Practice in material recognition |            | <p>Hydrogen      Lithium</p>                                                                                                                                                       | <p>Forest      Forest &amp; SeaLake</p>  <p>The same class shows stable curve pattern<br/>Slight variation caused by <math>L_l(\lambda)</math></p>                                                                                                                                                                                                                                                                                                                                                                                                                               | <p>Vegetation      Vegetation</p>  <p>Semantic ambiguity:<br/>two plants labelled the same</p> <p>Large variation caused by <math>L_l(\omega_l, \lambda)</math></p>                                                                                                                                                                                                                                                                |

**Figure 2:** Different settings for spectroscopy, remote sensing and computer vision, part of this figure made from [153, 110]. In spectroscopy, analysis primarily relies on emission or absorption characteristics, where the corresponding spectral response is typically sufficient for accurate interpretation and imaging. In remote sensing, the problem becomes more involved, as it requires accounting for the reflectance properties of various materials, which depend on both intrinsic characteristics and environmental factors. The complexity further increases in the computer vision domain. Here, illumination conditions are often inconsistent and uncontrolled, and the noise model becomes significantly more complex. As a result, the interaction between incident light and the surrounding environment plays a critical role in image formation. This makes computer vision scenarios the most challenging among the three, due to the combined effects of variable lighting, environmental interactions, and noise.

these methods on computer vision HSI datasets to demonstrate their generalization ability. However, the underlying noise models in remote sensing and computer vision scenarios are fundamentally different. Consequently, directly applying algorithms tailored for remote sensing to computer vision tasks may lead to suboptimal performance. In this view, we provide the first in-depth discussion of the differences between HSI in remote sensing and HSI in computer vision, highlighting distinctions in illumination conditions, materials, sensors, network architectures, and datasets.

- *First comparison between SOTA RGB networks and current HSI networks* Reviewing the state-of-the-art networks employed in RGB-based computer vision, we initially analyze the networks' generation gap between state-of-the-art RGB foundation models and current spectral foundation models, and identify key directions for future research.

The remainder of this survey is organized as follows:

- Section 2 systematically develops the fundamental theories of HSI from a computer vision perspective. We begin with the electromagnetic spectrum of light in natural scene scenarios and describe the various interactions between radiation and different media. The section concludes with an overview of existing HSI imaging models, providing the theoretical foundation for subsequent discussions.
- Section 3 reviews the existing literature on HSI in computer vision, categorizing studies into low-level and high-level vision tasks. In particular, we examine the most prevalent AI techniques applied in combination with HSI and their impact on vision applications.
- Section 4 summarizes existing HSI datasets used in computer vision, detailing their imaging configurations and collection preferences, as introduced in Section 2. This section also presents the performance of various algorithms on these datasets, referring to insights from Section 3.
- Section 5 benchmarks state-of-the-art spectral foundation models developed for remote sensing on three representative HSI datasets in computer vision. This evaluation assesses the current capabilities of spectral foundation models and highlights directions for advancing HSI in computer vision.
- Section 6 presents a systematic discussion of existing challenges and future opportunities for HSI in computer vision, organized around theory, data, and applications, based on the literature review and our evaluation results.
- Section 7 concludes the survey, summarizing key findings and providing perspectives for future research.

## 2. Physics and Sensing Systems

Light sources emit light, the light transmits within media, being more or less absorbed, and undergoes various interactions with different materials. Then the optical imaging systems collect incoming light, split it along its wavelengths, and respond specifically to different wavelengths. Finally, a 3-D HSI data cube is generated to represent the intensity distribution of its corresponding natural scene. The whole process is shown in Figure 1. In this section, we will model this process step by step, from the emission of light to the formation of the 3-D HSI data cube.

### 2.1. Physical fundamentals

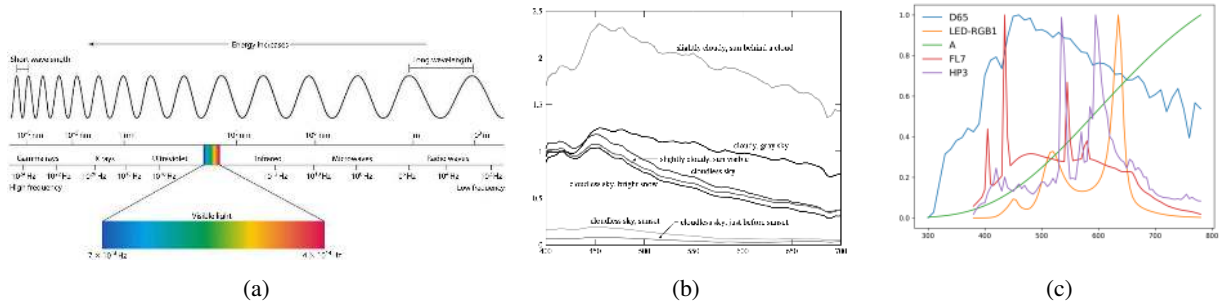
#### 2.1.1. Light sources and spectrum

*Electromagnetic radiation in visible spectrum* We commonly refer to the electromagnetic radiation whose wavelength varies from 400 nanometer (nm) to 700 nm as 'light' or 'visible light' because it can be perceived by human eyes. Newton's dispersion experiment [238] demonstrated that light can be dispersed into a continuous color band, called the spectrum. Modern spectroscopy defines the spectrum as a distribution over wavelengths [285]. Although electromagnetic radiation includes radiation of longer and shorter wavelengths, we mainly focus on visible light in the field of computer vision, as shown in Figure 3a. According to Planck's law, a heated black body reflects no light and emits electromagnetic radiation. It is a remarkable fact that the spectral power distribution of this radiation depends only on the temperature of the body. Assuming that the light source is in the visible region with radiance, it emits light with a distribution as:

$$E(\lambda) \propto \frac{1}{\lambda^5} \frac{1}{(\exp(hc/k\lambda T) - 1)}, \quad (1)$$

where  $\lambda$  denotes the wavelength,  $T$  is the temperature of the body in Kelvins,  $h$  Planck's constant,  $k$  for Boltzmann's constant,  $c$  for the speed of light.

**Natural illumination** Daylight is the holistic combination of the luminous characteristics of sunlight from direct solar radiation and skylight from diffuse solar radiation [154]. It is highly dynamic that changes within and across days, throughout the year, and with weather conditions. Figure 3b shows the spectral energy density of daylight measured at different times of day and under different conditions. The figure plots relative power against wavelength for wavelengths from 400 nm to 700 nm for a series of seven different daylight measurements, made by Jussi Parkkinen and Pertti Silfsten, of daylight illuminating a sample of barium sulfate (which gives a high reflectance white surface). A patch of surface outdoors during the day is illuminated both by light that comes directly from the sun and by the light from the sun that has been scattered by the air [83]. The intensity of daylight varies with the time of day and the season of year, as shown in Figure 3b. Generally, the intensity curves of daylight are relatively smooth or there exist only a global maximum.



**Figure 3:** (a) Electromagnetic spectrum. Courtesy of [231]; (b) Natural daylight exhibits a smooth and continuous Spectral Power Distribution (SPD); (c) Artificial illuminants include D65, A, FL7, LED-RGB1, and HP3 with distinct spectral characteristics.

**Artificial Illumination** An incandescent light contains a metal filament that is heated to a high temperature. The spectrum roughly follows the black-body law. A fluorescent light works by generating high-speed electrons that strike gas within the bulb. The gas releases ultraviolet radiation, which causes phosphors coating the inside of the bulb to fluoresce. Typically, the coating consists of three or four phosphors, which fluoresce in quite narrow ranges of wavelengths. Some fluorescent bulbs mimic natural daylight. There are various illuminant models. Fig. 3c shows the relative spectral power distribution of five standard CIE models, where **D65** mimics average daylight with a smooth and continuous spectrum; **A** represents a tungsten lamp, whose SPD is strongly weighted toward the red and infrared regions; **FL7** simulates a fluorescent light source and is characterized by spiky, discrete spectral peaks; **LED-RGB1** consists of three narrow peaks corresponding to red, green, and blue components; and **HP3** refers to high-pressure discharge sources, such as high-pressure sodium or metal halide lamps, which are commonly used in industrial environments.

### 2.1.2. Medium Interaction: absorption and reflection

*Noise model: From Spectroscopy to Remote Sensing and to Computer Vision*

Electromagnetic radiation can incur various interactions among media, referring to the mutual effect of photons and electrons of the micro-world. Spectroscopy is the study of the interaction between electromagnetic radiation and matter. By measuring how light is absorbed, emitted, or scattered by substances, it enables researchers to identify materials, determine their chemical composition, analyze molecular structures, and investigate molecular dynamics. Absorption and reflection are mostly studied.

When a light beam interacts with a surface, the incident spectral intensity  $E_0(\lambda)$  is partitioned into absorbed, transmitted, and reflected components:

$$E_0(\lambda) = E_a(\lambda) + E_t(\lambda) + E_r(\lambda). \quad (2)$$

According to the Beer–Lambert law [122],

$$E_a(\lambda) = E_0(\lambda) 10^{-\varepsilon Cl}, \quad (3)$$

where  $l$  is the optical path length and  $C$  is the absorber concentration.

Define absorbance, transmittance, and reflectance as

$$a(\lambda) = \frac{E_a}{E_0}, \quad T(\lambda) = \frac{E_t}{E_0}, \quad R(\lambda) = \frac{E_r}{E_0}, \quad a(\lambda) + T(\lambda) + R(\lambda) = 1. \quad (4)$$

where the final equality follows from energy conservation, indicating that all incident energy is partitioned into absorption, transmission, and reflection.

Even though being critical in chemical analysis and astronomy, the absorbance in macroscopic imaging applications can be neglected, while the transmittance is regarded as a variable that depicted the physical nature of the media (e.g. atmosphere) and the reflectance  $R(\lambda)$  is often considered an intrinsic property of the surface.

In remote sensing, solar illumination can be well approximated as a distant point source, producing nearly parallel rays with uniform irradiance across the scene. Under this assumption, the imaging process simplifies significantly: the bidirectional nature of reflectance collapses into a scalar quantity—spectral reflectance  $R(\lambda)$ —and the radiative transfer reduces to a product of illumination and surface properties. Ignoring atmospheric effects and denoting the incident radiance as  $L_i(\lambda)$ , the radiance recorded at the sensor can be expressed as

$$I(\lambda) = \int_{\Omega} f_r(\lambda) \cos \theta_i d\omega_i L_i(\lambda) S(\lambda), \quad (5)$$

where  $f_r(\lambda)$  is the bidirectional reflectance distribution function (BRDF),  $\theta_i$  is the incident angle, and  $S(\lambda)$  denotes the sensor response. Assuming Lambertian reflectance, where  $f_r(\lambda)$  is independent of direction, the hemispherical integral simplifies to

$$R(\lambda) = \int_{\Omega} f_r(\lambda) \cos \theta_i d\omega_i = \pi f_r(\lambda). \quad (6)$$

Substituting this into the imaging equation yields a simplified form:

$$I(\lambda) = R(\lambda) L_i(\lambda) S(\lambda). \quad (7)$$

Consequently, each pixel's spectral signature depends primarily on intrinsic material properties rather than viewing geometry, forming the fundamental basis of remote sensing hyperspectral imaging.

Moving into computer vision and close-range imaging, this geometric luxury disappears. The illumination is no longer distant and parallel — it may be near-field, area-based, or multiply bounced — and both the incident direction  $\omega_i$  and the exitant (viewing) direction  $\omega_r$  vary across the scene and must be tracked explicitly. The surface response is now governed by the Bidirectional Reflectance Distribution Function (BRDF), defined as the ratio of outgoing radiance to incident irradiance:

$$f_r(\lambda, \omega_i, \omega_r) = \frac{dL_r(\lambda, \omega_r)}{L_i(\lambda, \omega_i) \cos \theta_i d\omega_i}, \quad (8)$$

and the full imaging equation — integrating over the hemisphere  $\Omega$  of all incoming directions — becomes:

$$L_r(\lambda, \omega_r) = \int_{\Omega} f_r(\lambda, \omega_i, \omega_r) L_i(\lambda, \omega_i) \cos \theta_i d\omega_i. \quad (9)$$

Here  $f_r$  encodes everything about how the surface scatters light: a Lambertian surface has a constant BRDF ( $f_r = \frac{R}{\pi}$ , recovering the remote sensing case), while a glossy surface's BRDF peaks sharply around the specular direction, and a metallic surface's BRDF is both highly directional and strongly wavelength-dependent. The transition from remote sensing to computer vision is therefore a transition from a 1D spectral inversion problem to a 5D integral equation — where wavelength  $\lambda$ , two incident angles ( $\theta_i, \varphi_i$ ) and two exitant angles ( $\theta_r, \varphi_r$ ) all govern the measurement — making material appearance estimation, relighting, and scene reconstruction substantially harder, and substantially richer. Fig. 2 shows the different settings and imaging condition varying spectroscopy, remote sensing to computer vision.

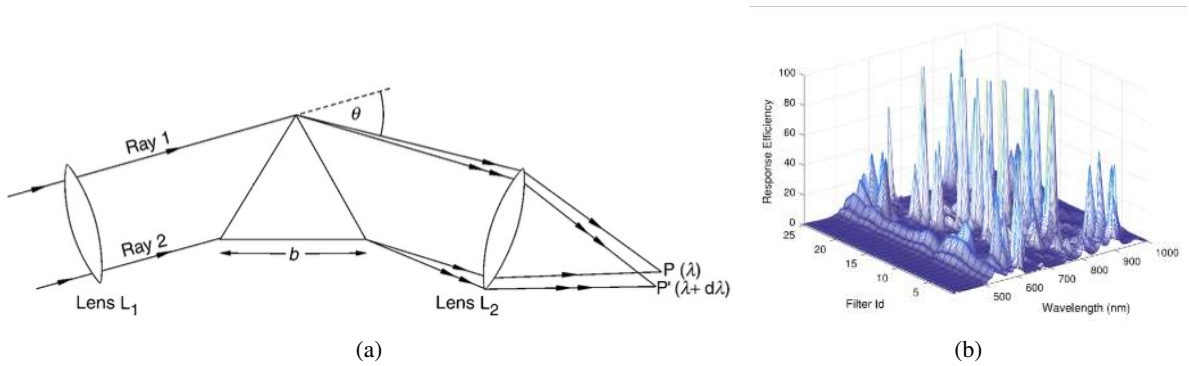
## 2.2. Imaging sensing systems

After interacting with media and the scene, the reflected light gets into a hyperspectral camera (or a spectrometer), where the incoming light is dispersed and collected so as to form a final HSI data cube. Such a process that captures the physical world by collecting scene light and generating a 3-D HSI data cube is called HSI imaging.

Generally, an HSI imaging system uses **dispersing elements** to split polychromatic light in the scene and chooses appropriate **optical sensors** to collect wavelength-wise light. The discrepancy between the requested and the available dimensionality of detectors has sparked different philosophies with respect to HSI imaging architectures. Depending on the process to generate a 3-D HSI data cube, imaging techniques can be sparsely divided into scanning-based methods and snapshot-based methods.

### 2.2.1. Dispersing elements and optical sensors

*Dispersing elements* Prism and diffraction gratings are commonly used as dispersing elements in spectrometers [122]. They are also used in combination to build a prism-grating-prism (PGP) structure in order to provide good image quality and resistance to harsh environmental conditions [121] as Fig. 4a. Interferometers and Fourier Transform Spectrometers (FTS) can also be used as alternatives in spectroscopy [253] but mostly explored in remote sensing [61, 42], not in computer vision yet due to their high cost and low temporal resolution. Besides, there are also hyperspectral fluorescence imaging for biological tissue optics [182], hyperspectral neutron tomography for material decomposition [16, 51], hyperspectral thermal measurements [65], interferometric hyperspectral imaging [184].



**Figure 4:** (a) Dispersion by a prism, courtesy of [122]; (b) Spectral response of the 25 spectral filters of the  $5 \times 5$  IMEC Snapshot Spectral Imaging camera in the 400–1000 nm range, courtesy of [292].

*Optical sensors* Being dispersed and split by wavelength, an optical sensor is designed to collect light of a specific band and generate a 3-D HSI data cube. In practice, the camera sensor then integrates the incoming spectral radiance weighted by its spectral response function  $S(\lambda)$ , as shown in Fig. 2. A typical spectral response curve for a hyperspectral sensor is shown in Fig. 4b.

### 2.2.2. Scanning-based methods

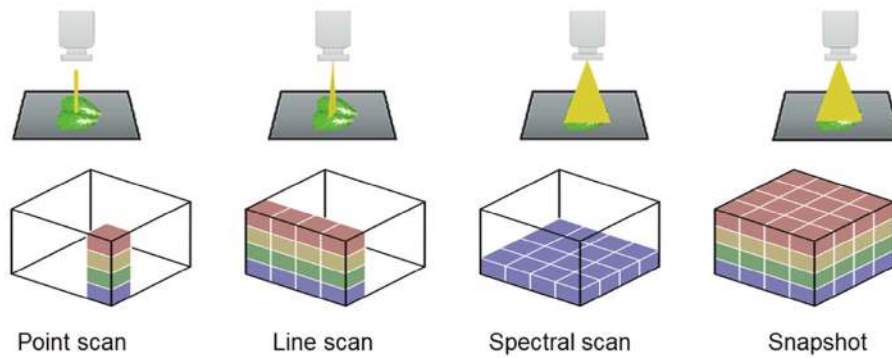
Once obtaining the intensity value of a spatial location at a specific wavelength, the next step is to generalize the point capture process and arrange these intensity values in order to generate a 3-D HSI data cube.

*Point/Line scanning methods* As shown in Tab. 1, there are point-scanning, line scanning, spectral scanning and snapshot imaging methods. The point-wise scanning method is also known as Whiskbroom Scan [257], which measures the spectral signature of a pixel in each scan and "whisk" the scene pixel-by-pixel. It offers a high degree of flexibility for spatial resolution and also shows the best spectral resolution, it is therefore widely used in chemistry and biomedical analysis [220]. But such a point-by-point scanning method suffers from high time cost, which is also described as low temporal resolution. An alternative to point-scanning, which shows a better balance between spectral and temporal resolution, is line-scanning. As the name suggests, it scans a spatial line horizontally or perpendicularly, and then spreads the captured spatial line on the spectral axis to get the spectral signature of all line points in a single shot, as shown in Fig.5. Owing to the development of remote sensing and industrial production, the line-scanning methods are widely used nowadays because they suit best the data collection process as airborne cameras scan the ground while

**Table 1**

Comparison of different imaging methods.

| Method                                                                              | Advantages                            | Drawbacks                                       | Application                                                 |
|-------------------------------------------------------------------------------------|---------------------------------------|-------------------------------------------------|-------------------------------------------------------------|
| Point scanning<br>[103, 94]<br>(Whiskbroom)                                         | Best spectral resolution              | Low temporal resolution                         | Chemical analysis<br>Biomedical diagnosis<br>Remote sensing |
| Line scanning<br>[19, 121, 6]<br>(Pushbroom)                                        | Balanced spectral/temporal resolution | Finite slit width causes overlap                | Remote sensing<br>Industrial inspection<br>Computer vision  |
| Spectral scanning<br>[96, 270, 67, 3]<br>(Filter-based scanning;<br>area scanning ) | Straightforward                       | Costly and limited spectral resolution          | Remote sensing<br>Computer vision                           |
| Snapshot<br>[296, 111, 161, 97, 291]                                                | Suitable for dynamic scenes           | Computationally heavy reconstruction algorithms | Remote sensing<br>Computer vision                           |

**Figure 5:** Imaging paradigm, courtesy of [8].

an aircraft flies forward, and industrial cameras scan products while the conveyor belt feeds materials in line. A recent promising study [94] improves conventional line scanning by adding an additional perpendicular scan that traverses the raster scan, eliminating temporal illumination variations, thereby increasing the temporal resolution to some extent.

**Spectral scanning methods** Different from point scanning and line scanning, spectral scanning methods collect 2-D spatial information within a specific wavelength at a time and generate a 3-D data cube by varying the wavelength. They are also known as 'wavelength-wise scanning', 'area-scanning' or 'filter-based method'. As such a kind of method can be achieved by putting an electronically controllable filter, e.g., Liquid Crystal Tunable Filter (LCTF) [96] or Acousto-Optic Tunable Filter (AOTF) [23] in front of the camera lens to control its spectral transmission so that the sensor can capture the scene under a specific band of light in a scan. LCTFs work from the VIS to NIR region while AOTFs operate at a broader wavelength range from UV to IR. However, LCTFs show better spatial performances and are more convenient and simpler to use [213, 3]. The spectral scanning systems are compact and operable, favoring precise analysis of spatial information, but the broad bandwidth limits its spectral resolution. Additionally, it can be seen that scanning-based imaging methods suffer from low temporal resolution and show poor performance under dynamic scenes based on our analysis above.

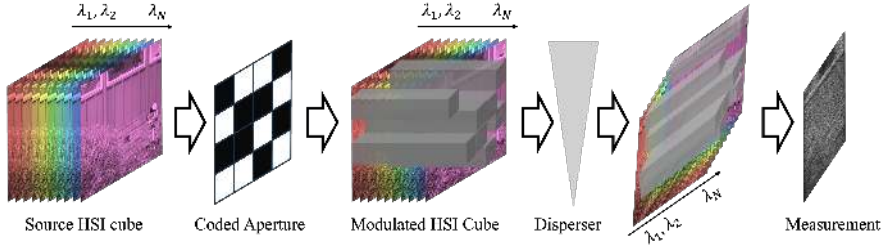
### 2.2.3. Snapshot-based methods

The aforementioned conventional scanning-based hyperspectral imaging techniques, either spatial scanning methods and spectral scanning methods, manage to get as high spectral and spatial resolution as possible at any time cost. In other words, they require a temporal sequential scanning along the spatial or spectral domain, taking a long time in data acquisition. However, the neglected time cost should also be taken into consideration in computer vision applications [99]. The need for fast imaging triggers the emergence of snapshot-based methods, which originate from compressive sensing theory [64] and capture the scene with high-dim images in a single snapshot using 2-D

detectors [227]. Such a technique generates high-dimensional HSIs with a few multiplexed measurements, thereby reducing data acquisition costs and complexity.

**Coded Aperture Snapshot Spectral Imaging** To improve the low temporal resolution and capture the scene in a snapshot, early Computed-Tomography Imaging Spectrometer (CTIS) [245, 56] projects the intensity distribution of the scene to multiple 2D images and reconstructs the original distribution using computed tomography algorithms. However, it sacrifices spatial resolution and is limited by the size of sensors.

Based on CTIS, Coded Aperture Snapshot Spectral Imaging (CASSI) [296] system is proposed and widely studied during the last decades. A typical CASSI system consists of 2 main components: a hardware encoder and a software decoder [352]. The hardware encoder conventionally consists of a series of refractive or reflective optical elements. Diffractive optical elements [15] are recently used to facilitate multi-modality imaging.



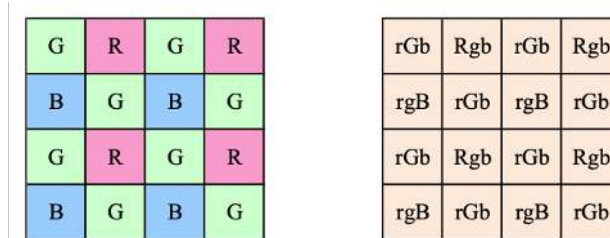
**Figure 6:** A schematic and the mathematically model of the CASSI system.

As shown in Fig 6, the imaging process of the original CASSI system with a single disperser design [296], which captures 2-D compressed measurements of realistic scenes, can be modeled as a process that the source signal pass through the coded aperture i.e. fixed mask being modulated and then a disperser shear the modulated cube along the  $y$ -axis before it can be captured in the detector plane and form the final 2-D CASSI measurement. Let  $X$  denotes the 3-D HSI data cube and  $\Phi$  denote the coded aperture (fixed mask) used for signal modulation, the measurement  $Y$  can be represented as

$$\mathbf{Y} = \Phi \mathbf{X} + G, \quad (10)$$

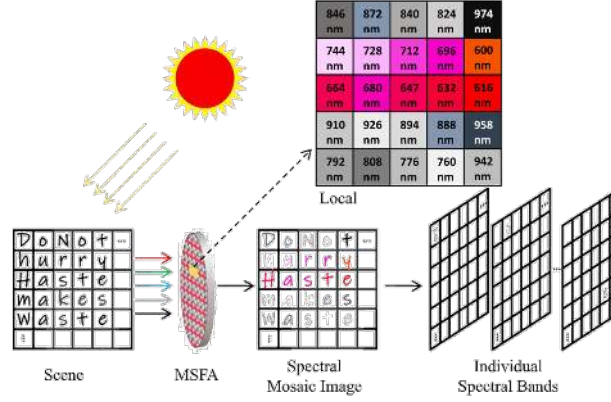
where  $G$  denotes the measurement noise. After capturing the measurement, the following task is given  $\mathbf{Y}$  (captured by the CASSI system) and  $\Phi$  (calibrated based on pre-design), solving  $\mathbf{X}$ . And it is achieved by the software decoder or reconstruction algorithm. The HSI reconstruction becomes an active research topic in computer vision, and we will elaborate it in Section 3.

**Fabry-Perot filters** The Bayer Filter Array (BFA) is the most commonly used color filter pattern in modern digital cameras. It places green filters over half of the sensor pixels, with red and blue filters covering the remaining ones. When capturing an image, each pixel records only one of the red, green, or blue components. The missing color values at each pixel are then estimated through interpolation, allowing every pixel to obtain a full color representation, as illustrated in Fig. 7.



**Figure 7:** Bayer Filter Array, where left figure shows the layout design and the right figure denotes interpolated pixel values, guessed values are shown as lower cases, courtesy of [285].

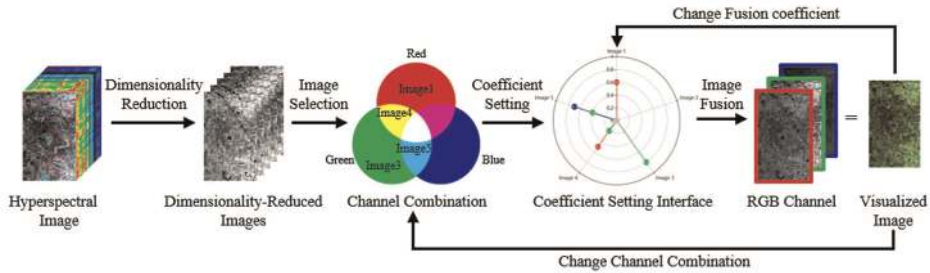
Similarly, Fabry–Perot-type band-pass filters, also known as Multispectral Filter Arrays (MSFAs), assign specific wavelengths to predefined spatial locations, resulting in distinct spectral bands being distributed across the spectral mosaic image. This process is illustrated in Fig. 8. The spectral mosaic image is then partitioned into the individual spectral-band images via masking based on the MSFA pattern or restored into a full spectral representation via a demosaicing process, more details in Sec. 3.1.5.



**Figure 8:** Multispectral Filter Array, courtesy of [332].

### 2.3. Visualization of HSIs

An HSI is commonly captured and processed as a high dimensionality data cube, which cannot be displayed directly in color space using traditional display strategies, since the existing display devices are designed for images with three channels for human visual perception. A conventional visualization method for HSIs is displaying multiple bands in different views simultaneously. However, this will deteriorate the visual interpretation as the number of views increases. Most existing remote sensing methods display HSIs in false color images, which is incompatible with natural scene scenarios in computer vision because it is not only hard to interpret when object colors are very different from their natural appearance, but also contradict with users' experience and expectations.



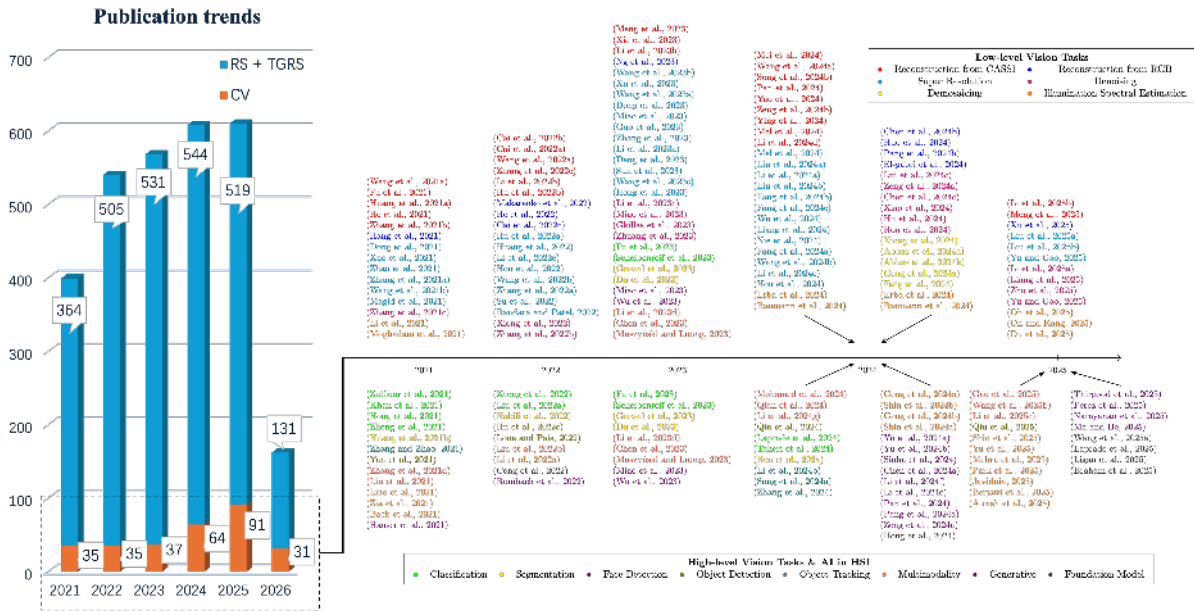
**Figure 9:** Visualization example, courtesy of [349].

In order to transform HSIs into RGB channels and then visualize them in color space, pioneer works visualize HSIs by either fusion of an HSI and a high-resolution color image via manifold alignment [194, 193], dimensionality reduction [146] or non-linear transformations that leverage domain knowledge [210]. Furthermore, [349] combined channel combination/fusion and dimensionality reduction method to realize color visualization of HSI data, as shown in Fig.9. Deep learning techniques are also exploited in HSI visualization. Apart from CNN-based methods [70, 256], generative adversarial networks are explored in [210] to estimate the inverse mapping from multispectral to RGB images. The recent BC4V [162] uses saliency detection to produce a composite image with three channels that convey rich visual information, striking a balance between information preservation and interpretability.

### 3. HSI in Computer Vision

A hyperspectral image (HSI) is a three-dimensional data cube consisting of two spatial dimensions that describe the surface structure of the scene and one spectral dimension that records detailed wavelength information. A 3-D HSI data cube can be obtained by expanding the conventional 3-channel RGB image to tens of even hundreds of channels so as to generate a 3D cube with 2 spatial dimensions and 1 spectral dimension.

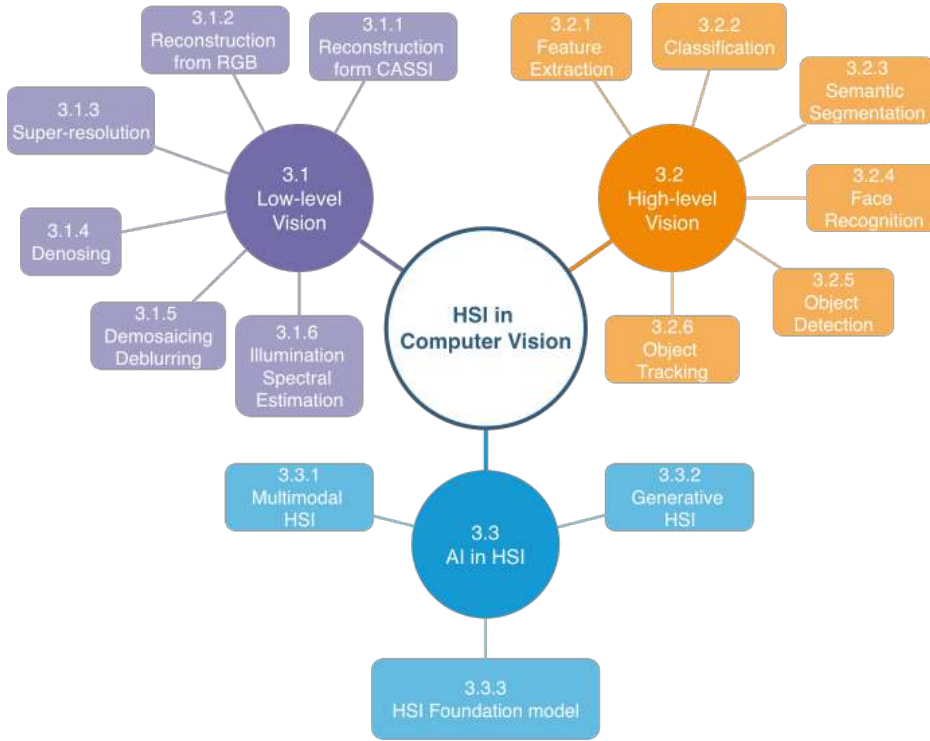
Even though hyperspectral imaging (HSI) has been widely used in remote sensing and geoscience—such as in Earth observation [251], environmental monitoring [280], agriculture [212, 155, 107] and ground object detection [254, 340, 209]—its applications in computer vision remain relatively underexplored. This trend is evident in Fig. 10, which compares the number of HSI-related publications in only two major remote sensing journals with those published across all top-tier computer vision conferences.



**Figure 10:** Left: Publication trends on HSI in remote sensing and HSI in computer vision, where RS denotes remote sensing journal, and TGRS denotes IEEE Transactions in Geoscience and Remote Sensing, CV denotes computer vision, where we count papers published on top conferences such as CVPR, ICCV, ECCV, NeurIPS, ICLR and ICML and top journals including IEEE TPAMI, IEEE TIP, IEEE TCI and IJCV. Data source: [semanticscholar.org](https://www.semanticscholar.org), [ieeexplore.ieee.org](https://ieeexplore.ieee.org) and [link.springer.com](https://link.springer.com). Right: HSI in computer vision, note that only some papers are listed here.

Despite substantial progress in remote sensing, the methods developed for remote sensing HSIs cannot be directly transferred to computer vision scenarios due to the difference in the noise model, as shown in Fig. 2. The first major difference lies in illumination conditions. Remote sensing scenes typically involve a single, distant light source (the sun), resulting in relatively consistent illumination across the scene. In contrast, computer vision environments often involve multiple illumination sources—such as lamps, reflections, and varying ambient lights—which invalidate the assumption of uniform illumination. Second, the imaging distance differs significantly. In remote sensing, sensors operate from high altitudes—on near-Earth orbits or airborne platforms—and therefore rely on a small field of view (FOV) to maintain adequate spatial resolution on the ground. In computer vision, however, hyperspectral cameras image scenes at much shorter distances and typically with a large FOV to capture as much contextual information as possible. An additional discussion of these differences can be found in Section 6.

In computer vision, the research on HSI has a twofold aim. Firstly, it should recover high quality data devoid of sensor degradation, environment noise, illumination variations, and scale change. Secondly, such research must explore the use of the imaging spectra for purposes of scene analysis and object material recognition. In recent years, the rapid



**Figure 11:** The organization of HSI in computer vision

advancement of artificial intelligence has also spread to HSI research [26], where multi-modal HSI, generative HSI, and HSI foundation models have become three major focal points. The organization of this section is shown in Fig. 11.

### 3.1. Low-level Vision

Raw HSIs captured by different sensors often exhibit various deficiencies. As discussed in Section 2, imaging systems inherently face trade-offs among spatial, spectral, and temporal resolution, which prevent the captured data from forming an ideal HSI cube. Consequently, it becomes necessary to enhance these imperfect data—characterized by low spatial resolution, limited spectral bands, noise, or illumination inconsistencies—to produce clearer, richer, noise-free, and illumination-agnostic representations. This enhancement process corresponds to low-level vision tasks in computer vision.

Following Section 2, once a raw HSI data cube is collected, various methods can be applied to enhance its quality from different perspectives. HSIs captured using different imaging systems may also introduce deficiencies to the data cube. For example, 2D CASSI measurements intrinsically require post-processing to reconstruct the 3D data cube, a process known as CASSI reconstruction [36, 352]. Sensors with limited spectral resolution necessitate spectral reconstruction [361], while those with low spatial resolution call for spatial enhancement methods, commonly formulated as hyperspectral super-resolution [311]. Additionally, the image acquisition process may introduce noise, blur, or mosaic artifacts; accordingly, denoising [364], deblurring [98], and demosaicing [81] techniques have been developed to address these issues. Beyond these sensor-induced degradations, the intrinsic composition of HSIs can be further analyzed through spectral unmixing [28].

In this section, we start from a review of different CASSI variants that are largely overlapped with HSI reconstruction task. We then focus on different low-level vision tasks improving HSI quality by either spectral or spatial enhancement, which are exactly spectral reconstruction, super-resolution, and denoising. Furthermore, other low-level vision tasks such as demosaicing and deblurring follow up.

**Table 2**

Optimization-based Hyperspectral Imaging.

| Prior Categories | Expression                                                                            | Pros                                 | Cons                                                      |
|------------------|---------------------------------------------------------------------------------------|--------------------------------------|-----------------------------------------------------------|
| Smooth Prior     | $\arg \min_{\mathbf{x}} \ \mathbf{y} - \Phi \mathbf{x}\ _2^2 + \gamma TV(\mathbf{x})$ | Spatial piecewise smoothness         | High-frequency loss; over-smoothing                       |
| Sparse Prior     | $\arg \min_t \ \mathbf{y} - \Phi \Psi t\ _2^2 + \gamma \ t\ _1$                       | Robust                               | High-dimensional structure loss                           |
| Low-rank Prior   | $\arg \min_{\mathbf{x}} \ \mathbf{y} - \Phi \mathbf{x}\ _2^2 + \gamma R(\mathbf{x})$  | Structural or contextual information | High computational cost and heavy parameter tuning burden |

### 3.1.1. Reconstruction from CASSI

*General HSI Spectral Reconstruction* aims to enrich more spectral details and to characterize spectral signatures more finely. Mathematically, the HSI imaging model is formulated as:

$$\mathbf{Y} = \Phi \mathbf{X} + \mathbf{G}, \quad (11)$$

where  $\mathbf{Y}$  denotes the 2-D CASSI measurement or the observed image,  $\Phi$  is the sensor matrix,  $\mathbf{X}$  is the HSI data cube and  $\mathbf{G}$  is the measurement noise. To estimate the original HSI by  $\hat{\mathbf{X}}$ , the optimization of HSI reconstruction could be modeled as:

$$\hat{\mathbf{X}} = \arg \min_{\mathbf{X}} \frac{1}{2} \|\mathbf{Y} - \Phi \mathbf{X}\|_2^2 + \eta J(\mathbf{X}), \quad (12)$$

Where  $J(\mathbf{X})$  denotes the regularizer term with parameter  $\eta$ .

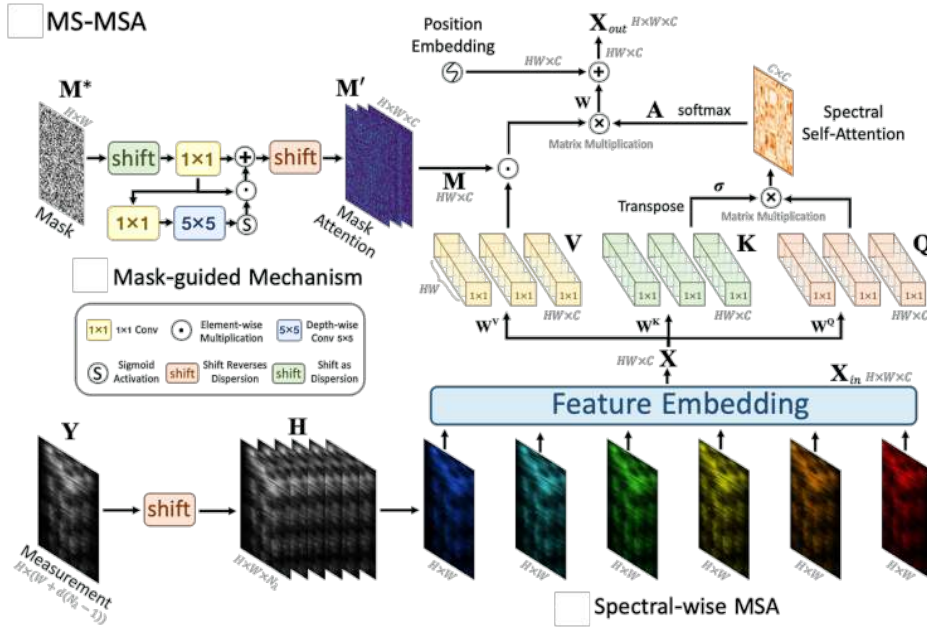
There are various classifications for enormous numbers of different CASSI methods [119, 205, 303, 118]. Apart from original single disperser CASSI [296], other variants include dual-camera design (DCD) [304], spatial-spectral compressive spectral imager (SSCSI), coded aperture compressive temporal imaging (CACTI) [208], and the most recent multiple snapshot imaging (MSIS). Low-level vision tasks mainly focus on the diverse software decoder techniques of the CASSI system. Basically, the decoders or CASSI reconstruction algorithms vary from optimization-based algorithms to deep learning-based algorithms [327].

*Optimization-based methods* Optimization-based methods typically consist of using priors to characterize the properties of HSI and solving the objective optimization problem. The most emphasized three kinds of priors include smooth prior, sparse prior and low-rank prior, as demonstrated in Table 2. *TwIST* [27] is a classic reconstruction algorithm to reconstruct HSI from its snapshot spectral imaging. For instance, TLPLN [365] proposes to learn the tensor low-rank prior of HSI in the feature domain to promote the reconstruction quality. Such methods usually handle only single snapshot CASSI.

*Deep learning-based methods* require a large amount of data, and the pre-trained networks are usually designed for specific CASSI systems. Three mainstream methods are summarized in Table 3. General deep learning methods leverage data in an *end-to-end* manner. Their development underwent the process from attention-free structures, soft-attention structures to the most fashionable self-attention mechanisms. For attention-free structures, [92] presents an effective CNN that learns the deep prior from the extra external dataset as well as the internal information from the input coded image.

With the advent of transformer, self-attention mechanism is introduced into HSI reconstruction and achieve strong performance [228, 226, 301, 33, 131, 300]. For instance, MST [34] aims at capturing spectral-wise similarity and long-range dependencies and makes use of the guidance effect of the mask to propose Mask-guided Spectral-wise Transformers. Besides, *Deep Unfolding Network (DUN)*, or Deep Unrolling Network, has achieved success in the snapshot compressive sensing field due to its great interpretability and performance. [134, 369, 227, 345, 169, 310, 178]

*Plug-and-Play (PnP)* methods bridge the gap between optimization-based methods and deep-learning-based methods. They plug off-the-shelf denoisers into optimization algorithms, but are still designed for single snapshot CASSI. Moreover, a recent PnP study [327] develops a framework applicable for any CASSI system, expanding from single snapshot systems to multi-shot systems.



**Figure 12:** Mask-guided Spectral-wise Multi-head Self-Attention (MS-MSA). Spectral-wise MSA treats each spectral feature as a token and calculates self-attention along the spectral dimension. Mask-guided Mechanism directs the Spectral-wise Multi-head Self-Attention to pay attention to spatial regions with high-fidelity HSI representations. Some components are omitted for simplification. Please refer to [34] for details.

### 3.1.2. Reconstruction from RGB

In addition to reconstruction from CASSI, which is closely related to compressive sensing, it is also well-explored to reconstruct HSIs from RGB images. Recent advance on reconstruction of HSIs from RGB images are partially devoted to improving **camera design**, exploring **spectral nature** and developing **network structures**.

Table 4: Comparison of representative hyperspectral reconstruction methods.

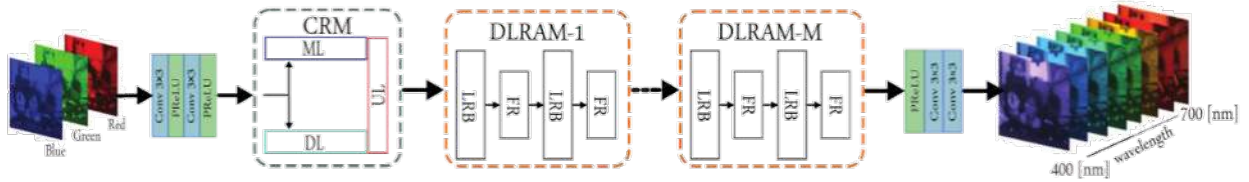
| Method | Category                 | Input             | Output                | Key Idea                                                                                     |
|--------|--------------------------|-------------------|-----------------------|----------------------------------------------------------------------------------------------|
| [4]    | Reconstruction from RGB  | RGB image         | HSI                   | Learn Gaussian processes under a representation model inspired by the Beta-Bernoulli process |
| [93]   | Camera design            | RGB image         | HSI                   | Joint optimization of camera spectral response and RGB-to-HSI reconstruction                 |
| [116]  | Low-rank prior           | Degraded HSI      | Restored HSI          | Assume a global spectral low-rank subspace shared across local patches                       |
| [219]  | Imaging system           | HSI               | Task-specific outputs | CMOS-compatible fast hyperspectral camera using metasurface optics                           |
| [35]   | Transformer              | RGB image         | Initial HSI           | Spectral-wise multi-head self-attention leveraging spectral self-similarity                  |
| [71]   | Low-rank prior           | MSI               | HSI                   | Local low-rank spectral and global contextual priors for spectral super-resolution           |
| [249]  | Low-rank prior           | Degraded HSI      | Clean HSI             | Unsupervised diffusion-based hyperspectral restoration                                       |
| [138]  | Self-supervised learning | RGB image         | Spectral reflectance  | Meta-auxiliary learning for test-time adaptation                                             |
| [334]  | Reconstruction from RGB  | HR-RGB and LR-MSI | HR-HSI                | Continuous coordinate-based spectral reconstruction                                          |

*Continued on next page*

| Method | Category                | Input                 | Output | Key Idea                                           |
|--------|-------------------------|-----------------------|--------|----------------------------------------------------|
| [156]  | Reconstruction from RGB | Single-shot RGB photo | HSI    | Co-design of color reference charts and algorithms |

*Camera design* [93] jointly selects the optimal camera spectral responses (CSR) and learns a mapping to recover HSI from a single RGB image captured with the selected camera. *Hyplex<sup>TM</sup>* [219] replaces traditional optical projectors with nanoscale metasurfaces as suitably defined projectors to compress high-dim spectral data into a low-dim space, and implements an optimal linear spectral encoder hardware for spectral reconstruction. Physpec [341] proposed to estimate camera spectral sensitivity (CSS) explicitly and apply orthogonal subspace decomposition.

*The nature of spectra* is assumed to exist in a subspace with a low rank. [143] shows that the spectra of natural scenes lie on an intrinsically low-dimensional manifold. [242] proposes to simulate the camera response as a hardware implementation of the convolution layer to learn the optimized response functions and hyperspectral reconstruction mapping. [116] explores the global spectral low-rank subspace of HSI and proposes a unified restoration paradigm to solve the restoration task. MST++ [35] firstly attempts to explore the potential of Transformer in spectral reconstruction task. It develops a spectral-wise multi-head self-attention based on the HSI spatially sparse while spectrally self-similar nature. DLRA-Net [71] is proposed to integrate local low-rank spectral and global contextual priors for improved spectral super-resolution. [249] decomposes 2 low-rank components to achieve restoration. FRN [223] treated spectral reconstruction as a progressive process, predicting from broad to narrow bands or employing a coarse-to-fine approach for predicting the next wavelength, which follows the low-rank property of spectral data.



**Figure 13:** Deep Local Residual Attention Network (DLRA-Net). Contextual-attention Refinement Module (CRM) learns complex contextual priors to remove the undesired artifacts, Dual Local Residual Attention Module (DLRAM) progressively build up the HSI by capturing local and global contexts. Courtesy of [71].

In addition to low-rank subspace, spectral correlation and physical relationship are also studied. PriNET+ [112] designed a new network by taking advantage of 2 intrinsic properties of HSI including spectral correlation and projection property to recover HSI from its corresponding RGB image. [334] points out that existing methods neglect the continuous nature of 3-D spectral signals and it achieves continuous reconstruction via a spectral profile interpolation module. [138] learns to recover spectral reflectance from RGB images, which uses meta-auxiliary learning (MAXL) to combine external information with internal information, integrating the physical relationship between the spectral reflectance and the corresponding RGB images. It also studies the effect of RGB captured under multiple illuminations. Diff-Spectra [342] integrated supervised physics-aware spectral estimation and unsupervised high-fidelity spectral regularization. MP-HSIR [321] released the specific degradation assumption and proposed a multi-prompt framework that integrates spectral textual and visual prompts to achieve universal HSI restoration across diverse degradation types and intensities. [90] systematically analyzed a category of data-driven spectral reconstruction methods from RGB images, it especially underscored the metamerism issue that has not yet been extensively evaluated.

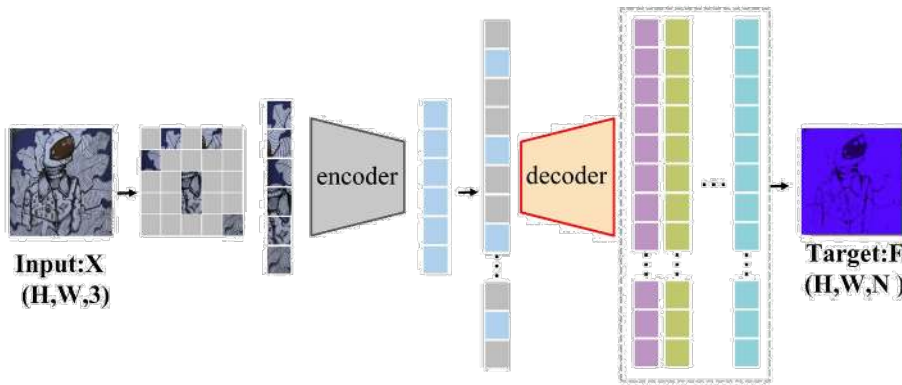
*Fusion of RGB and CASSI* Compared with conventional iterative methods, which update reconstructed HSIs step-by-step to explore the spatial-spectral priors of full HSIs, [118] analyzes the property of the low-dimensional subspace that HSIs are assumed to underlie, decomposing a HSI into two components: spectral basis and spatial coefficients. Accordingly, it proposes to utilize RGB measurement to estimate coefficients and adopt CASSI measurement to provide spectral basis.

*Network structures: CNN, 3-D conv, transformer, DUN, self-supervised* Compared with the fully Bayesian model [4] that recovers HSI from RGB images, deep neural networks are of special interest. ISTA-Net [359] casts the

**Table 3**  
Deep Learning-based HSI Reconstruction Methods.

| Category              | Algorithm                                                       | Advantages                                                            | Limitations                                                                       |
|-----------------------|-----------------------------------------------------------------|-----------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| End-to-End            | Attention-free:<br>HyperReconNet [305]                          | Simple structure                                                      | Lack of adaptive weight adjustment and low information utilization rate           |
|                       | Soft Attention:<br>DEIL [303, 92]                               | Improved information utilization                                      | Limited capability in modeling long-range dependencies and non-local correlations |
|                       | Self Attention:<br>TSA-Net [226], MST [34], CST [33], GST [301] | Effective long-range dependency modeling with large receptive fields  | High computational complexity and memory consumption                              |
| Unfolding / Unrolling | IST: ISTA-Net [359]                                             | Ability to model non-local correlations                               | Information transfer loss across stages and fixed iterative steps                 |
|                       | HQS: DSSP [303], TLPLN [365]                                    | Model-driven optimization framework                                   | Relatively complex optimization process                                           |
|                       | GAP: GAP-Net [225]                                              | Efficient iterative reconstruction                                    | Limited adaptability under diverse sensing conditions                             |
|                       | GSM: DGSM [134]                                                 | Flexible forward-model adaptation                                     | Increased training complexity                                                     |
| Plug-and-Play         | ADMM [39]                                                       | Good flexibility, interpretability, and robustness under limited data | High computational cost and sensitivity to initialization                         |
|                       | GAP [353]                                                       | Flexible integration of priors and denoisers                          | Limited real-time performance                                                     |

iterative shrinkage-thresholding algorithm into deep network form and inspires lots of subsequent works. [93] further proposes a CNN-based method to jointly select the optimal camera spectral response and learn a mapping to recover HSI from a single RGB image captured with the selected camera. *3-D conv* STAS [117] disentangles 3D convolution into lightweight 2-D spatial and spectral convolutions and builds a spectrum-aware search space for HSI restoration. Furthermore, it utilizes a neural architecture search strategy to learn the most efficient architecture. On the other hand, *DUNs attract equal interest*. HeorsNet [369] is based on DUNs' lack of cross-phase feature interaction and adaptive parameter adjustment. It includes several phases under the ISTA-unfolding framework. PADUT [171] combines deep unfolding frameworks and transformers to propose a deep unfolding transformer structure.



**Figure 14:** Masked Autoencoder-based Knowledge Transfer network (MAE-KTSR). Courtesy of [82].

MAE-KTSR [82] applied the concept of MAE to spectral reconstruction and propose an MAE-based knowledge transfer framework for RGB-to-HSI reconstruction.

[156] introduced a co-design of color reference charts and computational algorithms to extract high-resolution spectral information (1 – 2 nm) from a single-shot photo of an arbitrary scene or sample.

### 3.1.3. Super-Resolution

Hyperspectral image super-resolution differs from spectral reconstruction and aims to increase the spatial resolution of HSI. There are two benchmark super-resolution problems, including single hyperspectral image super-resolution and fusion-based hyperspectral image super-resolution. The former only utilizes a single low-resolution (LR) input to recover the high-resolution (HR) result, while the latter fuses the HS image with auxiliary images, such as panchromatic (PAN) images and multispectral images (MSI) .

*Single hyperspectral image super-resolution* does not utilize the auxiliary image, mining only information from the low-resolution image itself to recover the high-resolution image.

Existing methods for single HSI super-resolution includes two-step methods, group-based methods and other end-to-end methods. Two-step methods first super-resolves the bands just by spatial information using CNN [181, 128] or nonnegative matrix factorization (NMF) [354], and then incorporates a manual way to make a further utilization of the spectral information. Group-based methods concentrate on grouping the whole HS image along the spectral dimension into spectral subgroups for super-resolution by firstly dividing the bands of HS images into groups that are reconstructed by the branch network with shared parameters and finally merges the subgroups for processing to learn spectral correlation [144, 312], or by separating adjacent and extremely similar bands in each group and shuffling spectral band sequences [309] since the neighboring bands of the HS image are highly correlated in the spectral domain. With the prosperity of deep learning, end-to-end methods [376, 360, 127, 377] have been of special interest.

*Fusion-based hyperspectral image super-resolution* has the goal of producing the HSIs with finer spatial resolution than raw data. Panchromatic-based hyperspectral image super-resolution, also known as Pansharpening, is to obtain a high-resolution (HR) image from the fusion of an HR panchromatic (PAN) image and a low-resolution (LR) HSI [328, 63]. Panchromatic images are predominantly acquired by satellite imaging systems and employed in remote sensing applications [335, 165, 216]. However, their characteristics make them less compatible with natural scene computer vision tasks. To address this gap, researchers have explored adapting pansharpening techniques for hyperspectral image (HSI) fusion [59], as well as examining the generalization capability of such algorithms in computer vision scenarios [299].

We mainly focus on the hyperspectral and multispectral image fusion (HMIF). A typical HMIF process usually composes of a low resolution hyperspectral image (LR-HSI) and a high resolution multispectral image (HR-MSI).

Table 5: Comparison of different super-resolution methods.

| Method | Category             | Input                                       | Output                | Key Idea                                                                                           |
|--------|----------------------|---------------------------------------------|-----------------------|----------------------------------------------------------------------------------------------------|
| [75]   | Camera level         | HSI and MSI under different exposure levels | HR-HSI                | Proposed a deep unfolding structure with automatic exposure correction                             |
| [383]  | Prior exploration    | HR-MSI and LR-HSI                           | HR-HSI                | Get inspired by the classic wavelet decomposition                                                  |
| [338]  | Prior exploration    | HR-MSI and LR-HSI                           | HR-HSI                | Fully considers the spatial/spectral subspace low-rank relationships                               |
| [174]  | Prior exploration    | LR-HSI                                      | HR-HSI                | Explore a coarse and fine stage structure for HSI super-resolution                                 |
| [63]   | Framework innovation | HR-MSI and LR-HSI                           | HR-HSI                | Connect model-based research with model-guided design                                              |
| [219]  | Camera               | HSI                                         | task-specific outputs | a CMOS-compatible, fast hyperspectral camera that replaces bulk optics with nanoscale metasurfaces |

*Continued on next page*

| Method | Category                | Input                    | Output              | Key Idea                                                                                                                                                                |
|--------|-------------------------|--------------------------|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [127]  | Prior exploration       | LR-HSI                   | HR-HSI              | Formulate the embedding of spatial-spectral information of HSI from the probabilistic perspective                                                                       |
| [308]  | Prior exploration       | LR-HSI                   | HR-HSI              | Interval sampling of band groups to form new and more informative groups                                                                                                |
| [62]   | Probabilistic framework | HR-MSI and LR-HSI        | HR-HSI              | embeds the Bayesian inference with the Gaussian noise prior assumption to the deep neural network                                                                       |
| [229]  | SR                      | LR-HSI                   | HR-HSI              | Proposed a self-supervised deep diffusion spatio-spectral model                                                                                                         |
| [108]  | Prior exploration       | HR-MSI and LR-HSI        | HR-HSI              | the first work to explore the cooperative relationship between prior and likelihood in HSI-SR                                                                           |
| [302]  | Framework innovation    | LR-HSI                   | HR-HSI              | An effective knowledge-driven, end-to-end and data-dependent HS image super-resolution framework                                                                        |
| [363]  | Framework innovation    | LR-HSI                   | HR-HSI              | introduce the use of Vision Transformer for the single-HSI-SR task and present a novel self-attention method                                                            |
| [241]  | Framework innovation    | HR-MSI and LR-HSI        | HR-HSI              | a hybrid pixel-wise registration learning framework                                                                                                                     |
| [222]  | Framework innovation    | CTIS sensor measurements | High resolution HSI | A novel multi-scale learning framework for incremental image refinement                                                                                                 |
| [170]  | Framework innovation    | LR-HSI                   | HR-HSI              | A self-training framework where more accurate pseudo-labels and more accurate LR-HR relationships are generated                                                         |
| [76]   | Prior exploration       | HR-MSI and LR-HSI        | HR-HSI              | integrate the nonnegative matrix-vector tensor factorization into the deep image prior framework to jointly learn the abundance tensor and spectral feature matrix      |
| [202]  | Prior exploration       | LR-HSI                   | HR-HSI              | a dual-domain network with hybrid convolution and progressive upsampling to exploit both spatial-spectral and frequency information of the hyperspectral data           |
| [326]  | Framework innovation    | HR-MSI and LR-HSI        | HR-HSI              | embedded the linear mapping that links the HR-HSI to HR-MSI and LR-HSI                                                                                                  |
| [315]  | Framework innovation    | HR-MSI and LR-HSI        | HR-HSI              | designed a CNN-based variational probabilistic autoencoder framework                                                                                                    |
| [129]  | Framework innovation    | HR-MSI and LR-HSI        | HR-HSI              | the first attempt to apply the Transformer architecture to the fusion task                                                                                              |
| [284]  | Framework innovation    | HR-MSI and LR-HSI        | HR-HSI              | designed a dual spatial-spectral pyramid network with Transformer                                                                                                       |
| [55]   | Framework innovation    | HR-MSI and LR-HSI        | HR-HSI              | proposed a Shuffle-and-Reshuffle strategy and designed a window pyramid structure                                                                                       |
| [198]  | Framework innovation    | HR-MSI and LR-HSI        | HR-HSI              | provided an example of spatial-spectral transformer-based U-Net                                                                                                         |
| [74]   | Framework innovation    | HR-MSI and LR-HSI        | HR-HSI              | employed a novel multi-input multi-output spatial-spectral transformer structure to integrate the spectral information of LR-HSI and HR-MSI at different scales         |
| [203]  | Framework innovation    | HR-MSI and LR-HSI        | HR-HSI              | improved the spatial attention in Vision Transformer and utilized cross-attention mechanism to extract spatial and spectral features from LR-HSI and HR-MSI, separately |

*Continued on next page*

| Method | Category             | Input             | Output | Key Idea                                                                                                                                                                                                                                         |
|--------|----------------------|-------------------|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [307]  | Framework innovation | HR-MSI and LR-HSI | HR-HSI | developed a detail-preserving conditional invertible network, which successively converted image mapping to detail mapping, extracted auxiliary features directly from source images and learned the detail mapping based on auxiliary features. |
| [164]  | Framework innovation | HR-MSI and LR-HSI | HR-HSI | used a series of cascade-attention blocks to fuse high-spatial-resolution RGB images and to recover spectral sequential information.                                                                                                             |
| [168]  | Framework innovation | HR-MSI and LR-HSI | HR-HSI | proposed an unsupervised blind fusion network that operates on a single HSI and RGB image pair and requires neither known degradation models nor any training data.                                                                              |
| [319]  | Framework innovation | HR-MSI and LR-HSI | HR-HSI | used HR-MSI as high-frequency detail auxiliary input to introduce a new implicit neural feature fusion function                                                                                                                                  |
| [192]  | Framework innovation | HR-MSI and LR-HSI | HR-HSI | proposed a spatial and frequency implicit fusion function to capture high frequency information and expanding the receptive field                                                                                                                |
| [127]  | Framework innovation | HR-MSI and LR-HSI | HR-HSI | formulates HS embedding as an approximation of the posterior distribution of a set of carefully defined HS embedding events, and then incorporates the proposed feature embedding scheme into a source-consistent super-resolution process.      |
| [133]  | Framework innovation | HR-MSI and LR-HSI | HR-HSI | formulated deep hyperspectral image fusion as a differentiable optimization problem and proposed a novel regularization strategy to fully exploit the spatio-spectral dependency by a spatially adaptive 3-D filter                              |
| [124]  | Framework innovation | HR-MSI and LR-HSI | HR-HSI | decouples the input into common (or cross-sensor) and sensor-specific components and then blends them by a model-guided coupled spectral unmixing net                                                                                            |
| [333]  | Framework innovation | HR-MSI and LR-HSI | HR-HSI | proposed a deep blind hyperspectral image fusion method by unfolding model-based maximum a posterior (MAP) estimation into a network implementation                                                                                              |
| [336]  | Prior exploration    | HR-MSI and LR-HSI | HR-HSI | modeled low-rank visual data through matrix factorization by utilizing neural networks to parameterize the factor matrices                                                                                                                       |
| [89]   | Framework innovation | HR-MSI and LR-HSI | HR-HSI | combined conditional diffusion and constrained diffusion within the widely-used denoising diffusion probabilistic models framework                                                                                                               |
| [255]  | Framework innovation | HR-MSI and LR-HSI | HR-HSI | formulated the data fusion problem as the minimization of a mixed criterion with a convex differentiable regularization                                                                                                                          |
| [200]  | Prior exploration    | HR-MSI and LR-HSI | HR-HSI | guiding the spatial sampling process through mining the structural information correlations of spectra                                                                                                                                           |

#### Camera level, Prior exploration and network level

Camera exposure UHSR-AEC [75] focused on mitigating the effects of different exposure levels and proposed a deep unfolding HSI super-resolution with automatic exposure correction to generate a high-quality fused HSI-SR under imbalanced exposures.

**Prior exploration** Prior information can guide the fusion process, resulting robust fusion performances. PZRes-Net [383] got inspirations from classic wavelet decomposition-based image fusion and proposed a novel lightweight deep neural network-based framework. SSLRR [338] considered the spatial and spectral priors from high-level analysis, i.e., subspace low-rank relationships between HR-MSI / LR-HSI and latent HSI. DualSR [174] was motivated by the fact that adjacent bands of HSIs exhibit high similarity and it proposed a dual-stage coarse-to-fine super-resolution method. GSSR [308] designed the group shuffle to separate adjacent and extremely similar spectral bands and then compensated for the shuffle by the local spectral continuity constraint module. CS2DIPs [76] coupled spatial and spectral deep image priors by integrating the nonnegative matrix-vector tensor factorization into the deep image priors framework to jointly learn the abundance tensor and spectral feature matrix. SRDNet [202] captured inter-spectral self-similarity in the spatial domain and optimized the model in the frequency domain and improved the perceptual quality of HSI. [336] proposed to model low-rank visual data through matrix factorization by utilizing neural networks to parameterize the factor matrices. A recent work [200] proposed a heterospectral structure compensation sampling (HSC-sampling) mechanism to guide the spatial sampling process through mining the structural information correlations of spectra.

**Framework innovation** In the era of deep learning, innovation in network architectures has become one of the most active research topics, with extensive efforts devoted to designing a wide variety of neural network models. Early convolutional neural networks have several representatives, where MHF-Net [326] embedded the linear mapping that links the HR-HSI to Hr-MSI and LR-HSI, it also puts forth a consistent and a blind version of network to cope with the generalization ability. FusionNet [315] designed a CNN-based variational probabilistic autoencoder framework to fuse spatial and spectral information in an end-to-end and unsupervised manner. MoG-DCN [63] proposed an iterative algorithm to fuse a pair of images and unfold it into a novel model-guided deep convolutional network. DC-Net [124] devised a novel decoupled-and-coupled network to progressively fuse information from the pixel level to subpixel level and from the image level to feature level. DHIF-Net [133] formulated deep hyperspectral image fusion as a differentiable optimization problem and proposed a novel regularization strategy to fully exploit the spatio-spectral dependency by a spatially adaptive 3-D filter. Similarly, [255] formulated the data fusion problem as the minimization of a mixed criterion with a convex differentiable regularization and validated the improved ability of their semi-quadratic regularization via performance comparison.

With the rise of transformer backbone, lots of transformer-based fusion networks take place of CNN backbones. Fusformer [129] is the first attempt to apply the Transformer architecture to the fusion task. Different attention variants were implemented. CasFormer [164] used a series of cascade-attention blocks to fuse high-spatial-resolution RGB images and to recover spectral sequential information. Motivated by the idea that different attention can force the model to focus on the corresponding missing information, LRTN [203] improved the spatial attention in Vision Transformer and utilized cross-attention mechanism to extract spatial and spectral features from LR-HSI and HR-MSI, separately.

Experience accumulated from the CNN era in computer vision has also been leveraged in combination with transformer backbones, such as through multi-scale learning strategies. For instance, DSPnet [284] designed a dual spatial-spectral pyramid network with Transformer for HSI fusion. To address the quadratic computational complexity of Transformer, PSRT [55] proposed a Shuffle-and-Reshuffle strategy and designed a window pyramid structure, efficiently resolving the computational complexity issue. ESSAformer [363] proposes an iterative refinement process to extract features at different scales and an efficient self-attention based on the spectral correlation coefficient kernel. [222] proposed a novel multi-scale learning framework for incremental image refinement, which jointly reconstructs spectral information and spatial information. MIMO-SST [74] employed a novel multi-input multi-output spatial-spectral transformer structure to integrate the spectral information of LR-HSI and HR-MSI at different scales. U-shape network structure is also exploited in fusion task, SSTF-Unet [198] provided an example of spatial-spectral transformer-based U-Net.

In particular, Implicit Neural Representation (INR) recently receives a surge of interest [278, 230]. BUSIFusion [168] proposed an unsupervised blind fusion network that operates on a single HSI and RGB image pair and requires neither known degradation models nor any training data.  $INF^3$  [319] used HR-MSI as high-frequency detail auxiliary input to introduce a new implicit neural feature fusion function to solve the multispectral and hyperspectral image fusion problem. Inspired by the phenomena that the Fourier amplitudes of the HR-HSI latent code and LR-HSI are remarkably similar while their phases exhibit different patterns, FeINFN [192] proposed a spatial and frequency implicit fusion function to capture high frequency information and expanding the receptive field.

Apart from deep networks, other types of networks such as probabilistic frameworks and deep unfolding networks are also worth noticing. PDE-Net [127] formulates HS embedding as an approximation of the posterior distribution

of a set of carefully defined HS embedding events, and then incorporates the proposed feature embedding scheme into a source-consistent super-resolution process. BayeSR [62] proposed a noise prior knowledge informed Bayesian inference network for super-resolution, which reasonably embeds the Bayesian inference with the Gaussian noise prior assumption into the deep neural network. [333] leverages deep unfolding network (DUN) to realize HSI fusion. It proposes a deep blind hyperspectral image fusion method by unfolding model-based maximum a posterior (MAP) estimation into a network implementation. DAEM [108] proposed a positive cycle between fusion and degradation estimation under a new probabilistic framework to build a stable, interpretable and lightweight super-resolution system.

In addition to the above network frameworks, there are other variants. [302] combined vision transformer, 3-D convolution and recurrent neural network to build a knowledge-driven deep unrolling technique. DDS2M [229] utilizes diffusion models and infers the parameters of the variational spatio-spectral module using only the degraded HSI without any extra training data. [89] combined conditional diffusion and constrained diffusion within the widely-used denoising diffusion probabilistic models framework. DCINN [307] developed a detail-preserving conditional invertible network, which successively converted image mapping to detail mapping, extracted auxiliary features directly from source images and learned the detail mapping based on auxiliary features. [241] proposed a hybrid pixel-wise registration learning framework for fusion-based HSI SR to improve the generalization capacity. [170] developed a novel self-training framework to improve the quality of the synthesized samples and a new data augmentation method to increase the diversity of the training data at test time. [306] aimed at solving the lack of generalization in fusion models with fixed resolution scales, and proposed a novel continuous low-rank factorization by integrating two neural representations into the matrix factorization, capturing spatial and spectral information, respectively.

When fusing Hr-MSI and LR-HSI, the alignment of two components is a critical factor while it is usually overlooked, resulting in inaccuracy fusion and suboptimal performance. [157] coped with this issue via the combination of heterogeneous feature extraction, multistage feature alignment and attentive feature fusion. [371] constructed a two-stage image alignment with a synthetic generation pipeline in the image alignment module; propose a Feature Aggregation module and an Attention Fusion module.

Existing works rely on predefined or simplified degradation models during fusion training and overlook the degradation characteristics in designing the fusion networks. Recently, [313] proposed a generalized tensor formulation based on a Kronecker decomposition to handle any general spatial-degradation matrix. [189] introduced non-uniformity into the degradation processes of LR-HSI and HR-MSI and employed positional encoding to endow the fusion process with the position-aware attribute. [37] explored the combination of language and masks to guide image fusion tasks. [320] employed multilinear algebra to drive the development of a novel fully-connected self-attention (FCSA) method to fully exploit local and non-local domain-specific correlations among multi-source images. [336] modeled low-rank visual data through matrix factorization by utilizing neural networks to parameterize the factor matrices.

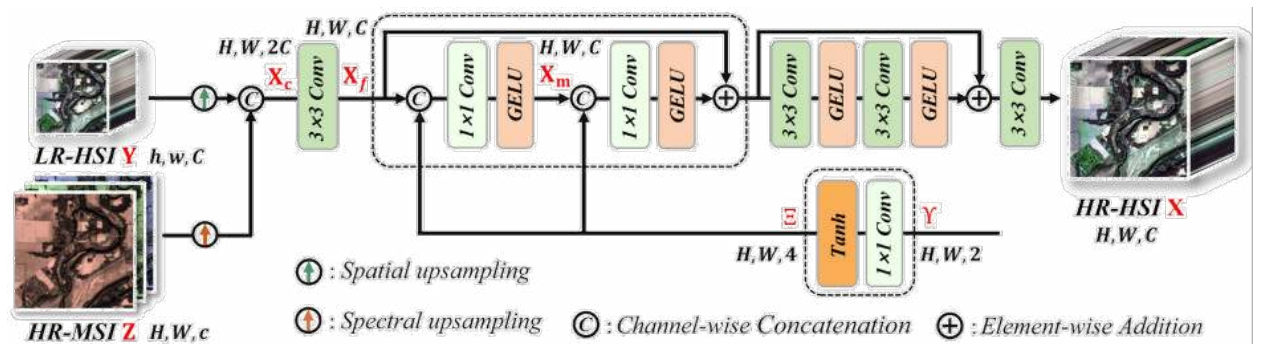


Figure 15: Position-aware fusion network. Courtesy of [189].

### 3.1.4. Denoising

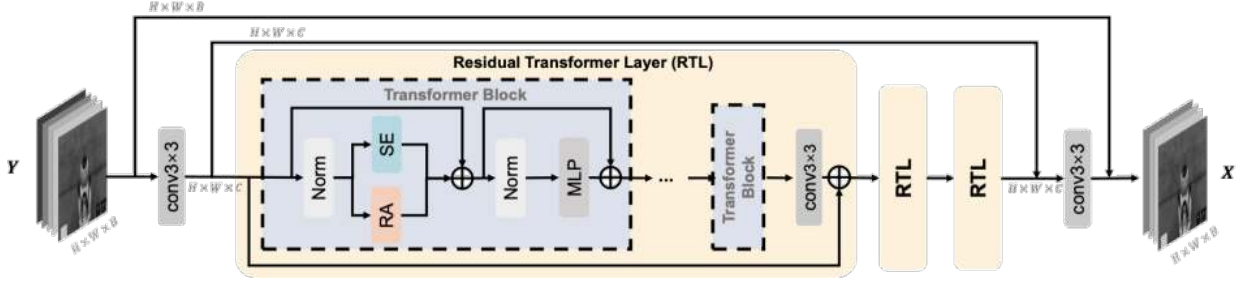
HSI denoising task removes noise from captured images to enhance clarity and improve the quality of raw HSIs. If we represent the noisy and clean HSI as  $Y \in \mathbb{R}^{H \times W \times B}$  and  $X \in \mathbb{R}^{H \times W \times B}$ , the task of HSI denoising can be formulated to maximize the conditional probability  $P(X|Y)$ .

Table 6: Comparison of different denoising methods.

| Method | Category           | Input            | Output                       | Key Idea                                                                                                                                                                                                                |
|--------|--------------------|------------------|------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [366]  | Network innovation | Noisy HSI        | Clean HSI                    | Realistic dataset and new model                                                                                                                                                                                         |
| [367]  | Denoising          | Noisy HSI        | Clean HSI                    | Propose a guided nonlocal HSI denoising network which can dynamically calculate spatial nonlocal correlation on cleaner mean image of spectral bands                                                                    |
| [331]  | Network innovation | Noisy HSI        | Denoised HSI                 | A model-guided interpretable network that leverages low-rank prior and sparsity                                                                                                                                         |
| [101]  | Prior exploration  | Noisy data block | reconstructed clean estimate | Bridge the sparse representation theory and deep learning modeling                                                                                                                                                      |
| [229]  | Diffusion model    | Noisy HSI        | Clean HSI                    | Proposed a self-supervised deep diffusion spatio-spectral model                                                                                                                                                         |
| [173]  | Diffusion model    | Noisy HSI        | Clean HSI                    | propose rectangle self-attention in the spatial domain to capture non-local spatial similarity and spectral enhancement module to leverage global spectral low-rank property                                            |
| [172]  | Diffusion model    | Noisy HSI        | Clean HSI                    | design multi-shape spatial rectangle self-attention module in both horizontal and vertical directions, and propose a spectral latent diffusion enhancement module for HSI low-rank vector extraction and representation |
| [44]   | Network innovation | Degraded HSI     | Clean HSI                    | Introduce a deep low-rank decomposition model into the optimization framework and incorporate traditional priors                                                                                                        |
| [356]  | Diffusion model    | Noisy HSI        | Clean HSI                    | Fusion of Spectral Unmixing and conditional abundance generation                                                                                                                                                        |
| [384]  | Network innovation | Noisy HSI        | Clean HSI                    | Formulate denoising process as an eigenimage denoising problem and propose a learning strategy to generate training samples                                                                                             |
| [324]  | Network innovation | Noisy HSI        | Clean HSI                    | Formulate denoising process as a sequence-to-sequence translation problem in an autoregressive fashion                                                                                                                  |
| [204]  | Network innovation | Noisy HSI        | Clean HSI                    | utilized a bidirectional state-space model along with self-attention mechanisms to consider both local and global spectral correlations                                                                                 |
| [190]  | Network innovation | Noisy HSI        | Clean HSI                    | designed a novel hybrid dual-path network to capture both local and non-local spatial details while suppressing noise                                                                                                   |

Recent trend of HSI denoising task mostly situates in **prior exploration** and new **structure proposal**. For instance, SMDS-Net [331] introduced a novel model-guided interpretable network for HSI denoising that fully considers the spatial redundancy, spectral low-rankness, and spectral-spatial correlations. [101] built a block-by-block architecture to bridge between classic sparse representation theory and modern deep-learning-based representation tools. Flex-DLD [44] leveraged the synergism of optimization-based methods and deep learning methods by introducing a plug-and-play deep low-rank decomposition model into the optimization for HSI denoising. Eigenimage2Eigenimage [384] turns the HSI denoising problem into an eigenimage denoising problem. Similarly, RAS2S [324] models the denoising process as a sequence-to-sequence translation problem in an auto-regressive fashion and builds a spatial-spectral region-aware module to distinctively perceive the semantic regions with different spatial-spectral representations. A salient work SERT [173] designs a rectangular self-attention in the spatial domain to explore non-local spatial similarity and proposes a spectral enhancement module to leverage the global spectral low-rank property to denoise HSIs. HSIDMamba [204] utilized a bidirectional state-space model along with self-attention mechanisms to consider both local and global spectral correlations, thereby enhancing the precision of denoising. Similarly, HSSD [190]

designed a novel hybrid dual-path network inspired by CNN and Transformer characteristics, capturing both local and non-local spatial details while suppressing noise.



**Figure 16:** Spectral Enhanced Rectangle Transformer (SERT). Courtesy of [173].

The success of diffusion models [268] also provoked interest of HSI denoising researchers. DDS2M [229] put forth a self-supervised diffusion model and inferred its parameters during the reverse diffusion process, which was robust to distribution shift. Diff-Unmix [356] proposed a generative unmixing framework to compensate for content loss and spectral deviation caused by noise. The derivative of SERT [173], LDERT [172] further made use of the diffusion model to enhance the rectangle transformer by utilizing both non-local spatial similarity and the inherent spectral low-rank property of degraded HSIs.

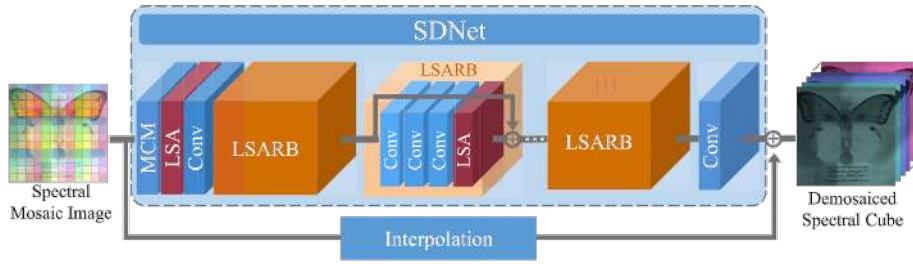
### 3.1.5. Demosaicing and Deblurring

Table 7: Comparison of different demosaicing and deblurring methods.

| Method | Category    | Input                                         | Output                                       | Key Idea                                                                                                                                            |
|--------|-------------|-----------------------------------------------|----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| [292]  | demosaicing | Spectral mosaic image                         | Fully defined spectral cube                  | formulate the problem in a novel framework of spectral measurement matrix completion and develop an efficient low-rank and graph regularized method |
| [81]   | demosaicing | Spectral mosaic image                         | Fully defined spectral cube                  | presents a comprehensive unsupervised spectral demosaicing framework                                                                                |
| [332]  | demosaicing | Spectral mosaic image                         | Reconstructed HSI                            | Couples a preliminary demosaicing network with a separate multi-stage progressive denoising network                                                 |
| [1, 2] | demosaicing | Spectral snapshot capture                     | Restored datacube, abundances and endmembers | perform demosaicing and unmixing jointly while combining low-rank matrix completion and factorization                                               |
| [98]   | deblurring  | Blurry SMHC raw frame and Noisy event signals | Recoverd sharp SMHC raw frame                | integrating Snapshot Mosaic Hyperspectral Cameras (SMHCs) with neuromorphic event cameras                                                           |

The process of interpolating the missing intensity values so as to have valid intensity values for all pixels is known as demosaicing [285]. Demosaicing RGB color images takes into account the presence of Bayer-structured filters for sampling the three primary colors at a 1:2:1 ratio, while for HSIs this process is generalized to a larger number of spectral bands with no preferential spectral sampling and a reduction in sampling frequency [292].

New software techniques keep raising actively. USD [81] presented a deep learning-based spectral demosaicing technique trained in an unsupervised manner. PDDNet [332] coupled a preliminary demosaicing network with a separate multi-stage progressive denoising network to ameliorate artifact issues which demosaicing algorithms commonly suffer. [1] improved demosaicing and unmixing from a two-stage process to a joint combination and proposed to process the raw compressed snapshot spectral imaging data without original post-processing.



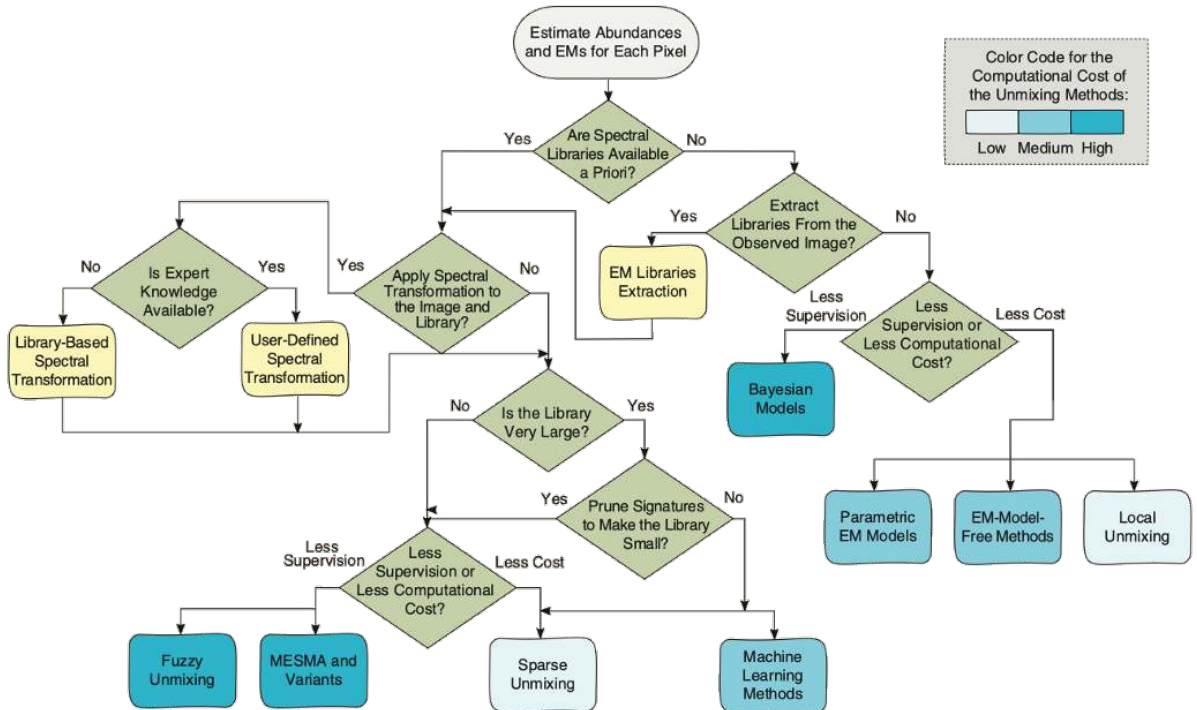
**Figure 17:** The schematic diagram of the full network structure. Courtesy of [81].

The hardware innovation can be seen in [98], which augments snapshot mosaic hyperspectral cameras with the neuromorphic event cameras, resulting in a hybrid camera system. The incorporation of event cameras provides additional motion clues to mitigate blur.

[385] seeks to explore intrinsic and extrinsic reflectance properties in a monocular HSI such as spectral chromatic and spatial defocus aberration for relative depth estimation.

### 3.1.6. Spectral Unmixing

Spectral variability is ubiquitous in hyperspectral images: the pixels in regions composed of a single material can contain very different spectral signatures (or endmembers), more pixels are actually a mixture of several different pure materials. This mixing process conceals important information about the pure materials and their distribution in an image. The Spectral Unmixing (SU) task aims to address the complex spectral mixtures in hyperspectral data by decomposing each pixel's spectral signature into pure spectral signatures (endmembers) and their corresponding proportions (abundances) [28, 30].



**Figure 18:** Spectral unmixing taxonomy, courtesy of [30]

As shown in Fig. 18, according to whether or not a spectral library or endmember set is known a priori, spectral unmixing methods family can be coarsely divided into two main categories. Furthermore, it is proceeded to different specific SU methods based on the tradeoffs they offer regarding the need for user supervision and computational cost. In the case that spectral library is unknown, endmember sets have to be extracted from the data using endmember extraction algorithms. Whether the endmembers are assumed known or not, SU remains a (semi-)blind or (un/semi-)supervised source separation problem [66]. Unfortunately, there is no open source natural scene HSI spectral unmixing dataset except for a laboratory-created dataset [375].

Recent works have demonstrated that engaging spectral unmixing block can enhance surface reflectance modeling and facilitate downstream tasks such as super-resolution [386], illumination spectra estimation [244] and hyperspectral novel view synthesis [252]. Additionally, spectral unmixing is also applied on isolating signals as proposed in [113].

### 3.1.7. Illumination Spectra Estimation

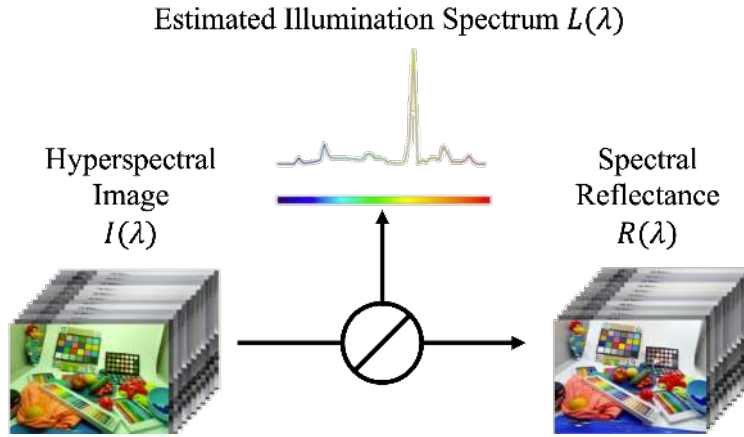
Different light sources can cause different perceptive effects on the perception of a scene, though human vision has the natural tendency to correct for these effects of the color of the light source, it is critical to correct the images so that they visually appear to be the same in the field of computer vision. Therefore, the process to explicitly correct an image for the color of the light source and get a transformed version of the input image is called color constancy [100]. To achieve color constancy we need illuminant estimation. In hyperspectral imaging, temporal illumination variations of the scene exist in general and cause different effects on the spectral signature of pixels. By estimating the illumination spectra, these negative effects can be mitigated and so the spectral signature can be correctly measured, as shown in Fig. 19.

Table 8: Comparison of different illumination spectra estimation methods.

| Method | Category    | Input                                              | Output                                                    | Key Idea                                                                                                                                                                                   |
|--------|-------------|----------------------------------------------------|-----------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [378]  | calibration | HSI data                                           | illumination and reflectance                              | used low-rank matrix factorization to handle the separation problem                                                                                                                        |
| [151]  | calibration | HSI data                                           | estimate of the illuminant                                | Extend computational illuminant estimation algorithms from RGB imaging to hyperspectral imaging                                                                                            |
| [281]  | calibration | input image and the initial illumination spectrum  | reflectance image and estimated illumination              | Enable the decomposition of the illumination and reflectance components from a single hyperspectral image by utilizing the specular independent subspace                                   |
| [267]  | calibration | image patch taken at three different scales        | illuminant power spectrum                                 | presented a convolutional neural network to achieve pixel-wise illuminant estimation for both RGB and hyperspectral images                                                                 |
| [180]  | calibration | Real spectral reflectance and illumination spectra | Estimated spectral reflectance and estimated illumination | cast the ISE into a constrained matrix factorization problem and solve it by a deep unrolling network constructed from the alternating direction method of multipliers (ADMM) optimization |
| [232]  | calibration | Damaged bands                                      | Restored bands                                            | Restore the damaged HSI bands caused by illumination changes instead of directly estimate the illumination                                                                                 |
| [73]   | calibration | Color image and spectral image                     | Color illuminant or spectral illuminant                   | combine multispectral and color information together to estimate illuminants                                                                                                               |
| [21]   | calibration | raw hyperspectral image                            | corresponding white reference image                       | replaces tedious manual calibration with a 3D-convolutional neural network                                                                                                                 |

*Continued on next page*

| Method | Category    | Input                         | Output                                                | Key Idea                                                                                                                                    |
|--------|-------------|-------------------------------|-------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| [69]   | calibration | Hyperspectral Intensity Image | Hyperspectral Reflectance Image                       | Firstly propose to estimate the illumination spectrum using learning-based method so that spectral reflectance can be calculated indirectly |
| [243]  | calibration | Hyperspectral Intensity Image | Hyperspectral Reflectance Image                       | exploit physical characteristics of spectral images via incorporating illuminant prior information                                          |
| [244]  | calibration | HSI                           | Reconstructed HSI and Estimated Illumination Spectrum | employ spectral unmixing to enhance surface reflectance modeling and utilize cross-attention to fuse spectral and spatial features          |

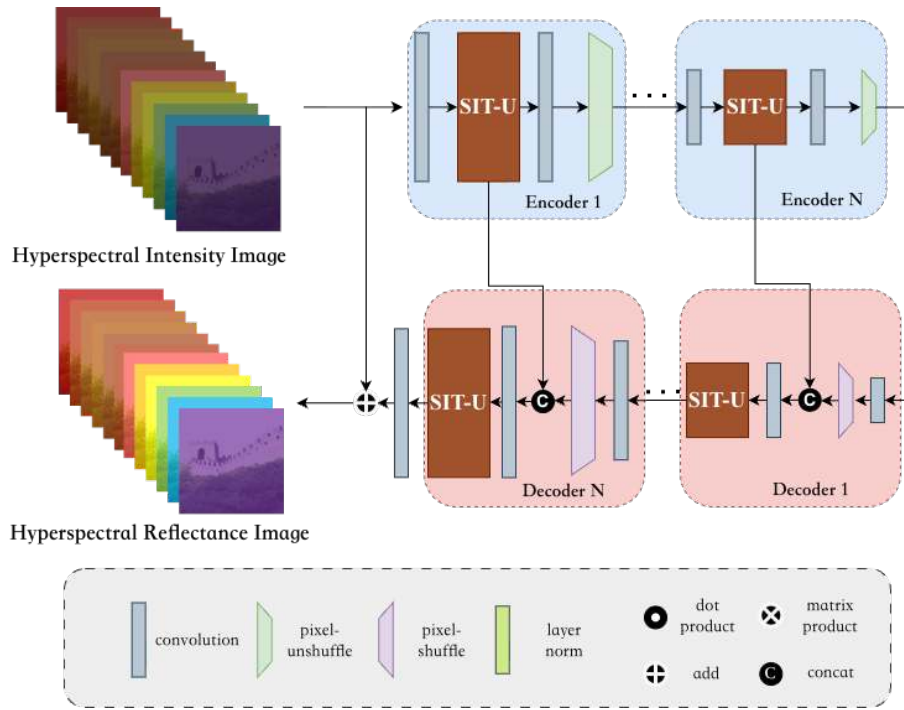


**Figure 19:** Illumination Spectral Estimation, image made from [244].

This problem is usually addressed manually using the famous GrayWorld algorithm [32]. In the past decade, learning-based approaches emerged as they automatically recalibrated hyperspectral images and precluded the tedious process of white reference-based recalibration.

Early work addressed the illumination spectra estimation problem as the separation of illumination and reflectance [378], and used low-rank matrix factorization to handle the separation problem. [150] reviewed the illuminant estimation methods which are based on highlights in images and identified the potential methods which can be extended to multispectral imaging. DS-Net [274] was an early attempt to achieve RGB color constancy through illuminant estimation. Experiments were implemented on NUS-8 dataset [47]. [151] illustrated the potential benefits of using multispectral imaging in computer vision applications, but also acknowledging that multispectral imaging can still be sensitive to changes in illumination. [281] proposed a general framework to estimate the spectrum of the illumination from global specular information in a single hyperspectral image. It enabled the decomposition of the illumination and reflectance components from a single hyperspectral image by utilizing the specular independent subspace. In the same year, [267] presented a convolutional neural network to achieve pixel-wise illuminant estimation for both RGB and hyperspectral images.

The network was able to predict the illuminant per-pixel using locally supported multiscale information. [180] formulated the problem of multispectral illumination estimation as a matrix factorization task, and used the alternating direction method of multipliers optimization algorithm to solve it, by unrolling it into a multi-stage network. Recently, DivIII [232] used CNN layers and residual blocks to address reflectance abnormalities caused by illumination variations in indoor scenes. [73] introduced a method that enhances RGB color constancy accuracy by combining neural network and k-means clustering techniques. It incorporated both the RGB and spectral domains, instead of considering either one of the two, to capture a more comprehensive set of features related to the illuminant, thus improving the accuracy of the estimation. To maintain the generalization capability to unseen settings, [21] designed a novel 3-D convolutional neural network to estimate the white tile measurement for a given scene rather than directly predicting the recalibrated



**Figure 20:** Spectral Illumination Transformer (SIT) framework and its unit (SIT-U) with Spectral and Illumination Attention branches for improved HSI illumination spectra estimation. The Illumination Attention branch mimics the Gray-World method to capture illumination features. Courtesy of [69].

image. Inspired by the success of U-shape transformer network architecture applied on HSIs, ASC [69] proposed an automatic spectral calibration algorithm. Recently, [243] proposed to exploit the physical characteristics of spectral images for accurate illumination spectrum estimation via incorporating illuminant prior information, while [244] employed spectral unmixing techniques and applied Gaussian Mixture Model to estimate illumination spectrum. [140] proposed a novel method for separating the direct and indirect reflection components of a scene from a single hyperspectral image. PhySpec [341] designed a dynamic illumination estimation module that estimates image-specific illumination descriptors implicitly without requiring prior knowledge of illumination conditions.

### 3.2. High-level Vision

After raw HSIs are separated from illumination, restored and augmented to some extent, they can be used in vision tasks involving human knowledge, referring to high-level vision. With supervisions, high-level vision tasks include classification [176, 12, 179, 80], semantic segmentation [104], face recognition [370], object detection [191, 254, 209, 132] and object tracking [259].

#### 3.2.1. Feature Extraction

The goal of feature extraction is to discard useless redundancy from the raw data while retaining the most relevant information and to represent that information in a lower-dimensionality space to reduce computational complexity. The large number of spectral bands in hyperspectral data carries intrinsic characteristics of the natural scene, but it may also decrease the performance of algorithms due to the curse of dimensionality. Moreover, the reflectance measured at a given wavelength may be significant for a specific task while useless for another. Therefore, it is important to perform feature extraction to obtain the most task-relevant information and process the dataset more efficiently and accurately. An overview of the evolution of hyperspectral feature extraction backbone can be seen in Figure 21.

As the most remarkable characteristic of HSI, the dimension of spectral bands is of great significance, while incorporating the spatial information has been proven to be distinguishable. An early classification survey [176] categorized pioneering neural networks for HSI feature extraction into spectral feature, spatial feature, and spectral-spatial feature in terms of their exploitation of information. While [158] proved that it is best to treat the spectral

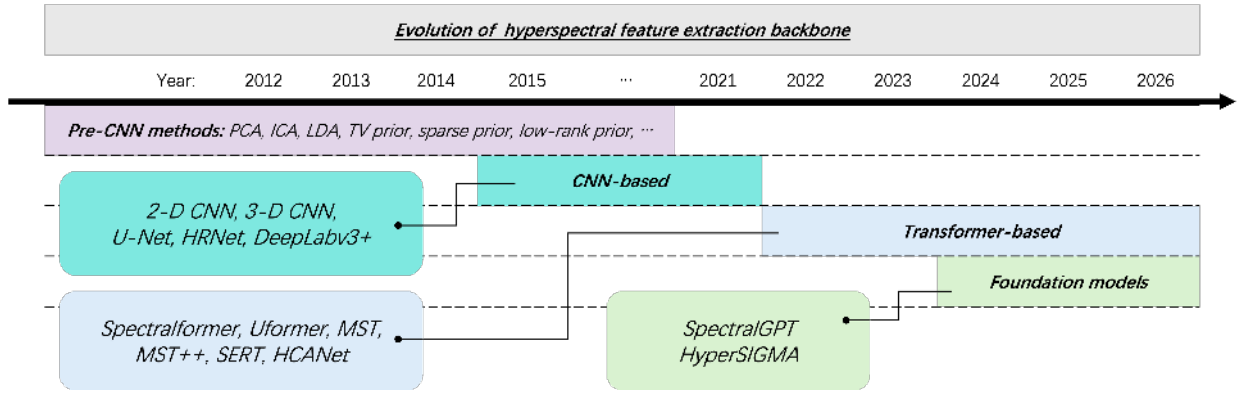


Figure 21: Encoder network evolution.

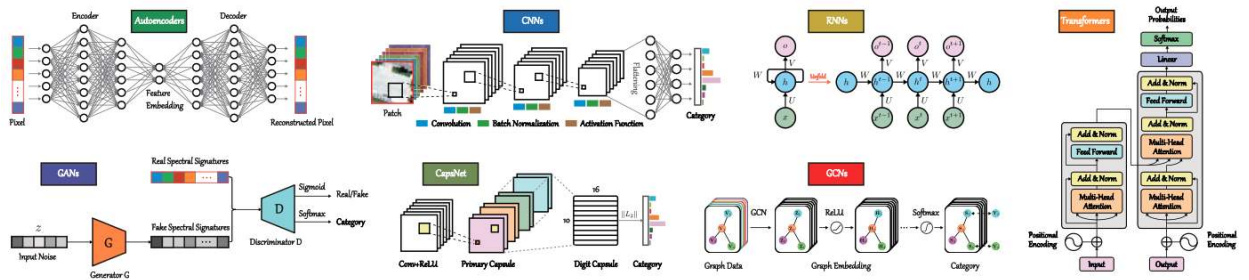


Figure 22: Backbone networks for HSI classification task (which is also for feature extraction), courtesy of [123].

dimension in the same manner that the spatial dimensions are treated. A recent work [215] systematically compared eight unsupervised dimensionality reduction algorithms but limited for earth observation application.

A naive idea to exploit the information in HSI effectively is to reduce its dimensionality and handle correlated bands. The most widely used dimensionality reduction method is principal component analysis (PCA). PCA reduces redundant information in the bands of hyperspectral imagery while preserving as much of the variance in the high-dimensional space as possible [258]. Other PCA variants include ICA [295] and LDA [18]. These simple linear processing methods existing in the linear models are hard to handle the complex spectral properties of HSIs. Similarly, nonlinear dimensionality reduction techniques, e.g, manifold learning [143], are explored but still limited to known prior of hyperspectral dataset and certain spectral response curves.

**Pre-CNN methods** Before the prevalence of deep learning, a plethora of pioneering handcrafted priors are explored in low-level vision tasks, such as total variation (TV), sparse prior, low-rank prior. Other early methods also leveraged the traditional approaches including matrix decomposition [147], Bayesian model [5], non-local sparse tensor factorization [58].

Inspired by classic SIFT algorithm, [7] proposes spectral-spatial scale invariant feature transform (SS-SIFT) to extract spectral and geometric transformation invariant features. In view of texture analysis, [52] defines texture as joint probability of spectral and spatial distribution. It then establishes a physically relevant and directly interpretable feature named relative spectral difference occurrence matrix. The proposed feature achieves good performance in texture classification, image retrieval and land cover classification tasks. [126] proposes to modify 3-D convolution's topology to enhance the rank upper-bound, thereby yielding a rank-enhanced spatial-spectral symmetrical convolution set to learn diverse and powerful feature representations.

However, these hand-crafted features are usually task-specific and depend on expert knowledge in the parameter setup phase, which hinders their applicability in complex scenarios and further improvement of performance. Thus, extracting more discriminative HSI features is considered a crucial procedure.

**CNN and its variants** The rise of deep learning has led to major breakthroughs in many computer vision tasks due to its non-linear representation ability. Motivated by this, deep learning techniques such as convolutional neural network, deep belief network and stacked autoencoder are explored to extract informative and efficient representation from raw HSIs.

Promising results notwithstanding, these approaches model HSI by learned multichannel or band-wise 2-D convolutions, which sacrifice either the flexibility with respect to the spectral dimension [40] (hence requiring retraining the network to adapt to HSIs with mismatched spectral dimension), or the model capability to extract global spectral correlation knowledge.

For the input HSI  $\mathbf{X}$ , 2D CNN [218] has been employed for spectral and spatial feature extraction as

A complete 2-D CNN layer contains a convolutional layer and a pooling layer. The 2-D convolutional layer is obtained by the extension of (1). The value of a neuron  $v_{ij}^{xy}$  at position  $(x, y)$  of the  $j$ th feature map in the  $i$ th layer is denoted as follows

$$v_{ij}^{xy} = g \left( b_{ij} + \sum_m \sum_{p=0}^{P_i-1} \sum_{q=0}^{Q_i-1} w_{ijm}^{pq} v_{(i-1)m}^{(x+p)(y+q)} \right) \quad (13)$$

where  $m$  indexes the feature map in the  $(i - 1)$  th layer connected to the current ( $j$ th) feature map,  $w_{ijm}^{pq}$  is the weight of position  $(p, q)$  connected to the  $m$ th feature map,  $P_i$  and  $Q_i$  are the height and the width of the spatial convolution kernel, and  $b_{ij}$  is the bias of the  $j$ th feature map in the  $i$ th layer.

Deep Belief Network (DBN) [45] spectral feature network to extract spectral-spatial features for HSI classification by choosing 50 % of the tagged samples as training set.

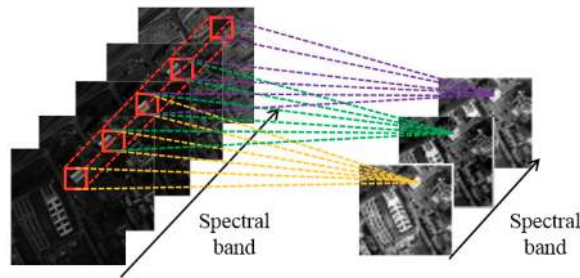
Stacked Autoencoder (SAE) [355] segmented the spectral domain of HSI into different regions, to which the stacked autoencoder (SAE) is applied individually.

However, SAE and DBN required the input training samples to be flattened, which neglected the spatial information of HSI data. In addition, there are too many parameters in hidden layers of SAE and DBN, though CNN has the advantage of local connections and shared weights that can reduce computational cost. Besides, 2D CNN-based methods cannot integrally deal with 3-D spectral-spatial structure in HSIs.

Different from 2-D convolution, 3-D CNN can naturally better deal with HSI, as it can simultaneously exploit the 3-D spectral-spatial structure in HSI. Formally, 3-D CNN-based methods use 3-D kernel filters

$$\mathbf{F}_{x,y,b,\lambda} = \mathbf{X} \otimes \mathbf{C}_{x,y,b,\lambda}, \quad (14)$$

where  $\mathbf{C}_{x,y,b,\lambda}$  refers to the 3-D filter with the size of  $x \times y \times b$  and channel of  $\lambda$ . Without separating spatial and spectral information, 3-D CNN embodies the natural advantage for downstream tasks [183].



**Figure 23:** 3D convolution operation [183].

A successful application of 3-D CNN is QRNN3D [317], which provided a unique approach to hyperspectral image denoising by embedding quasi-recurrent units in 3-D convolutional operations. The network is composed of two main operations. One is 3D convolution, which is applied to extract spatial and spectral correlation in an HSI, and the other is quasi-recurrent pooling function, which can capture the global correlation along spectra.

Nevertheless, there are still some limitations. QRNN3D focuses mainly on local spatial correlation of the HSI while overlooking the global contextual information. Furthermore, 3D convolution used in QRNN3D leads to a sharp increase in the number of parameters.

*Transformer* The advent of transformer structures has revolutionized the deep learning field thanks to their self-attention mechanism and long-term dependency representation ability.

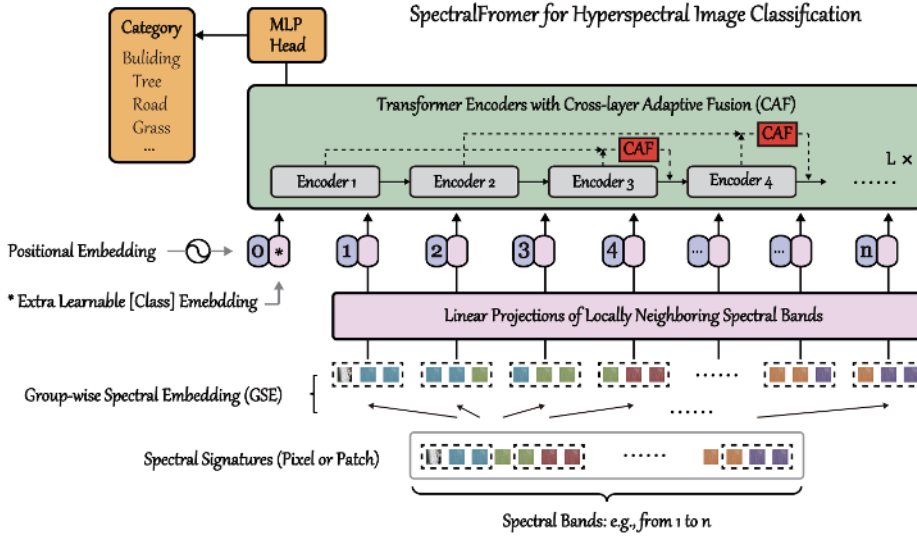


Figure 24: Paradigm of SpectralFormer [123].

[381] proposed an architecture search framework to find a suitable network consisting of spectral and spatial Transformer for HSI classification. Spectralformer [123] is a milestone work that adapts the classic transformer backbone into hyperspectral imaging. Specifically, it enabled group-wise spectral embeddings to replace the original band-wise embeddings, and it developed a cross-layer skip connection to progressively convey memory-like components from shallow to deep layers, thus overcoming the drawbacks of transformers. DSS-TRM [197] utilized deep transformer along the spectral dimension and the spatial dimension respectively to make full use of the self-attention mechanism, that is to use transformer layer instead of convolution layer.

Different from these works that tend to directly employ existing transformer blocks to other tasks like pansharpening [282, 357], methods that in CST and MST [33, 34] solve the HSI reconstruction problem with task-oriented transformer blocks under the guidance of degradation mask. HDNet [131] developed a high-resolution spatial-spectral attention module and provided continuous and fine pixel-level features. A 3D quasi-recurrent and Transformer network was presented in HyperTransformer [17] for hyperspectral image denoising, which combined the 3D quasi-recurrent layer with Swin blocks.

Uformer [316] developed a u-shaped transformer network consisting of locally enhanced window (LeWin) transformer block and multiscale restoration modulator. After its success, HCANet [130] further developed the U-shape transformer structure by designing a hybrid convolutional and attention network, which is able to simultaneously capture the long-range dependencies and neighborhood spectral correlations.

The combination of convolution and transformer structure was also explored, for instance, CL-MGNet [91] proposed a spectral weight learning module with a category consistency loss to achieve the enhancement of information in important spectral bands and the mining of category-specific properties, and it further proposed a multiscale backbone to extract the spatial information at different scales and the cross-channel attention via multiscale convolution and a grouping attention module.

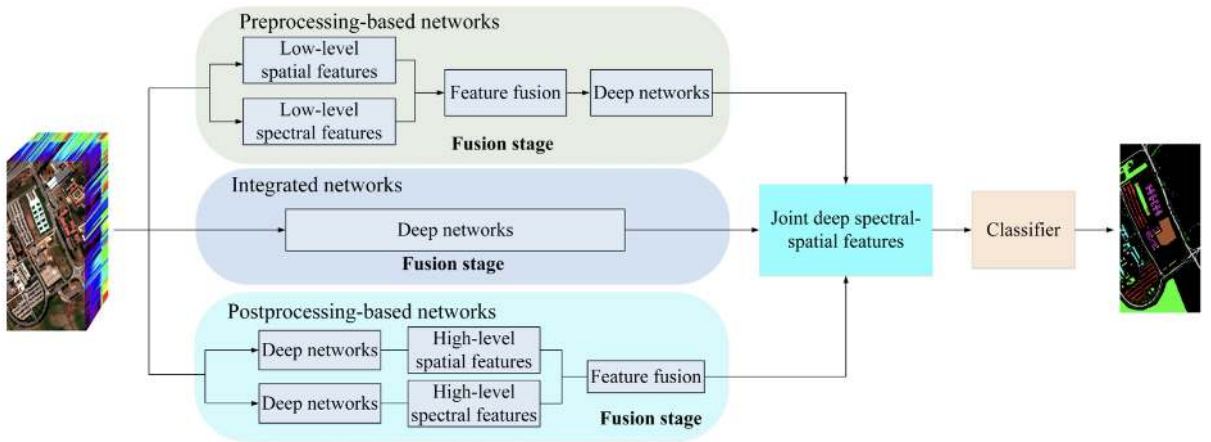
*Mamba and foundation models* Mamba [105] utilizes a state-space model (SSM) to capture global semantic information with low computational complexity. Unlike the Transformer, it has linear complexity, allowing it to adaptively adjust denoising strategies based on the noise characteristics of the data through parameterized SSM and self-attention

mechanisms. HSiDMamba [204] utilized a bidirectional state-space model along with self-attention mechanisms to consider both local and global spectral correlations, thereby enhancing the precision of denoising. Furthermore, MUnet [382] designed a Mamba-based context interaction module and constructed a U-shaped hierarchical encoder-decoder structure network (MUNet). The network takes pixel scale as input to maximize image information retention, enhancing the modeling of global and local features through a Mamba-based contextual interaction module and convolution for better denoising. Sp3ctralMamba [224] exploited Mamba's implicit attention mechanism and integrates frequency domain knowledge as well as physical priors to enhance reconstruction quality.

SpectralGPT [125] emerges as the first hyperspectral foundation model and has shown its outstanding performance in several computer vision tasks including classification, segmentation and change detection. It is obviously a powerful feature extraction tool, while its representation ability and further application remain to be explored. More details will be discussed in Section 3.3.3

### 3.2.2. Classification

HSI classification is one of the most active research areas in the hyperspectral imaging community. In the field of remote sensing, it typically refers to assigning each pixel to a specific class based on its spectral characteristics [176], with the primary goal of per-pixel image classification (i.e., one label per pixel). In contrast, the potential of hyperspectral imaging for full-image classification and regression tasks (i.e., one label per image) has been relatively underexplored. Beyond conventional classification settings, methods with improved generalization capabilities—such as cross-scene classification [135] and few-shot open-set classification [322]—have attracted increasing research attention in recent years.



**Figure 25:** Paradigms of spectral-spatial networks for classification task [176]

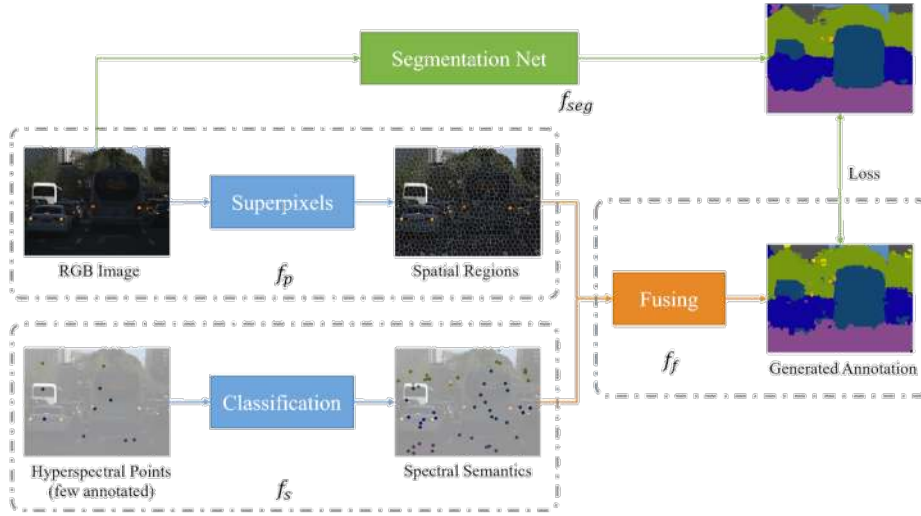
On the other hand, various specific applications of HSI full image classification in natural scene scenarios can be found recently, such as minced meat classification [13], rice seed varieties classification [286], wheat leaf classification [159] and flower classification [379], bloodstain identification [388], and adulterated chili identification [152].

### 3.2.3. Semantic Segmentation

Semantic segmentation aims to assign a semantic label to each pixel in an image [104], which seems to be quite similar with HSI classification. In fact, what sets the natural scene HSI semantic segmentation task apart is the special interest in the interdisciplinary research of hyperspectral imaging and autonomous driving. In the field of autonomous driving and advanced driving assistance systems, the use of HSIs has been widely recognized to improve scene understanding and system robustness. The underlying idea is that the incorporation of richer spectral information can provide a distinct spectral fingerprint for each entity in the image.

Hyperspectral images have higher spectral resolution, they can provide more distinctive features for semantic segmentation since different materials have different spectra. As a result, hyperspectral information can be treated as additional prior knowledge. The semantic segmentation task usually needs pixel-level annotations as supervision,

which are expensive, time-consuming and mistake-prone. Utilizing hyperspectral information to realize accurate pixel-level classification has been proven to be promising in the field of remote sensing [176].



**Figure 26:** Overview of the method proposed in [266]. First, RGB image is segmented in an unsupervised manner into several regions, each of which share similar spatial features. Second, a HSI classification network is trained to extract spectral semantic information. Third, the outputs are fused to generate more accurate annotations. Finally, a semantic segmentation network is trained with generated annotations in a supervised manner.

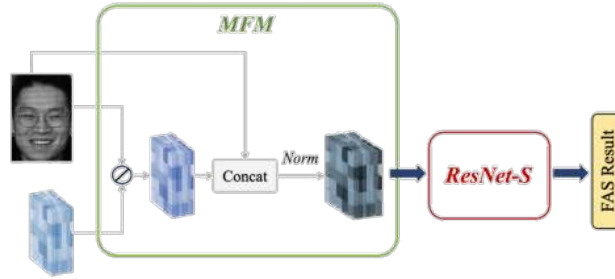
However, the case of natural scenes can be different as there is a huge number of objects, large-scale variations and multiple occlusions. To alleviate this problem, preliminary researches proposed weakly-supervised segmentation frameworks that either directly used pixel-level annotations [346], or fused spectral prior and coarse annotations to generate refined labels as supervision [136, 266] and improve the semantic segmentation performance. Specifically, [136] proposed a semantic segmentation framework that firstly generated a spectral prior from coarse label supervision via HSI classification, then fused the spectral prior and coarse label to generate more detailed and accurate refined label as the final supervision of segmentation. [266] adopted a similar idea, and it combined spatial features from RGB images and spectral semantic information extracted from a few annotated hyperspectral points - supervised HSI classification network to ultimately generate pixel-level refined and semantically correct annotations.

### 3.2.4. Face Recognition

Among the various recognition tasks that computers are asked to perform, face recognition is the one where they have arguably had the most success [285]. The most active topic and successful application of HSI recognition is hyperspectral-cube-based mobile face recognition. In face recognition, visual images approach human perception but suffer from illumination change, pose variation, and partial occlusion. Infrared images have good properties to resist ambient light interference and harsh atmospheric conditions while struggling with low resolution and poor texture, and are vulnerable to temperature, state of health, and mood [370]. Face images from hyperspectral cubes have the abilities to capture distinct personal identification patterns and handle the difficulties in traditional face recognition.

HSI recognition can explore the intrinsic spectral information of the skin at many subtle wavelengths adding to spatial information [294], and spectral features of the same subject remain stable even if large changes in face orientation, while significant spectral variability appears in the case of different subjects under different face orientations. Hyperspectral cubes in the near-infrared spectrum are illumination-invariant and suitable for face recognition under different illumination conditions. And there is an apparent difference between the reflectance property of the human skin and other materials under a near-infrared spectrum [373], which can easily solve the problem of *in vivo* detection of face recognition. Typical occlusions like eyeglasses and sunglasses widely exist in face detection. Fortunately, it can be clearly found that in the case of longer wavelength bands, glasses basically have no interference effect.

Even though hyperspectral cubes circumvent the demerits of visual and infrared images, there are relatively inadequate research works and datasets [57, 49, 370]. In 2024, Snapshot Spectral Imaging Face Anti-spoofing Challenge <sup>1</sup> has created the first snapshot spectral face anti-spoofing dataset, HySpeFAS, which contains 6760 hyperspectral images reconstructed from SSI images by TwIST algorithm [27]. Accordingly, one of the resulting winner works [166] introduced a unique data processing to avoid the model's misjudgment of materials such as medical masks, and then proposed a model ensemble strategy to combine the prediction results of multiple models trained with different augmentations to improve accuracy and robustness. Another first-class [279] harmonized class-level contrastive learning with data resampling and a real-face oriented reweighting technique to mitigate dataset imbalances and reduce identity-related biases. In addition to these two workshop papers, SSINet [368] directly treats the snapshot spectral imaging hardware as an optical encoder and designs the end-to-end Snapshot Spectral Imaging Network (SSINet) to decode the optical encoding results to obtain FAS results, instead of reconstructing from snapshot spectral images as traditional methods did.



**Figure 27:** The overall structure of SSINet [368]. It first uses Matrix Fusion Module to fuse snapshot spectral image and broadband modulation matrix, and then it uses ResNet-S to extract a 512-dimensional feature vector. Finally the feature vector is passed through the MLP and Sigmoid layer to obtain the Face Anti-Spoofing (FAS) results.

### 3.2.5. Object Detection

*Definition and HSI advantages* Object detection is a task of estimating and locating a visual object of a certain class precisely in a digital image [387]. It is a fundamental problem in computer vision and has been studied extensively in low-dimensional 3-channel RGB images. The auxiliary information provided by HSI can detect objects even in a similar background like *camouflage* [209]. The continuous spectral curves that HSI is producing are able to collect the small changes in the spectrum leading to better performance and results than traditional images, and to distinguish objects under *metamerism* - the phenomenon of lights that elicit the same response from the RGB sensory system (like human eyes) but have different radiant power distributions over the visible spectrum [86]. Despite these advantages, HSI object detection remains a prevalent topic in remote sensing [48, 254] and has not been explored much with respect to natural scene scenarios.

Hyperspectral object detection methods can be finely divided into three parts: *anomaly detection*, *supervised object detection* and *salient object detection*. A taxonomy can be seen in Figure 28. **Anomaly detection** identifies pixels whose spectral characteristics are abnormal. **Supervised object detection** has prior knowledge of the object to detect. **Salient object detection** looks for the most noticeable segment.

*Supervised Object Detection* In Supervised Object Detection, the prior knowledge of the object to be detected is usually known. Thus, the spatial coordinates and expanse of the object will be returned if the object is present in the image. However, insufficient labeled data, high dimensionality of HSI and its consequent Hughes phenomenon - decrease in classification accuracy as the number of features (dimensions) increases beyond an optimal point, given a fixed number of training samples [137] - pose great challenges.

One of the key limitations of HSI is its hypersensitivity to small changes in illumination. To address this issue, many mathematical or statistical measures of spectral signature are transformed in a way that they become invariant to a range of spectral variations, e.g. applying mathematical transformations on the original object spectral signature to develop an elaborated set of object signatures or using variants of spectral signature like mean as an object signature. Furthermore,

<sup>1</sup>Source: [https://codalab.lisn.upsaclay.fr/competitions/17908#learn\\_the\\_details-overview](https://codalab.lisn.upsaclay.fr/competitions/17908#learn_the_details-overview)

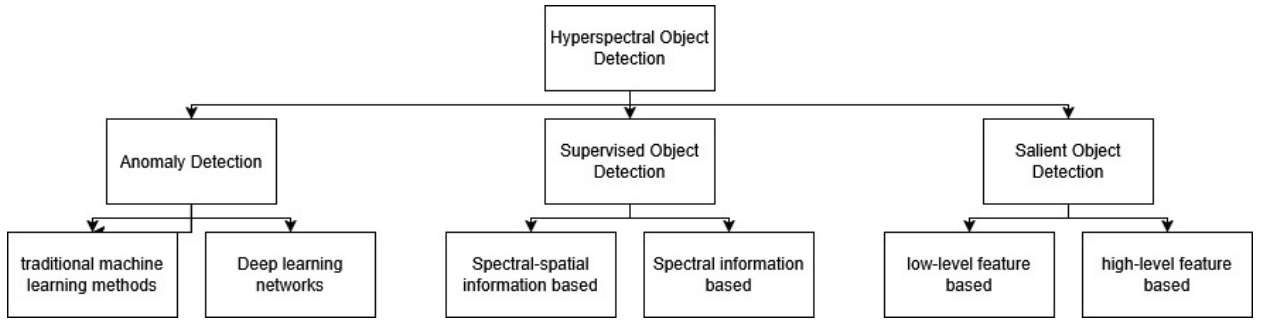


Figure 28: Hyperspectral Object Detection Taxonomy.

to include spatial information will enable the detection algorithms with sensitivity of the shape of object. Such spectral-spatial information based approaches benefit from image segmentation, where spatially adjacent pixels are grouped into spectrally homogeneous regions and then matched with region as minimum detection units. Meanwhile, this approach inherits two limitations of image segmentation and suffers from under-segmentation and over-segmentation, where extracted features do not represent real objects properties and can reduce accuracy if not chosen appropriately [209].

The supervised object detection algorithms can be compared according to various indexes including Precision(P), Recall(R), F-measure(F), Overall Accuracy(OV), Average Accuracy(AA), Kappa coefficient, receiver operating characteristic curve (ROC), area under ROC curve (AUC or AUROC).

**Anomaly Detection** The purpose of anomaly detection is to identify data that does not match normal data or that is significantly different from expected data [132]. Hyperspectral anomaly detection (HSI-AD) refers to the identification of pixels whose spectral characteristics in an image are significantly different from adjacent or global background pixels. Also, the task can be named hyperspectral abnormal object detection, where a feature shows differences from the spectral curve of its surrounding background environment will be regarded as ‘abnormal’.

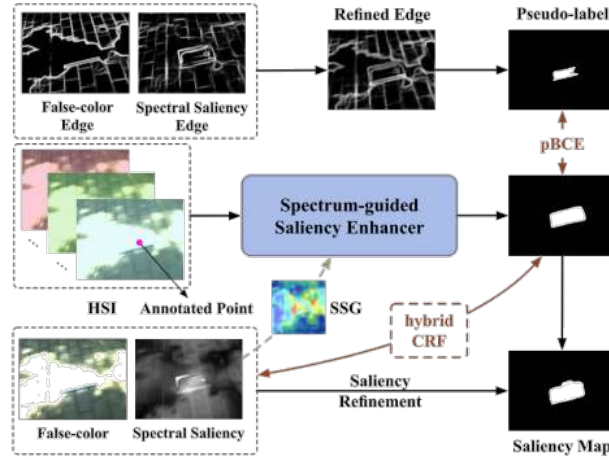
Due to the difference in the absorption and reflection of different wavelengths of light by different object material components as we demonstrate in Section 2.1.2, a certain feature of the object will be very prominent in the image at a specific wavelength, but less prominent in the image at other wavelengths. Therefore, the spectral information can fully reflect the difference in the physical structure and chemical composition of the object. In practice, HSI-AD can be used as a preliminary screening work for object detection, to quickly screen out suspected target features from a large scene, and then match with the prior information of the object spectrum to identify the object features.

According to the survey by [132], two important characteristics can be used to distinguish the anomaly from its surrounding background: i) abnormal spectral features show distinguishable differences in the spectral domain, ii) abnormalities usually exist in small areas. Thus, it can be detected by either establishing a reference background model or suppressing the background. The evolution of HSI-AD algorithms underwent traditional machine learning methods, reaching modern era of deep learning. Classic networks, including CNN, SAE, GAN, DBN, RNN, LSTM, etc., are explored in [132]. A most recent work [293] proposed a self-supervised masked graph autoEncoder to reconstruct the background of hyperspectral images and identifies anomalies by comparing the reconstructed features with the original features.

**Salient Object Detection** This Salient Object Detection originates from humans’ attention mechanism, which prioritizes the selected data to process in real-time because of the need to deal with a very high amount of input data at each moment. It can be defined as a sparse noise in certain feature space where the non-salient background is assumed as a low-rank matrix and salient object as sparse noise [132].

In order to handle the large number of bands in HSI, salient object detection approaches use low-level or high-level features. The low-level features, such as edges, corners, gradients, etc., can be used to construct the saliency maps and the saliency maps are integrated to get the final salient object. The high-level features are always related to particular objects, as they need supervision. Thus, they benefit from supervised detection and image segmentation methods, or even known object property prior. The low-level features are easy to extract but are semantically less meaningful, while high-level features are hard to extract but bring more semantic information. Precisely, [191] presents a material-based salient object detection method which can effectively distinguish objects with similar perceived color but different

spectral responses. It extracts explicit material properties embedded in the spectral response of end members and estimates the spatial distribution of endmembers. [340] designed a deep neural network-based method for object-based hyperspectral detection and created a high-quality hyperspectral image dataset for object-level target detection. SPSPD [201] introduced point supervision into HSOD to get rid of pixel-level annotation dependence that existing methods necessitate, the network structure is shown in Fig. 29.



**Figure 29:** Spectrum-oriented point-supervised saliency detector (SPSPD). SPSPD utilizes spectral saliency as a foundational element, aiding in the generation of edge-refined pseudolabels, model supervision, and saliency refinement. The spectrum-transformed spatial gate (SSG) is employed to guide our proposed Spectrum-guided Saliency Enhancer model, enhancing the representation of saliency features. Courtesy of [201].

### 3.2.6. Object Tracking

Hyperspectral object tracking is aimed at taking advantage of the spectral and spatial information in visual object tracking. The biggest challenge is that RGB features cannot effectively deal with similar color between the target and its neighborhood [259].

Early hyperspectral object tracking methods employ hand-crafted features to represent the target [260, 329, 330, 372], while deep learning-based hyperspectral trackers mainly employ the pretrained backbone networks on color images/videos to represent the target [206, 188]. For example, SiamHYPER [207] used an RGB-based Siamese tracker module and additionally designed a hyperspectral target-aware module to mine the spectral information during the target prediction and introduced a spatial-spectral cross-attention module to further fuse the deep spatial and spectral features. [233] extends a siamese tracking pipeline which showcased superior performance in the RGB image domain to extract knowledge from hyperspectral modalities. HyA-T [95] extracts spectral information directly from the hyperspectral images, which prevents the loss of the spectral information that exists in fine-tuning pre-trained RGB-based object tracking networks for hyperspectral tasks.

Additionally, as the intrinsic of HSIs, the band correlation was also explored in hyperspectral object tracking. To fully interact with the band information while maintaining tracking speed, SPIRIT [43] proposed a novel spectral awareness module to learn band contributions with consideration of nonlinear and global interactions between HS bands. SEE-Net [186] proposed deeply learned features to describe objects. It establishes a spectral self-expressive model to learn the band correlation so the prior knowledge of bands is transformed into a learnable network architecture. Emphasizing the material identification capability of hyperspectral videos, MMF-Net [187] employs hyperspectral unmixing to extract false-color, hyperspectral, and material information, which are then integrated to enhance the object's representation.

Building on the success of the SOTA RGB tracker OSTRack [344]—a one-stream tracking framework that jointly learns features and relational representations via self-attention—HELIOS [234] further exploits hyperspectral characteristics by introducing a band-selection algorithm and enforcing forward-backward temporal consistency. Another derivative of OSTRack is DaSSP-Net [185], a universal representation learning framework for multi-domain hyperspectral tracking. DaSSP-Net leverages domain adaptation to benefit from the OSTRack foundation model

pre-trained on RGB tasks and incorporates a band-invariant spatial-spectral prompt module to guide the model toward learning a domain-agnostic and broadly transferable representation. Similarly, BihoT [298] manifests that existing hyperspectral object tracking datasets are visually distinguishable, and most existing algorithms overlook the advantages of hyperspectral information. It provides a new hyperspectral camouflaged object tracking task, which focuses on tracking objects that share the same visual information but differ in spectral characteristics. Besides, recent multispectral object tracking is also proposed [177], which could also inspire the development of hyperspectral object tracking.

### 3.3. Artificial Intelligence techniques used in HSI research

In recent years, the rapid advancement of artificial intelligence (AI) has significantly influenced hyperspectral imaging (HSI) research [26]. Deep learning techniques have enabled more effective exploitation of the rich spectral-spatial information contained in hyperspectral data, leading to substantial improvements across a wide range of HSI applications. Among the emerging research directions, three topics have attracted particular attention: multimodal hyperspectral learning, generative hyperspectral modeling, and hyperspectral foundation models.

#### 3.3.1. Multimodal HSI

Hyperspectral 3-D imaging combines spectrum imaging system with depth imaging system, aiming to capture both depth and spectrum per pixel [15, 163, 275, 276], which is different from simple depth estimation [385]. To achieve this goal, researchers either used DOE [15], combined RGB camera and projector [163], or combined a conventional trichromatic camera with a trichromatic projector and a diffraction-grating file mounted in front [275]. To make the system faster and more affordable, [276] followed up and designed spectrally multiplexed dense dispersed structured light patterns.

And also, the utilization of other accessible modalities are explored. For example, [99] combined coded aperture snapshot spectral imager and a spike camera to involve temporal information, [98] incorporates neuromorphic event cameras to provide additional motion clues to mitigate blur, [195] used an ultrasound sensor to detect sound waves generated by the optically excited targets. [347] combined an event camera with a dynamic illumination strategy for real-time capture with low bandwidth in dynamic scenes.

The infrared modality captures thermal signatures that maintain reliability under harsh weather conditions [221], or in poor illumination [24], and it has widely been used and small target detection [11] in autonomous driving [141, 250].

Precisely, [221] conducted the first comparative experimental study evaluating the detection and segmentation performance of two short-wave infrared (SWIR) technologies in harsh weather conditions. [24] designed a Filtered Multi Modal Cross Attention Fusion (FMCAF) that combined a frequency domain filtering block and the architecture enhanced the fusion of RGB and infrared inputs.

Thermal cameras with long-wave infrared (LWIR) imaging enable reliable perception in low-light conditions, adverse weather, and night-time scenarios where RGB sensors fail. Javidnia et al. [141] introduced a lightweight U-Net that converts raw 16-bit thermal frames into detector-ready 8-bit images while avoiding excessive loss of quantization levels for low-emissivity targets. IVFormer [250] introduces a novel cross-attention mechanism that dynamically switches the roles of query and key/value features between the infrared and visible domains. This role-reversal is governed by a scene-classification module that determines whether the input corresponds to daytime or nighttime conditions.

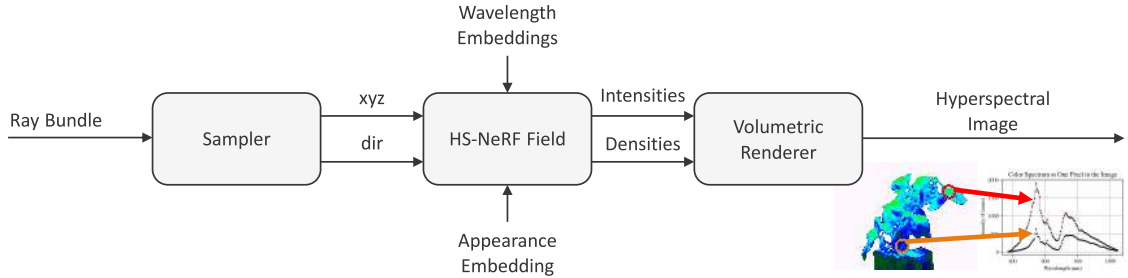
TY-RIST [11] framework, built on YOLOv12n, introduces several enhancements on infrared small target detection (IRSTD) task: a stride-aware backbone for better localization, a high-resolution tiny-object detection head, cascaded coordinate attention to reduce missed detections, improved bounding-box regression, and architecture pruning for efficiency. TY-RIST achieves substantial gains in mAP, precision, and recall across four major benchmarks, runs in real time at 123 FPS, and shows strong cross-dataset generalization.

#### 3.3.2. Generative HSI

*Novel View Synthesis* Synthesizing novel views from hyperspectral data, or Hyperspectral Novel View Synthesis (HNVS), presents significant challenges due to its high dimensionality and the requirement for spectral consistency across different perspectives for each pixel. Existing literatures mainly explore neural radiance field (NeRF) [230] and 3D gaussian splatting (3D-GS) [149] techniques to address this problem.

HS-NeRF [41] has explored HNVS by adapting NeRF architectures to hyperspectral data but suffered from unstable training dynamics, slow rendering times and inefficiencies in handling high-dimensional. [217] proposed a

neural radiance field method employing a multi-channel volume density distribution function to enable the generation of neural radiance fields from hyperspectral data, which can help discriminating objects more effectively than conventional RGB methods.

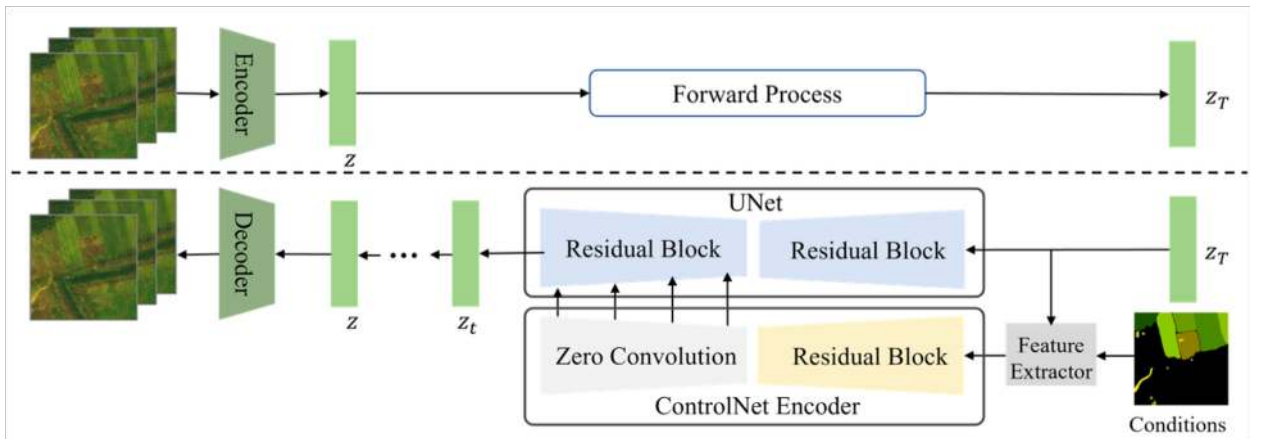


**Figure 30:** HS-NeRF overview, courtesy of [41].

With the emergence of 3DGS in the RGB domain, SpectralGaussians [277] and HS-GS [290] try to accommodate high-dimensional HSI data types.

**HSI Synthesis** Generative AI has achieved tremendous success across diverse fields, yet research on generative models for hyperspectral imaging (HSI) remains relatively limited. An early effort, SHS-GAN [114], proposed generating synthetic hyperspectral cubes in the visible spectrum using a small set of natural hyperspectral cubes, an unlimited number of RGB images, and a Generative Adversarial Network. More recently, Denoising Diffusion Probabilistic Models (DDPMs) have emerged as a compelling alternative, primarily applied to low-level HSI tasks such as reconstruction [247], denoising [229, 356], and super-resolution [229, 318].

UBF [351] was the first to address the HSI synthesis problem in remote sensing. It incorporates unpaired RGB images covering similar scenes and decomposes both unpaired HSIs and RGB images into fixed endmembers and varying abundances, enabling the fusion and generation of new HSIs that closely resemble real-world data. Similar in spirit, Unmixdiff [350] trains a generative model on abundance maps extracted from real HSIs. However, both UBF and Unmixdiff generate only abundance maps rather than full hyperspectral cubes, making their performance heavily dependent on the underlying unmixing network.



**Figure 31:** HSIgene overview, where the ControlNet encoder incorporates the information of the various conditions and hyperspectral images are generated with the diffusion process, courtesy of [248].

To mitigate this limitation, HSiGene [248] performs spatial super-resolution on existing real HSIs and supports multi-condition controllable generation by adopting ControlNet [362] as its backbone.

### 3.3.3. HSI Foundation Model

The topic of foundation models has achieved tremendous success across a wide range of fields [29, 79]. In the domain of RGB imagery, numerous large-scale foundation models [337, 358, 362, 246, 199] have demonstrated exceptional performance across diverse computer vision tasks. This prosperity has also begun to influence hyperspectral imaging (HSI) research, although the development of HSI foundation models remains in its early stages. SatMAE [53] is among the first attempts to design a foundation model for satellite imagery, with its core approach involving the grouping of adjacent spectral bands together with distinct spectral positional encodings. Building on this direction, [271] introduced masked vision transformers for hyperspectral image classification, while SpectralEarth [31] incorporated spectral adapters into backbone networks to better capture the unique spectral characteristics of hyperspectral data. PhySwin [288] integrates physical priors with computational efficiency. Alliance [374] proposed the All-in-One Spectral-Spatial-Frequency Awareness Foundation Model to cope with the limitations in fully utilizing frequency characteristics that capture hidden features.

Notably, all existing HSI foundation models are trained exclusively on remotely sensed multispectral or hyperspectral data. To adapt these models across diverse downstream tasks, several parameter-efficient fine-tuning strategies—such as Low-Rank Adaptation (LoRA), Kronecker-product-based adaptation (KronA), and Low-Rank Kronecker adaptation (LoKr)—have been introduced to effectively fine-tune remote sensing spectral foundation models [196]. While these approaches make such models well suited for RS applications, they leave a substantial gap in natural scene understanding. Although a recent study proposed a general-purpose spectral foundation model based on a masked autoencoder architecture that explicitly targets proximal spectral imaging—that is, spectral imaging captured at distances typical of everyday human visual perception [160]—this gap remains largely unaddressed.

SpectralGPT [125] and HyperSIGMA [297] are two representative spectral foundation models that will be further discussed separately.

*SpectralGPT* SpectralGPT introduces a novel 3D masking strategy built upon the masked autoencoder (MAE) framework [115]. The foundation model is pretrained from scratch in a progressive manner using a curated dataset comprising one million spectral remote sensing images. The overall pipeline is illustrated in Fig. 32.

*HyperSIGMA* HyperSIGMA [297] also employs MAE [115] for pretraining. As shown in Fig. 33, the foundation model adopts a two-branch architecture: the upper spatial branch extracts spatial features, while the lower spectral branch focuses on spectral feature extraction. To improve the modeling of diverse contextual information, the spectral branch additionally incorporates sparse sampling attention. The spatial and spectral features are then fused within a purpose-built Spectral Enhancement Module (SEM) before being passed to task-specific heads for various downstream remote sensing tasks.

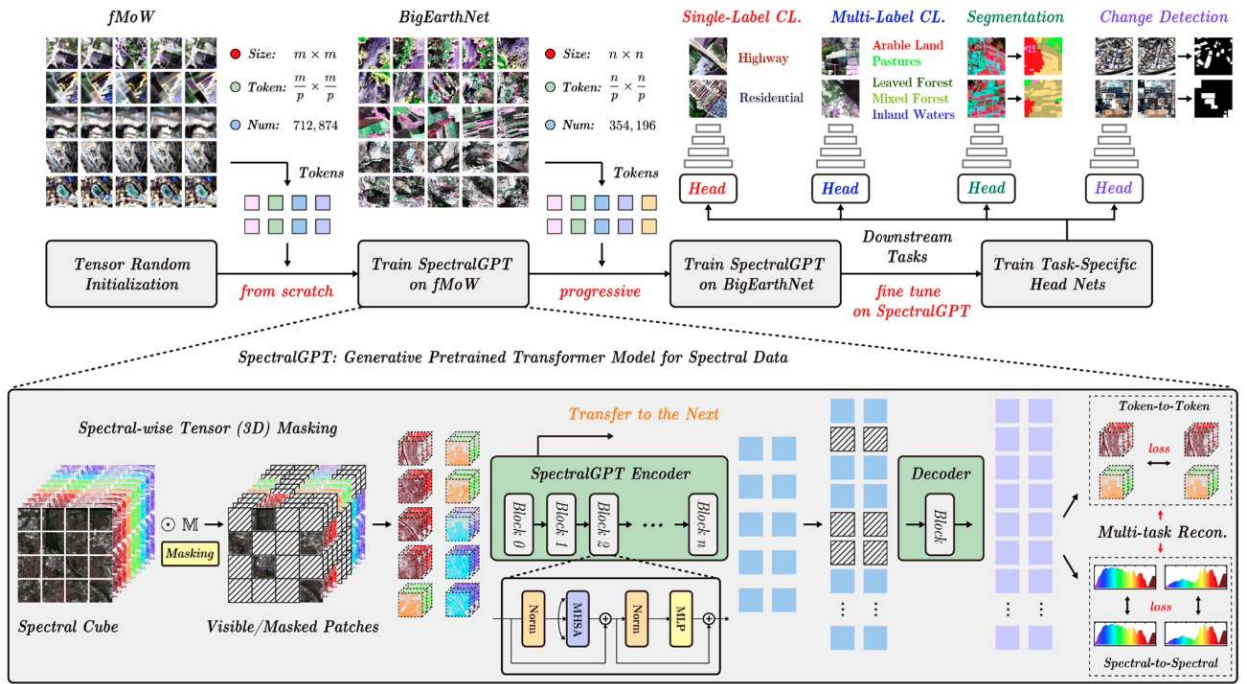


Figure 32: Pipeline of spectral foundation model SpectralGPT [125].

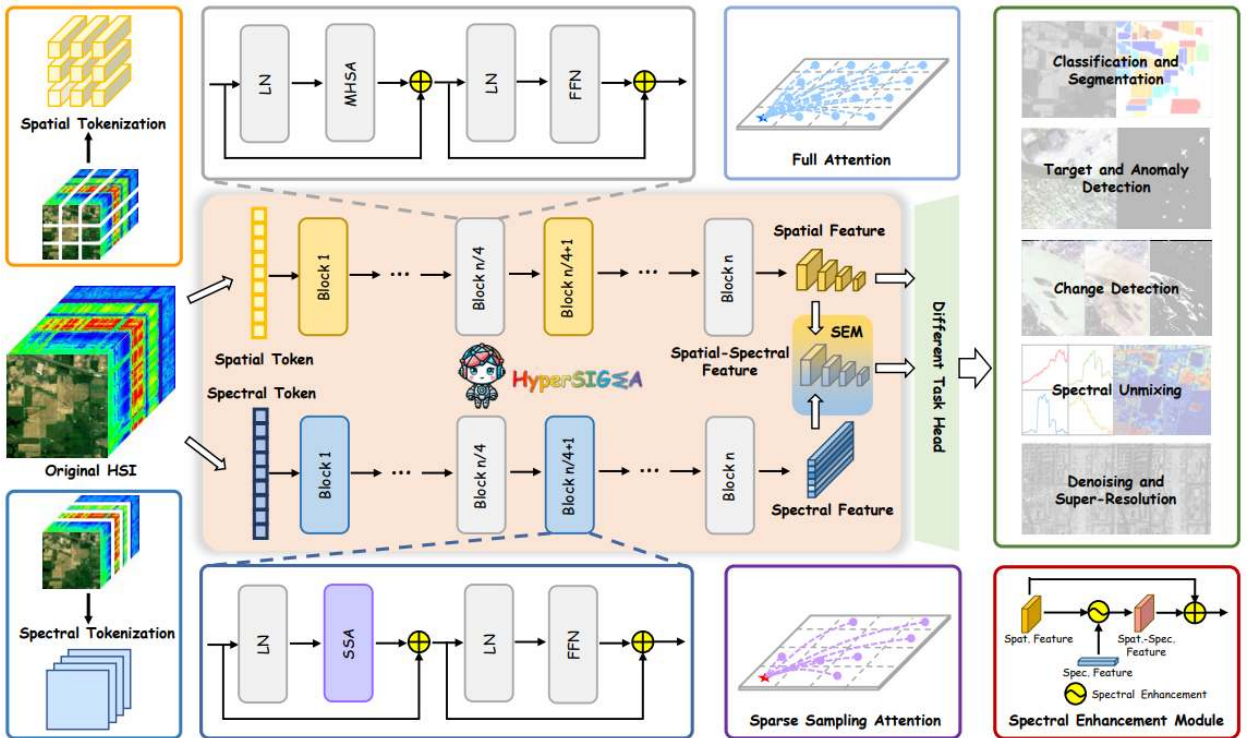


Figure 33: Pipeline of spectral foundation model HyperSIGMA [297].

**Table 9**

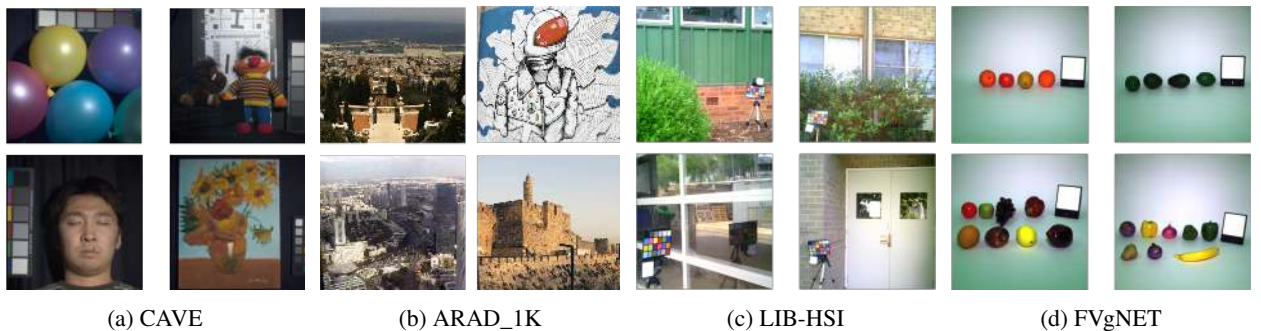
Summary of various HSI datasets collection configurations.

| Dataset          | Year      | Number     | Bands | Spectral Range (nm) | Spatial Resolution | Bit Depth | Techniques | Illuminant                  |
|------------------|-----------|------------|-------|---------------------|--------------------|-----------|------------|-----------------------------|
| Foster [237]     | 2002      | 30         | 31    | 400–720             | various            | 10        | LCTF       | daylight                    |
| CAVE [343]       | 2010      | 32         | 31    | 400–700             | $512 \times 512$   | 16        | LCTF       | CIE Standard Illuminant D65 |
| Harvard [38]     | 2011      | 50         | 50    | 400–700             | $1392 \times 1040$ | 12        | LCTF       | daylight                    |
| NUS [240]        | 2014      | 77         | 31    | 400–700             | $1392 \times 1300$ | 12        | line scan  | halide lamp LED             |
| ICVL [9]         | 2016      | 201        | 31    | 400–700             | $1392 \times 1300$ | 12        | line scan  | daylight                    |
| KAIST [50]       | 2017      | 30         | 28    | 400–720             | $2704 \times 3376$ | 14        | LCTF       | plasma lamp                 |
| HS-SOD [139]     | 2018      | 60         | 81    | 380–780             | $1025 \times 768$  | 10        | LCTF       | daylight                    |
| HSICityV2 [273]  | 2019/2020 | 1306       | 128   | 450–950             | $1422 \times 1889$ | 12        | snapshot   | daylight                    |
| HSIDrive [20]    | 2021/2023 | 752        | 25    | 600–975             | $2048 \times 1088$ | 12        | snapshot   | daylight                    |
| HSIDwRD [366]    | 2021      | 62         | 256   | 375–1075            | $696 \times 520$   | 12        | line scan  | daylight & lamp             |
| KAUST [180]      | 2021      | 400        | 204   | 400–1000            | $512 \times 512$   | 12        | line scan  | simulated illumination      |
| NTIRE2022 [10]   | 2022      | 1000       | 31    | 400–700             | $480 \times 512$   | 12        | line scan  | daylight                    |
| LIB-HSI [110]    | 2022      | 513        | 204   | 400–1000            | $512 \times 512$   | 12        | line scan  | daylight                    |
| FVgNET [219]     | 2022      | 317        | 204   | 400–1000            | $512 \times 512$   | 12        | line scan  | indoor lab LED              |
| HFD100 [379]     | 2023      | 10738      | 256   | 375–1075            | $696 \times 520$   | 12        | line scan  | daylight                    |
| BJTU-UVA [69]    | 2024      | 765 (7650) | 204   | 400–1000            | $512 \times 512$   | 12        | line scan  | daylight                    |
| HOTC [330, 259]  | 2024      | 197 videos | 25    | 470–620             | $512 \times 256$   | 10        | snapshot   | daylight                    |
| Beyond RGB [102] | 2024      | 1680       | 36    | 380–730             | $2584 \times 1936$ | -         | snapshot   | daylight & lamp             |
| HySpeFAS [265]   | 2026      | 6760       | 30    | 450–750             | -                  | -         | snapshot   | daylight & LED              |

## 4. Hyperspectral Image Datasets in Computer Vision

### 4.1. Dataset Collection

In this section, we will report the collection configurations of some HSI datasets used in computer vision. Recall that according to Section 2, utilizing different image sensing equipments can obtain different HSI datacubes, which differ from each other in spectral range, band number and spatial resolution. Scanning-based imaging methods, either LCTF or line-scan methods, and snapshot imaging methods will result in different intrinsics when capturing HSIs. Furthermore, different illumination conditions will also affect the characteristics of HSIs. A summary can be seen in Table. 9. We will report more details for each of these widely used HSI datasets.



**Figure 34:** Visual Examples of Hyperspectral Datasets in Computer Vision: (a) CAVE [343]; (b) ARAD\_1K [10]; (c) LIB-HSI [110]; (d) FVgNET [219].

#### 4.1.1. CAVE

CAVE<sup>2</sup> [343] dataset is composed of 32 hyperspectral images at a spatial size of  $512 \times 512$ . It captured 31-band multispectral images (400–700 nm, at 10 nm intervals) of several static scenes using a tunable filter (VariSpec Liquid Crystal Tunable Filter) and a cooled CCD camera (Apogee Alta U260,  $512 \times 512$  pixels), under the CIE Standard Illuminant D65. The images include a wide variety of objects and materials, ranging from textiles, skin, hair, real and fake fruits to vegetables, candy, drinks, paints, etc. And the images are stored in 16-bit PNG format.

#### 4.1.2. ICVL

ICVL [9] dataset of hyperspectral natural images is acquired using a Specim PS Kappa DX4 hyperspectral camera and a rotary stage for spatial scanning. 100 images were captured from a variety of urban (residential/commercial), suburban, rural, indoor and plant-life scenes but the database is designed to grow progressively. All images are  $1392 \times 1300$  in spatial resolution and 519 spectral bands (400–1,000 nm at roughly 1.25 nm increments).

#### 4.1.3. KAIST

KAIST [50] dataset consists of 30 hyperspectral images at a spatial size of  $2704 \times 3376$ . The setup of the dataset consists of a VeriSpec VIS LCTF filter with the effective spectral range from 420 nm to 720 nm, a Coastal Optics 60 mm apochromatic lens, a Monochromatic PointGrey Grasshopper camera (GS3-PGE-91S6M-C) and a solid-state plasma lamp illuminant (Thorlab HPLS-30-04).

#### 4.1.4. Hyperspectral City

Hyperspectral City [346] was proposed as a semantic segmentation challenge for urban autonomous driving scene which utilized the LightGene camera sensor. In particular, the camera can provide hyperspectral images with the size  $1400 \times 1800 \times 125$ . The dataset was collected in Shanghai for three days in June. A variety of environment including crowded traffic area, famous buildings and structures, CBD, highways, quiet suburbs, overpasses and underground parking were covered. The weather condition encompassed sunny and cloudy days. And the lighting condition varies from day, night to sunset. Color calibration was implemented by using standard color board. The dataset contains manually selected 367 (training) plus 55 (testing) hyperspectral images, where 300 images of the training dataset were coarsely labeled by different people in a quick and rough fashion. The author compressed the image using H.264 encoder and decoder with a quality setting at 90%. The compression ratio is smaller than 2% and the final dataset size is around 40 Gigabytes. In 2021, an extended new version HSIcityV2 [273] was introduced with 1054 HSI frames with coarse labeling for training and 200 frames with fine labeling for testing.

#### 4.1.5. NTIRE2022

NTIRE 2022 Spectral Recovery Challenge released a dataset termed the “ARAD 1K Natural Hyperspectral Image Dataset” [10] for the reconstruction of whole-scene hyperspectral (HS) information from a 3-channel RGB image. The 1,000 images included in the dataset were divided as follows: 900 training images, 50 validation images, and 50 test images. These images were captured with the Specim IQ camera, a stand-alone, battery-powered, compact, push-broom spectral imaging system that provides RAW  $512 \times 512$  pixel images with 204 spectral bands in the 400–1000 nm range. For the purpose of this challenge, manufacturer-supplied radiometric calibration was applied to the RAW images, and the images were resampled to 31 spectral bands in the visual range (400–700 nm).

#### 4.1.6. LIB-HSI

This Light Industrial Building HSI (LIB-HSI) dataset [110] consists of 513 hyperspectral images and their corresponding RGB images, which can be divided into 9 categories and 44 classes. Different classes of material/context were labelled across the images. Images were taken of building facades in a light industrial environment. Data augmentation was performed to increase the size of the dataset, resulting in over 15768 patches of size  $256 \times 256$  pixels for training. For testing, the test set is kept in its original form. The train/validation/test splits consist of 393/45/75 images, separately.

#### 4.1.7. FVgNET

FVgNET dataset [219] consists of 317 scenes showing fruits and vegetables, both natural and artificial, taken indoors under controlled lighting conditions and covering the 400 nm to 1000 nm range. There exists a near equal

<sup>2</sup>Source: <https://cave.cs.columbia.edu/repository/Multispectral>

number of natural and artificial objects in the scenes, 80% of the images are with segmentation masks and the rest with labels only. The dataset contains 12 classes of objects, such as potato, onion, apple, orange, etc. Each HSI has a spatial resolution of  $512 \times 512$  and 204 spectral bands.

#### 4.1.8. HOTC

Hyperspectral Object Tracking Challenge (HOTC), also named WHISPERS competition dataset in some literatures, consists of three types of video sequences, including hyperspectral video data, false-color video data and RGB video data. Wherein RGB video data was shot at the same time with almost the same angle of view as the hyperspectral video, and the false-color video data was synthesized from a hyperspectral video sequence. Hyperspectral sequences and RGB sequences were co-registered by the Xiong [330] dataset so that they describe almost the same scene. A XIMEA SSM4x4VIS snapshot mosaic hyperspectral camera was used to capture the whole dataset [259], under daylight illumination.

The original dataset contains 40 sets of videos for training and 35 sets of videos for testing. Every video is labeled with the associated challenging factors out of 11 attributes, and each frame was manually marked with the bounding boxes of the targets by [330]. The labels for the hyperspectral and RGB videos were generated independently and those for the hyperspectral videos can be used directly on the false-color videos. The dataset is extended and enlarged every year. In the most recent challenge in 2024<sup>3</sup>, the dataset contains 182 training videos, 89 validation videos, and 75 unlabeled hyperspectral videos as a testing set for ranking purpose.

#### 4.1.9. Beyond RGB

Beyond RGB [102] contains 1680 scenes, of which 472 are collected in the field and 1208 are collected in the lab. The field scenes consist of 235 indoor scenarios and 237 outdoor scenarios at many different locations, hours of the day and contain diverse subject matters. The illuminants present in the field scenes can be expected for natural light. The scenes collected in the lab are comprised of 13 well-designed distinct scenarios including a wide gamut of colors.

The illumination in the viewing booth is spatially uniform, and 12 different light sources can be set on or off, allowing for generating many distinct mixtures. In practice, 93 different illuminations are generated. Illuminant spectrum ground truth are provided, making the dataset suitable for illumination spectral estimation, color constancy and color accuracy tasks.

#### 4.1.10. HS-NeRF

HS-NeRF [41] is a dataset containing nearly 2000 hyperspectral images across 8 scenes and 2 cameras. With the Surface Optics camera, it collected a dataset of 4 scenes of 48 images each, with 2 of the scenes exhibiting intricate plant geometry ("Rosemary" and "Basil") and the other two exhibiting several objects with wavelength-dependent transparency and radiance/reflection ("Tools" and "Origami"). And with the BaySpec camera, it collected a dataset of 4 scenes of 433 images each, all exhibiting intricate plant geometry ("Anacampseros" and "Caladium" made of plastic, a "Pinecone", and a "Buttercrunch Lettuce").

#### 4.1.11. Other task-specific datasets

Apart from the aforementioned datasets, there are some other task-specific datasets. For instance, GHIFVD [72] aimed at detecting 42 fruits and vegetables, [106] introduced a hyperspectral dermoscopy image dataset for melanoma detection, Powder100 [380] acted as a material recognition benchmark, MUSIC [148] was released as a multispectral computed tomography dataset in the area of segmentation for material differentiation, HSIroad [214] is a dataset for road segmentation that comprises images taken both in urban and rural scenes, HSIfoodIngr [323] was proposed for food-centric tasks, Hyper-Skin [239] can be used to recognize different human skin, and an recent agricultural wheat leaves classification dataset HyperLeaf [159].

Dr. David H Foster released several hyperspectral image datasets in natural scenes in 2002 [237], 2004 [87], 2016 [85, 236] and 2022 [88, 84]. All these datasets can be found on his personal webpage<sup>4</sup>. Harvard [38], NUS [240] dataset, KAUST multispectral reflectance image dataset [180]. BJTU-UVA dataset [69]. HSIIDwRD dataset [366]

HS-SOD [139]. Recently, a follow-up work introduced the HSOD-BiT dataset [262], an updated version—HSOD-BiT-V2 [263]—has also been released and made open-source for further research.

<sup>3</sup>Source: <https://www.hsitracking.com/>

<sup>4</sup>Source: <https://personalpages.manchester.ac.uk/staff/d.h.foster//default.html>

**Table 10**

Comparison of various hyperspectral image datasets and supported tasks.

| Dataset         | Low-level Vision Tasks |    |           |     | High-level Vision Tasks |             |           |          |
|-----------------|------------------------|----|-----------|-----|-------------------------|-------------|-----------|----------|
|                 | Reconstruction         | SR | Denoising | ISE | Segmentation            | Recognition | Detection | Tracking |
| Foster [237]    |                        |    |           | ✓   |                         |             |           |          |
| CAVE [343]      |                        | ✓  | ✓         | ✓   |                         |             |           |          |
| Harvard [38]    |                        |    | ✓         |     |                         |             |           |          |
| NUS [240]       |                        | ✓  |           |     |                         |             |           |          |
| ICVL [9]        | ✓                      | ✓  | ✓         |     |                         |             |           |          |
| KAIST [50]      | ✓                      |    | ✓         |     |                         |             |           |          |
| HS-SOD [139]    |                        |    |           |     |                         |             | ✓         |          |
| HSICityV2 [273] |                        |    |           |     | ✓                       |             |           |          |
| HSIDrive [20]   |                        |    |           | ✓   |                         |             |           |          |
| HSIDwRD [366]   |                        |    | ✓         |     |                         |             |           |          |
| KAUST [180]     |                        |    |           | ✓   |                         |             |           |          |
| NTIRE2022 [10]  |                        | ✓  |           |     |                         |             |           |          |
| LIB-HSI [110]   |                        |    |           |     | ✓                       |             |           |          |
| FVgNET [219]    |                        |    |           |     | ✓                       |             |           |          |
| HFD100 [379]    |                        |    |           |     | ✓                       |             |           |          |
| BJTU-UVA [69]   | ✓                      | ✓  | ✓         | ✓   |                         |             |           |          |
| HOTC [330, 259] |                        |    |           |     |                         |             |           | ✓        |
| BeyondRGB [102] |                        |    |           | ✓   |                         |             |           |          |
| HySpeFAS [265]  |                        |    |           |     |                         | ✓           |           |          |

HSI-Drive [20] Version v2.0 [109] contains 752 images, incorporating new images taken during fall and winter. hyperspectral flower dataset (HFD100) [379] for flower classification, The HySpeFAS dataset [265] for face anti-spoofing.

## 4.2. Dataset Utilization

As analyzed in the above section, different datasets are collected under different conditions so that they suit distinct task evaluations. An overview of diverse datasets applications is listed in Table 10. We will discuss these specific evaluations in details.

### 4.2.1. Low-level Evaluation

Since hyperspectral image (HSI) restoration tasks—such as reconstruction, super-resolution, and denoising—are commonly formulated by first degrading raw data cubes and then restoring the degraded HSIs, datasets captured using scanning-based techniques are generally applicable to these tasks. Among them, CAVE, Harvard, ICVL, and KAIST are the most widely used classic datasets and have been extensively adopted for evaluating both general HSI restoration and remote sensing algorithms [328, 129, 124, 384, 284, 377, 55, 43, 187, 74]. However, these datasets suffer from several inherent limitations: their image quantities are small, scene diversity is limited, and the imaging equipment used for acquisition is relatively outdated.

As a consequence, algorithmic performance on these datasets has gradually converged. Widely adopted evaluation metrics, including PSNR, SSIM, and SAM, are becoming insufficient to fairly and effectively differentiate advanced methods. For instance, [334] reported reconstruction results of approximately 0.1 MRAE and 0.02 RMSE on the CAVE dataset; [363] achieved super-resolution performance of up to 50 PSNR and as low as 2.64 SAM on Harvard; [172] reported denoising results of 48 PSNR and SSIM close to 0.99 on ICVL; and [310] achieved an average reconstruction performance of 44 PSNR and 0.99 SSIM on the KAIST simulation dataset.

To address these limitations, several new datasets—HSIDwRD, KAUST, and NTIRE2022—have been released in recent years. These datasets contain a larger number of HSIs and cover more diverse scenes, thereby posing more realistic and challenging benchmarks for existing algorithms and enriching evaluation protocols. Nevertheless, their

potential has not yet been fully explored. For example, [348] reported their best denoising performance of 27 PSNR, 0.8422 SSIM, and 4.312 SAM on the HSIDwRD dataset, which is significantly lower than the performance claimed on traditional benchmarks. Similarly, [341] achieved a reconstruction performance of 33 PSNR on the real NTIRE2022 dataset. On the same dataset, [71] reported demosaicing results of 37 PSNR and 0.97 SSIM.

It is also worth noting that illumination spectral estimation remains a long-standing and challenging problem in hyperspectral imaging. Although Foster released pioneering datasets more than two decades ago, this task has regained considerable attention in recent years, leading to the release of two new public datasets: BJTU-UvA and BeyondRGB. Xie et al. [325] evaluated their reconstruction method on BJTU-UvA and achieved a maximum of 39.535 PSNR and 0.9689 SSIM; however, they acknowledged that their approach still suffers from local detail loss under illumination shadows. Meanwhile, [145] reported an auto white balance result with the lowest color difference of 0.489 on the BeyondRGB dataset. On the same dataset, [339] addressed the illumination estimation task by achieving a minimum mean angular error of  $2.01^\circ$  and an average color difference of 4.72 in terms of  $\Delta E_{2000}$ .

#### 4.2.2. High-level Evaluation

Despite the abundance of RGB datasets for high-level computer vision tasks and the abundance of satellite image datasets in remote sensing, the development of hyperspectral image (HSI) datasets targeting high-level computer vision tasks has been relatively limited and slow, only a handful of public HSI datasets are currently available.

Early studies on hyperspectral salient object detection (HSOD) relied on datasets that were not specifically curated for this task, often suffering from low data quality and limited applicability [263]. The first dedicated HSOD dataset, HS-SOD [139], was small in scale and restricted to common scenes. Subsequently, HSOD-BIT [262] and its improved version HSOD-BIT-V2 [263] were released, incorporating larger-scale data with higher quality and greater scene diversity.

The first hyperspectral object tracking dataset was introduced in 2019 [329]. Building upon this effort, the Hyperspectral Object Tracking Challenge (HOTC) has released and updated benchmark datasets on a yearly basis. However, the hyperspectral images in these datasets are generally limited to fewer than 25 spectral bands, and hyperspectral object tracking datasets with wider spectral coverage remain scarce.

In the area of face recognition, although several hyperspectral face datasets have been reported in the literature [57, 49, 370], most of them have not been made publicly available due to privacy concerns. Recently, the Snapshot Spectral Imaging Face Anti-spoofing Challenge at CVPR 2024 introduced a new dataset, HySpeFAS [368, 166, 279], and its conditional version has also been released [265], providing new opportunities for research in this domain.

For segmentation tasks, relatively more HSI datasets are available. Representative examples include the building façade segmentation dataset LIB-HSI [110], the real and synthetic fruit segmentation dataset FVgNET [219], and the urban scene segmentation dataset HSiCityV2 [273]. It is worth noting that these datasets are primarily designed for pixel-level segmentation, whereas the HFD100 dataset [379] can be applied to image-level segmentation, despite being limited to flower categories.

#### 4.2.3. AI Technique Evaluation

Unfortunately, there is no large-scale natural scene HSI dataset except for [160], which combined 11 HSI datasets to aggregate a 10k-level dataset for both proximal and remote sensing spectral imaging. However, it still lagged behind its counterparts in remote sensing, which pretrained their foundation models with 450k HSI patches [297] or with 1.5M satellite images. Furthermore, it did not notice the difference between natural scene HSIs and remote sensing HSIs, therefore, the involvement of remote sensing data may prevent the foundation model from achieving computer vision tasks.

For HNVS task, algorithms are evaluated on HS-NeRF datasets [41], such as [235, 290, 252], before its open-source, literatures generally captured their own datasets and evaluated onward [175, 167, 217].

### 4.3. Performance Indicator

The performance evaluation varies across different tasks. For illumination spectra estimation, metrics can be divided into reconstruction-based and spectrum-based categories. The former includes PSNR, SSIM, SAM, and ERGAS, while the latter commonly adopts Mean Angular Error (MAE) and CIE  $\Delta E_{2000}$  [272]. The MAE measures the angular difference between ground-truth and estimated spectral vectors, whereas  $\Delta E_{2000}$  quantifies perceptual color differences. For classification tasks, overall accuracy (OA), average accuracy (AA), and the Kappa coefficient are widely used, where OA reflects the overall correctness, AA averages per-class accuracy, and Kappa evaluates

the agreement between predictions and ground truth. In semantic segmentation, pixel-level accuracy and mean Intersection over Union (mIoU) are the primary metrics. For face recognition, in addition to recognition accuracy, Attack Presentation Classification Error Rate (APCER), Bona Fide Presentation Classification Error Rate (BPCER), and Average Classification Error Rate (ACER) are commonly reported. In object detection and salient object detection, Precision (P), Recall (R), F-measure (F), and AUC are frequently employed, with object detection also considering ROC curves and AUROC. Finally, hyperspectral object tracking is typically evaluated using success plots and precision plots under the one-pass evaluation (OPE) protocol, where the area under the success curve (AUC) and a 20-pixel threshold precision are commonly used as standard indicators.

## 5. Benchmarking Foundation Models for CV Tasks

Now that spectral foundation models are available, can we use them to solve computer vision tasks? If they perform well, how satisfactory can the results be? If their performance falls short, what factors limit their transferability? Bearing these ideas in mind, we design our benchmarking as follows. To the best of our knowledge, at the time of this survey's submission, no existing work has examined how spectral foundation models perform on computer vision tasks. Our experiments lead to a new benchmark and open up new possibilities for applying foundation-model-based HSI methods to computer vision scenarios.

### 5.1. Experimental settings

In order to evaluate the capabilities of existing spectral foundation models, we conduct high-level semantic segmentation experiments on two state-of-the-art spectral foundation models, SpectralGPT [125] and HyperSIGMA [297]. Three representative datasets with human annotations, LIB-HSI [110], FVgNET [219] and HSICityV2 [273] are chosen as evaluation datasets. For LIB-HSI [110], the dataset was split into train, validation and test set, we can thus follow the common machine learning process and report the evaluation metrics on the test set. However, for FVgNET [219] and HSICityV2 [273], we report the validation statistics as did in the literatures.

We quantitatively assess the performance of foundation models in terms of overall accuracy (OA) and mean intersection over union (mIoU) for semantic segmentation task. Utilizing the computational power of 4 NVIDIA A100 GPUs, we meticulously elaborate our benchmarking settings, aiming at exploring the possibilities and weaknesses of spectral foundation models under natural scene scenarios.

For SpectralGPT [125], we follow its original settings and implement the Adam optimizer [211] with a foundational learning rate of  $10^{-4}$ , coupled with a linear warm-up and a polynomial decay schedule. For HyperSIGMA [297], the Adam optimizer is also employed, while the learning rate is set to  $6 \times 10^{-5}$ , coupled with a layer-wise half-cycle cosine learning rate decay, where the weight decay is set to 0.05.

Because existing spectral foundation models i.e. both SpectralGPT [125] and HyperSIGMA [297] are designed for Sentinel-2 spectral satellite images. To be specific, the spatial size of the input image is limited to  $128 \times 128$  and they can handle 10 spectral bands. However, HSIs captured in natural scene scenarios typically have much larger spatial and spectral resolutions. For instance, FVgNET [219] and LIB-HSI [110] capture HSIs with size of  $512 \times 512 \times 204$  and HSICityV2 dataset contains cityscape HSIs of  $1422 \times 1889 \times 128$ . To address this mismatch, we reduce the spatial and spectral dimensions of these HSIs for compatibility to enable compatibility with spectral foundation models.

#### 5.1.1. Spatial Dimension reduction strategies: overlapped patchifying

Considering that the HSIs in FVgNET and LIB-HSI are both captured using the Specim IQ camera, they share the same spatial dimensions. Therefore, we preprocess them by dividing each HSI of size  $512 \times 512$  into  $128 \times 128$  patches with a 50% overlap. For the HSICityV2 dataset, we also split the raw HSI into  $128 \times 128$  patches but without overlap, since the original spatial resolution of each image ( $1422 \times 1889$ ) is sufficiently large. In this case, overlapping patches would not provide additional benefits for the model but would instead increase the computational burden during training.

#### 5.1.2. Spectral Dimension reduction strategies: PCA & Pseudo Sentinel Bands

Generally, the principle component analysis (PCA) technique is usually used to reduce the dimension of spectral bands while maintaining critical information. [213, 12, 104, 286] We act similarly and pre-process the dataset by employing PCA as dimension reduction as did in the literatures. Apart from PCA, we additionally design another dimension reduction strategy that select corresponding bands manually. In Sentinel-2 images, the chosen spectral bands

**Table 11**  
Band selection.

| Band Index | Remote sensing<br>(Sentinel-2) | LIB-HSI & FVgNET<br>(Specim IQ) | HSICityV2<br>(LightGene) |
|------------|--------------------------------|---------------------------------|--------------------------|
| 1          | B1: 443 nm                     | 443.56 nm                       | 450 nm                   |
| 2          | B2: 490 nm                     | 490.06 nm                       | 490 nm                   |
| 3          | B3: 560 nm                     | 560.3 nm                        | 562 nm                   |
| 4          | B4: 665 nm                     | 663.81 nm                       | 662 nm                   |
| 5          | B5: 705 nm                     | 705.57 nm                       | 706 nm                   |
| 6          | B6: 740 nm                     | 738.53 nm                       | 738 nm                   |
| 7          | B7: 783 nm                     | 783.67 nm                       | 782 nm                   |
| 8          | B8A: 865 nm                    | 865.5 nm                        | 886 nm                   |
| 9          | B11: 1610 nm                   | 1000.49 nm                      | 942 nm                   |
| 10         | B12: 2190 nm                   | 1003.58 nm                      | 946 nm                   |

**Table 12**  
Comparison of off-the-shelf and fine-tuned foundation models on LIB-HSI.

| Foundation model | SpectralGPT |      | SpectralGPT+ |      | SpatSIGMA |      | HyperSIGMA |      |
|------------------|-------------|------|--------------|------|-----------|------|------------|------|
|                  | OA          | mIoU | OA           | mIoU | OA        | mIoU | OA         | mIoU |
| Off-the-shelf    | 4.7         | 1.0  | 11.6         | 2.7  | 13.2      | 1.5  | 10.6       | 1.2  |
| Fine-tuned       | 88.7        | 75.3 | 90.0         | 77.9 | 85.8      | 70.0 | 85.3       | 69.0 |

are B1, B2, B3, B4, B5, B6, B7, B8A, B11 and B12, whose corresponding wavelengths are 443 nm, 490 nm, 560 nm, 665 nm, 705 nm, 740 nm, 783 nm, 865 nm, 1610 nm and 2190 nm<sup>5</sup>. Considering that the spectral range of Specim IQ hyperspectral camera is limited to 400-1000 nm, we employ a Sentinel-2-aligned band selection strategy, referred to as pseudo-Sentinel (pSentinel) in analogy to pseudo-RGB (pRGB) [60, 289]. Specifically, bands of wavelength 443.56 nm, 490.06 nm, 560.3 nm, 663.81 nm, 705.57 nm, 738.53 nm, 783.67 nm, 865.5 nm, 1000.49 nm and 1003.58 nm are chosen as the simulated counterparts of the Sentinel-2 inputs of SpectralGPT [125] and HyperSIGMA [297], details can be seen in Table 11.

### 5.1.3. Class imbalance addressing method

Different from remote sensing, where different semantic classes share similar pixel numbers, HSIs captured in natural scene may have objects whose sizes shift severely. In other words, natural scene HSI datasets suffer from severe class imbalance problem, shown in Figure 35a. This case results in satisfactory performance on classes (like Glass window, Vegetation, Metal) which have large number of pixels but poor performance on classes (like Timber) whose pixel number is relatively few.

In response to this problem, we adapt Median Frequency Balancing [14] strategy to calculate the weights to mitigate the influence caused by class imbalance.

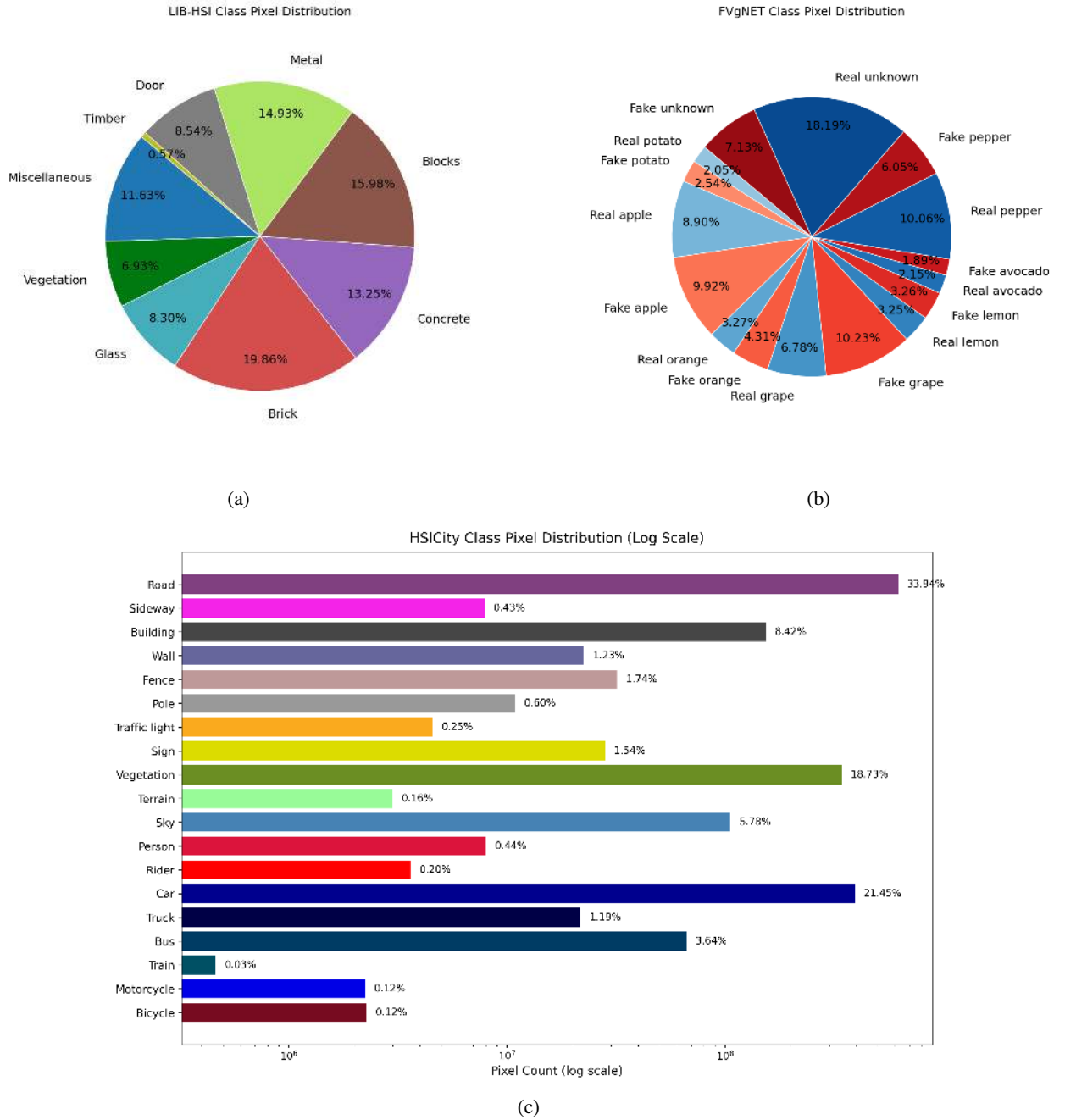
HSICityV2 dataset suffers from severe class imbalance problem. As shown in Fig.35c, several semantic classes such as road, building, vegetation, and car account for the majority of the pixels where others like terrain, person, rider, train, motorcycle and bicycle are relatively few. This poses challenges on segmentation algorithm.

## 5.2. Quantitative Results

### 5.2.1. Evaluation on LIB-HSI

Table 12 compares the performance on the validation set between off-the-shelf spectral foundation models and their finetuned version. Neither SpectralGPT or HyperSIGMA can achieve satisfactory results in segmentation of building facade. Even though 150 epochs fine-tuning can largely enhance the ability of all spectral foundation models in identifying different facade, they are still far from foundational enough to cope with more complex real natural world.

<sup>5</sup>Source:<https://custom-scripts.sentinel-hub.com/custom-scripts/sentinel-2/bands/>



**Figure 35:** Different Class Label Distribution of (a) LIB-HSI, (b) FVgNET and (c) HSiCityV2.

Table 13 lists the performance of different models on LIB-HSI dataset. The author of LIB-HSI provided segmentation results employing FCN network using RGB images and hyperspectral (HS) images. We adapt a general U-Net network that can handle 10-band input as a pre-foundation model, and we evaluate 4 spectral foundation models. By selecting manually the corresponding bands whose wavelengths are close to the pSentinel, pre-foundation and foundation models outperform the original paper [110]. This demonstrates the ability of spectral foundation models in extracting and identifying different materials from data with multiple spectral bands.

**Table 13**

Dimension Reduction Performance on LIB-HSI.

| Dimension Reduction | LIB-HSI            |                   |       |             |              |           |            |
|---------------------|--------------------|-------------------|-------|-------------|--------------|-----------|------------|
|                     | FCN (RGB)<br>[110] | FCN (HS)<br>[110] | U-Net | SpectralGPT | SpectralGPT+ | SpatSIGMA | HyperSIGMA |
| pSentinel           | 44.3               | 54.7              | 73.5  | 75.3        | <b>77.9</b>  | 70.0      | 69.0       |
| PCA                 |                    |                   | 65.0  | 66.0        | 59.0         | 50.7      | 48.1       |

**Table 14**

Per-class performance on LIB-HSI.

| Model / Class  | Miscellaneous | Vegetation | Glass window | Brick | Concrete | Block | Metal | Door  | Timber |
|----------------|---------------|------------|--------------|-------|----------|-------|-------|-------|--------|
| IoU-SpecGPT    | 0.693         | 0.875      | 0.764        | 0.844 | 0.802    | 0.899 | 0.826 | 0.782 | 0.610  |
| IoU-SpecGPT+   | 0.731         | 0.886      | 0.805        | 0.862 | 0.819    | 0.913 | 0.844 | 0.766 | 0.610  |
| IoU-SpatSIGMA  | 0.656         | 0.852      | 0.673        | 0.820 | 0.730    | 0.834 | 0.794 | 0.695 | 0.541  |
| IoU-HyperSIGMA | 0.632         | 0.840      | 0.651        | 0.798 | 0.750    | 0.817 | 0.760 | 0.679 | 0.507  |
| F1-SpecGPT     | 0.819         | 0.933      | 0.866        | 0.916 | 0.890    | 0.947 | 0.905 | 0.878 | 0.758  |
| F1-SpecGPT+    | 0.845         | 0.940      | 0.892        | 0.926 | 0.900    | 0.955 | 0.915 | 0.867 | 0.757  |
| F1-SpatSIGMA   | 0.792         | 0.920      | 0.804        | 0.901 | 0.844    | 0.909 | 0.885 | 0.820 | 0.703  |
| F1-HyperSIGMA  | 0.774         | 0.913      | 0.789        | 0.888 | 0.857    | 0.899 | 0.864 | 0.809 | 0.673  |

**Table 15**

Comparison of off-the-shelf and fine-tuned foundation models on FVgNET.

| Foundation model | SpectralGPT |      | SpectralGPT+ |      | SpatSIGMA |      | HyperSIGMA |      |
|------------------|-------------|------|--------------|------|-----------|------|------------|------|
|                  | OA          | mIoU | OA           | mIoU | OA        | mIoU | OA         | mIoU |
| Off-the-shelf    | 2.3         | 0.1  | 1.6          | 0.6  | 0.0       | 0.0  | 3.3        | 0.5  |
| Fine-tuned       | 96.2        | 78.3 | <b>97.6</b>  | 84.9 | 96.7      | 78.5 | 96.7       | 78.0 |

Table 14 presents the detailed per-class segmentation performance of four variants of spectral foundation models on the LIB-HSI dataset. It can be observed that, in terms of both IoU and F1-score, SpectralGPT+ achieves the best performance among all building façade classes except Door, while the original SpectralGPT ranks second. Among the nine building façade classes, Vegetation, Brick, and Block are relatively easy to segment, as nearly all foundation models achieve IoU scores above 0.8 and F1-scores above 0.9. In contrast, Miscellaneous, Glass Window, and Timber are more challenging to recognize due to greater object variability, disturbances caused by specular reflections, and the relatively limited number of annotated training samples.

### 5.2.2. Evaluation on FVgNET

As introduced in section 4.1.7, the FVgNET [219] labeled 12 kinds of real and fake fruits, resulting in 25 different classes in total with background. The corresponding label masks were also provided by the author, the class distribution is shown in Fig. 35b. However, the author reported their segmentation results under the specific condition where onion, plant, banana, miscellaneous and starfruit were clustered as a single 'unknown' class. To make a fair comparison, we follow this setting and conduct our experiments in segmenting 17 classes of real and fake fruits, results are listed in table 16. Additionally, we also evaluate the segmentation performance of raw 25 classes as in table 17.

Table 18 presents the detailed 17-class segmentation performance of four variants of spectral foundation models on the FVgNET dataset. It can be observed that the algorithm proposed in [219] achieves the best IoU and F1-score for most real and artificial fruit identification tasks. However, this algorithm completely fails to recognize Artificial Lemon and Real Avocado, whereas all four foundation models successfully identify these classes. SpectralGPT+ achieves the highest IoU for Artificial Apple, Real Grape, Real Lemon, Real Avocado, and Artificial Unknown, and obtains the second-best IoU performance across 11 classes overall. In terms of F1-score, either SpectralGPT or SpatSIGMA achieves the second-best performance, each covering seven classes.

**Table 16**

Dimension Reduction Performance on FVgNET across 17 classes.

| Dimension Reduction | FVgNET (17 classes) |            |       |             |              |           |            |
|---------------------|---------------------|------------|-------|-------------|--------------|-----------|------------|
|                     | FVgNET(RGB)         | FVgNET(HS) | U-Net | SpectralGPT | SpectralGPT+ | SpatSIGMA | HyperSIGMA |
| pSentinel           |                     |            | 80.0  | 78.3        | <b>84.9</b>  | 78.5      | 78.0       |
| PCA                 | 68.8                | 81.2       | 41.0  | 41.3        | <b>46.0</b>  | 27.7      | 39.8       |

**Table 17**

Dimension Reduction Performance on FVgNET across 25 classes. A dash (–) indicates results or code that are unavailable and thus cannot be reproduced.

| Dimension Reduction | FVgNET (25 classes) |            |       |             |              |           |            |
|---------------------|---------------------|------------|-------|-------------|--------------|-----------|------------|
|                     | FVgNET(RGB)         | FVgNET(HS) | U-Net | SpectralGPT | SpectralGPT+ | SpatSIGMA | HyperSIGMA |
| pSentinel           |                     |            | 75.2  | 75.5        | <b>77.7</b>  | 69.8      | 60.3       |
| PCA                 | –                   | –          | 37.8  | 33.7        | <b>43.3</b>  | 33.5      | 36.6       |

**Table 18**

Per-class performance on FVgNET.

| Model / Class  | background | real potato | artificial potato | real apple | artificial apple | real orange | artificial orange | real grape | artificial grape | real lemon | artificial lemon | real avocado | artificial avocado | real pepper | artificial pepper | real unknown | artificial unknown |
|----------------|------------|-------------|-------------------|------------|------------------|-------------|-------------------|------------|------------------|------------|------------------|--------------|--------------------|-------------|-------------------|--------------|--------------------|
| IoU-FVgNET     | 0.9940     | 0.9626      | 0.9655            | 0.9522     | 0.9021           | 0.9786      | 0.9541            | 0.8294     | 0.8971           | 0.8673     | 0.0009           | 0.0365       | 0.9436             | 0.9586      | 0.9426            | 0.9243       | 0.7016             |
| IoU-SpecGPT    | 0.9610     | 0.7920      | 0.8740            | 0.8600     | 0.8110           | 0.8470      | 0.8370            | 0.6660     | 0.6820           | 0.8050     | 0.8320           | 0.7250       | 0.7550             | 0.8360      | 0.7990            | 0.6600       | 0.5780             |
| IoU-SpecGPT+   | 0.9870     | 0.9390      | 0.9480            | 0.9480     | 0.9270           | 0.9400      | 0.9160            | 0.8540     | 0.8860           | 0.9260     | 0.9380           | 0.9150       | 0.9170             | 0.9390      | 0.9120            | 0.8750       | 0.8180             |
| IoU-SpatSIGMA  | 0.9660     | 0.7500      | 0.8470            | 0.8250     | 0.7430           | 0.8210      | 0.7950            | 0.7430     | 0.7850           | 0.7370     | 0.7990           | 0.7590       | 0.7720             | 0.8290      | 0.8060            | 0.7360       | 0.6270             |
| IoU-HyperSIGMA | 0.9660     | 0.7310      | 0.8330            | 0.8280     | 0.7580           | 0.8050      | 0.7440            | 0.7590     | 0.7650           | 0.7180     | 0.7850           | 0.8110       | 0.7640             | 0.8380      | 0.7880            | 0.7570       | 0.6110             |
| F1-FVgNET      | 0.9970     | 0.9809      | 0.9824            | 0.9754     | 0.9485           | 0.9891      | 0.9764            | 0.9067     | 0.9457           | 0.9289     | 0.0018           | 0.0704       | 0.9709             | 0.9788      | 0.9704            | 0.9606       | 0.8246             |
| F1-SpecGPT     | 0.9800     | 0.8840      | 0.9330            | 0.9250     | 0.8960           | 0.9170      | 0.9110            | 0.8000     | 0.8110           | 0.8920     | 0.9080           | 0.8410       | 0.8600             | 0.9110      | 0.8880            | 0.7950       | 0.7330             |
| F1-SpecGPT+    | 0.9740     | 0.8860      | 0.9010            | 0.9020     | 0.8640           | 0.8860      | 0.8440            | 0.7460     | 0.7960           | 0.8620     | 0.8830           | 0.8440       | 0.8460             | 0.8860      | 0.8390            | 0.7770       | 0.6920             |
| F1-SpatSIGMA   | 0.9830     | 0.8570      | 0.9170            | 0.9040     | 0.8520           | 0.9020      | 0.8860            | 0.8520     | 0.8790           | 0.8490     | 0.8880           | 0.8630       | 0.8710             | 0.9060      | 0.8920            | 0.8480       | 0.7710             |
| F1-HyperSIGMA  | 0.9830     | 0.8450      | 0.9090            | 0.9060     | 0.8620           | 0.8920      | 0.8530            | 0.8630     | 0.8670           | 0.8360     | 0.8790           | 0.8950       | 0.8660             | 0.9120      | 0.8810            | 0.8620       | 0.7580             |

**Table 19**

Comparison of off-the-shelf and fine-tuned foundation models on HSICityV2.

| Band selection | HSICityV2 |             |              |           |            |
|----------------|-----------|-------------|--------------|-----------|------------|
|                | U-Net     | SpectralGPT | SpectralGPT+ | SpatSIGMA | HyperSIGMA |
| pSentinel      | 32.5      | 39.1        | 42.4         | 33.4      | 34.0       |
| PCA            | 33.1      | 34.2        | 37.4         | 28.2      | 28.3       |

### 5.2.3. Evaluation on HSICityV2

HSICityV2 contains much more complex scene and fine-grained semantic categories so that it is more challenging to correctly segment the street objects.

As shown in Table 20, SpectralGPT+ achieves the best performance for almost all street scene classes, followed by SpectralGPT, which ranks second overall, except for the Terrain class, where SpectralGPT and SpectralGPT+ exchange their rankings. It is worth noting that all four foundation models fail to accurately segment the Train class. We attribute this limitation to the extremely small number of labeled pixels for this category (only 0.03%, as illustrated in Fig. 35c).

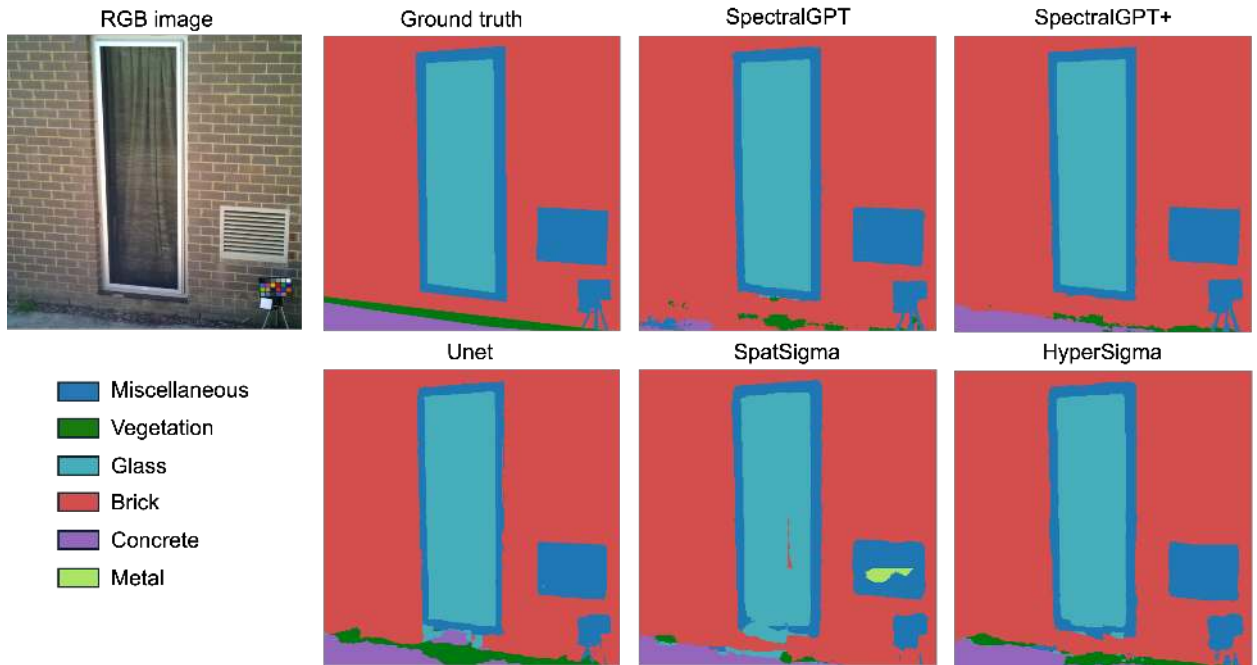
## 5.3. Qualitative Results

**Visualization of LIB-HSI** Fig. 36 shows a building façade sample from the LIB-HSI dataset, featuring brick and glass window structures. Both Unet and the four foundation models produce accurate predictions for this image, with

**Table 20**  
Per-class performance on HSICityV2.

| Model / Class  | Road  | Sidewalk | Building | Wall  | Fence | Pole  | Traffic Light | Sign  | Vegetation | Terrain | Sky   | Person | Rider | Car   | Truck | Bus   | Train | Moto  | Bike  |
|----------------|-------|----------|----------|-------|-------|-------|---------------|-------|------------|---------|-------|--------|-------|-------|-------|-------|-------|-------|-------|
| IoU-SpecGPT    | 0.879 | 0.344    | 0.638    | 0.435 | 0.092 | 0.288 | 0.416         | 0.456 | 0.791      | 0.146   | 0.869 | 0.149  | 0.174 | 0.732 | 0.359 | 0.395 | 0.000 | 0.145 | 0.125 |
| IoU-SpecGPT+   | 0.894 | 0.366    | 0.668    | 0.462 | 0.097 | 0.345 | 0.484         | 0.523 | 0.812      | 0.142   | 0.883 | 0.185  | 0.204 | 0.767 | 0.413 | 0.456 | 0.000 | 0.189 | 0.163 |
| IoU-SpatSIGMA  | 0.832 | 0.223    | 0.574    | 0.389 | 0.052 | 0.255 | 0.351         | 0.440 | 0.755      | 0.107   | 0.813 | 0.108  | 0.104 | 0.627 | 0.296 | 0.283 | 0.000 | 0.080 | 0.067 |
| IoU-HyperSIGMA | 0.847 | 0.257    | 0.578    | 0.399 | 0.043 | 0.258 | 0.298         | 0.417 | 0.754      | 0.115   | 0.827 | 0.138  | 0.117 | 0.647 | 0.258 | 0.312 | 0.000 | 0.097 | 0.097 |
| F1-SpecGPT     | 0.936 | 0.512    | 0.779    | 0.606 | 0.169 | 0.447 | 0.587         | 0.626 | 0.883      | 0.254   | 0.930 | 0.260  | 0.297 | 0.845 | 0.529 | 0.566 | 0.000 | 0.254 | 0.222 |
| F1-SpecGPT+    | 0.944 | 0.536    | 0.801    | 0.632 | 0.176 | 0.514 | 0.653         | 0.687 | 0.896      | 0.248   | 0.938 | 0.313  | 0.339 | 0.868 | 0.585 | 0.626 | 0.000 | 0.318 | 0.281 |
| F1-SpatSIGMA   | 0.908 | 0.365    | 0.730    | 0.560 | 0.099 | 0.406 | 0.519         | 0.611 | 0.860      | 0.193   | 0.897 | 0.194  | 0.189 | 0.771 | 0.457 | 0.441 | 0.000 | 0.148 | 0.126 |
| F1-HyperSIGMA  | 0.917 | 0.409    | 0.732    | 0.570 | 0.083 | 0.411 | 0.459         | 0.589 | 0.859      | 0.206   | 0.905 | 0.243  | 0.209 | 0.786 | 0.410 | 0.476 | 0.000 | 0.177 | 0.177 |

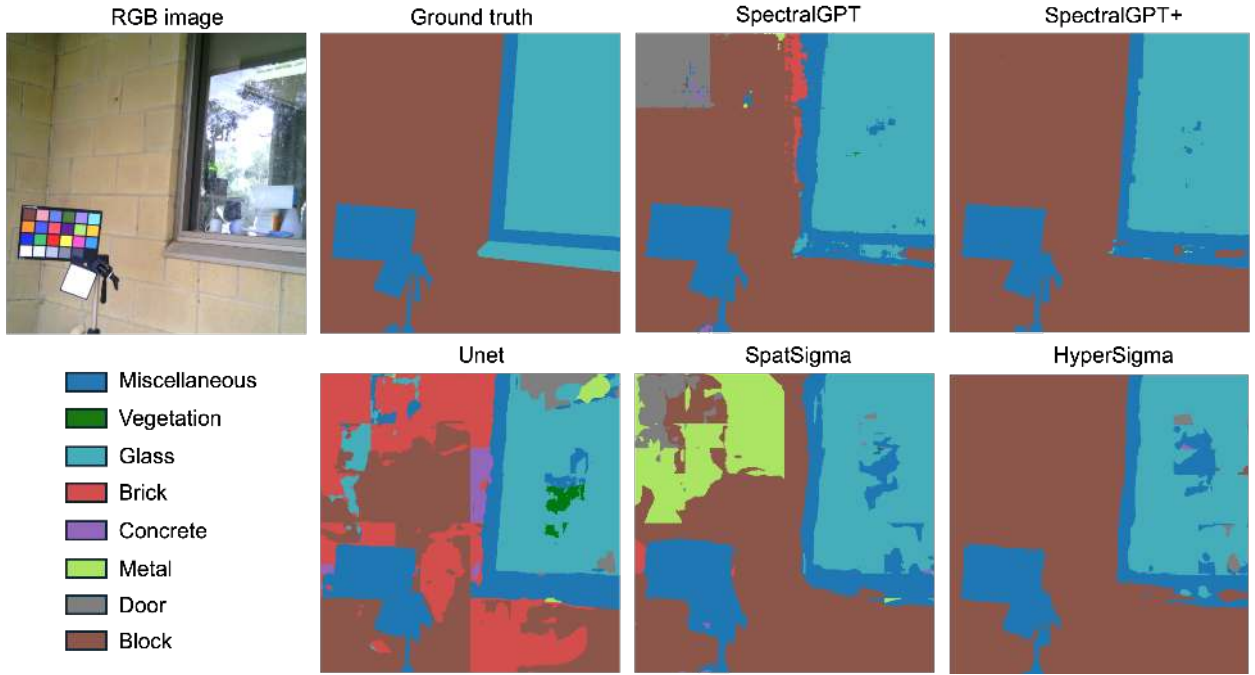
only the edges of vegetation and concrete appearing slightly ambiguous. Relatively, SpectralGPT and SpectralGPT+ better capture the shapes of the tripod and panel, whereas SpatSIGMA and HyperSIGMA provide clearer delineation of vegetation and concrete edges.



**Figure 36:** Visualization of a sample in LIB-HSI predicted using different models.

Fig. 37 illustrates another LIB-HSI sample featuring a block structure and glass window with strong specular reflections. U-Net misclassifies part of the block background as Brick, resulting in poor visualization quality. Among the four foundation models, SpectralGPT incorrectly labels the upper-left corner of the image as Door, while SpatSIGMA misclassifies the same region as Metal. In contrast, SpectralGPT+ and HyperSIGMA produce visually superior segmentation results, with SpectralGPT+ achieving the best performance by minimizing noise in the segmentation of the glass under heavy specular reflection.

Fig. 38 demonstrates additional visualization results: Sample 1 shows a metal background, where a green plant is located at the center of the scene. All models are able to capture the plant; however, SpectralGPT and SpectralGPT+ produce the clearest boundary between the metal and concrete regions; Sample 2 depicts a main door scene, where a window-like object appears in the upper-left corner. This object is not an actual glass window but is labeled as a miscellaneous class in the ground truth. Unet incorrectly predicts brick within this region, while SpectralGPT predicts it as a window, suggesting that the foundation model has learned semantic priors—such as windows commonly appearing



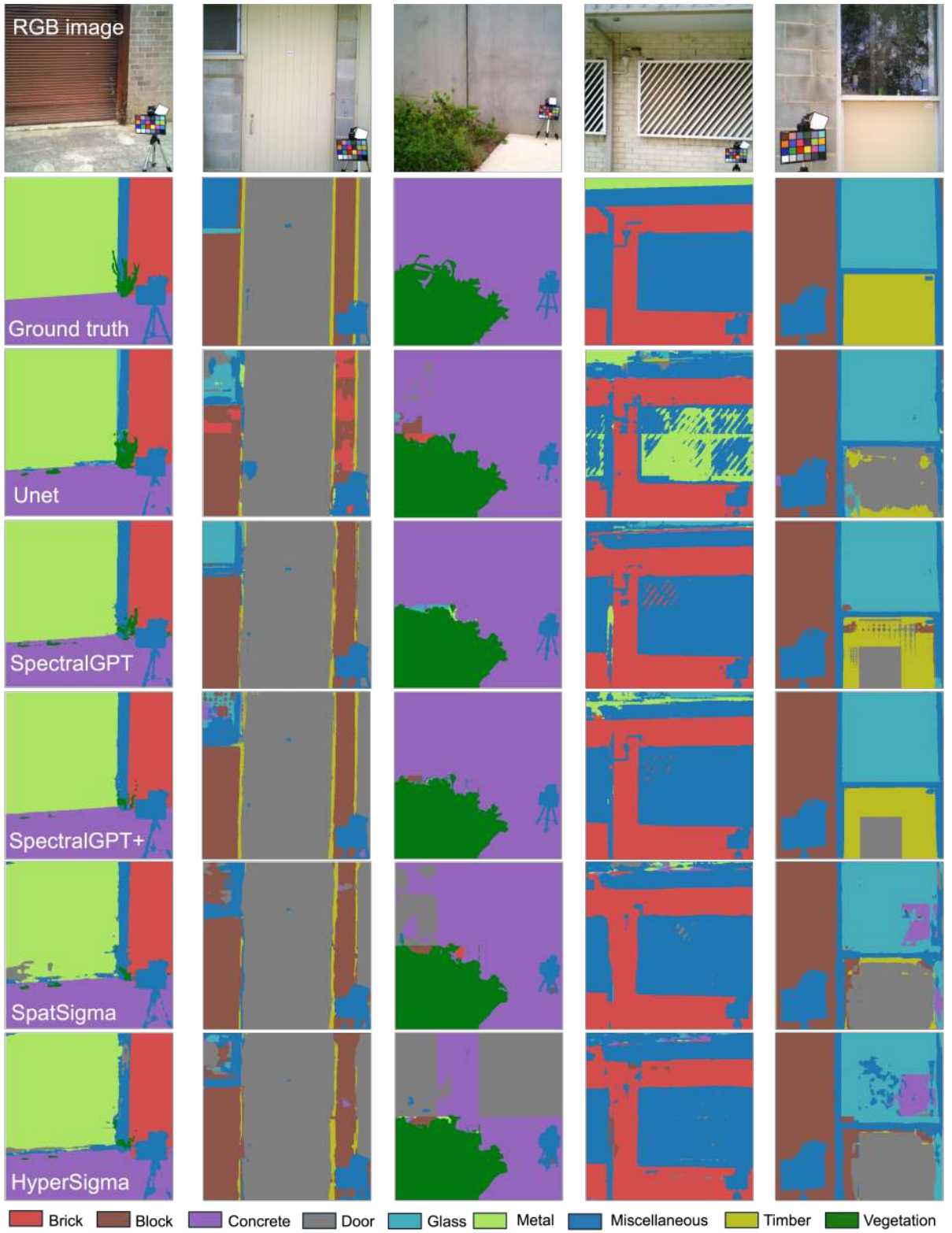
**Figure 37:** Visualization of a sample in LIB-HSI predicted using different models.

near doors; Sample 3 consists of a vegetation foreground and a concrete background. SpatSIGMA and HyperSIGMA fail to correctly classify the concrete background, mislabeling the upper-left and upper-right regions as the Door class. In contrast, SpectralGPT+ produces the clearest boundary for the foreground vegetation; Sample 4 shows a rectangular window with white diagonal slats, likely made of metal material. Consequently, Unet predicts the main region as metal, even though it is labeled as Miscellaneous in the ground truth; Sample 5 presents a structure composed of both glass and timber. Unet predicts the entire structure as glass but fails to capture the timber component. SpectralGPT and SpectralGPT+ successfully identify both glass and timber, with SpectralGPT+ achieving the best performance. In contrast, SpatSIGMA and HyperSIGMA produce incomplete glass and timber regions: the glass area is contaminated by concrete predictions, and the timber region is partially misclassified as the Door class.

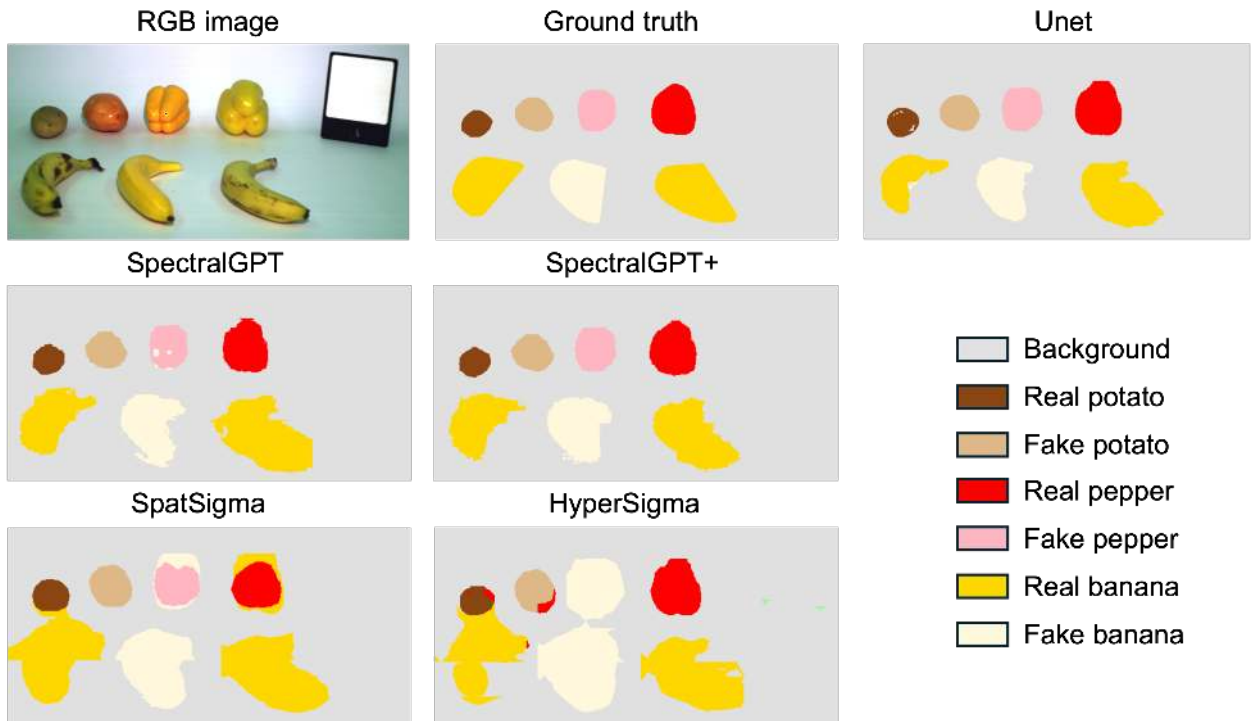
*Visualization of FVgNET* As shown in Fig. 39, different real and artificial fruits can be successfully segmented and correctly classified when employing two spectral foundation models, while the baseline U-Net predicts the real potato as real pepper, and some of the background regions are incorrectly classified as real unknown objects, which can be unexpected noise in segmentation. The visualization result validates the effectiveness of spectral foundation models. Within four variants of spectral foundation models, SpectralGPT predict the example with correct semantic classes but blurry edges exist around every fruit. As a comparison, SpectralGPT+ model segments all fruits with clearer edges. SpatSIGMA failed in segmenting the real potato in the scene by recognizing half of it as real unknown, while HyperSIGMA succeeded.

Fig. 40 presents additional results for real and fake fruit recognition. Almost all fruits are segmented smoothly from the background using either the pre-foundation model or the foundation models. Most real and fake fruits are correctly identified; however, some errors remain. Specifically, part of the first apple is ambiguous, part of the first fake orange is misclassified as real, the second fake lemon is not recognized and is treated by SpectralGPT as Fake Unknown, and part of the last fake avocado is misclassified as real.

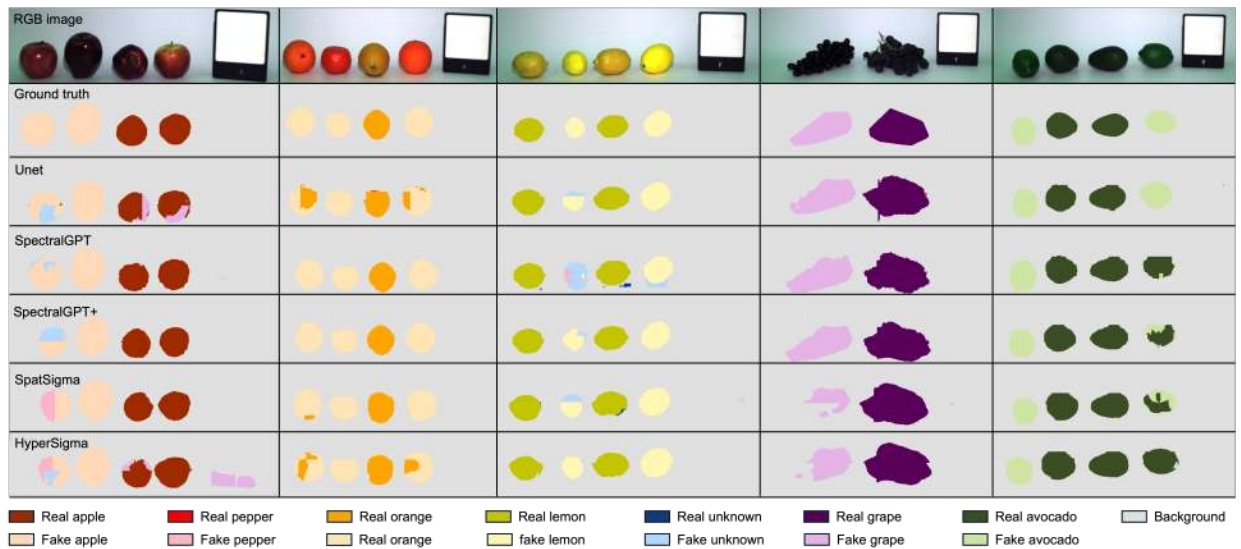
*Visualization of HSiCityV2* Fig. 41 presents qualitative results for urban scene segmentation. It can be observed that under more complex scenes and fine-grained semantic categories, all remote sensing foundation models exhibit degraded performance except for SpectralGPT+. U-Net, as a model originally designed for RGB imagery, outperforms most HSI-oriented foundation models in this setting. These results further validate our analysis that discrepancies



**Figure 38:** Visualization of 5 samples in LIB-HSI predicted using different models.

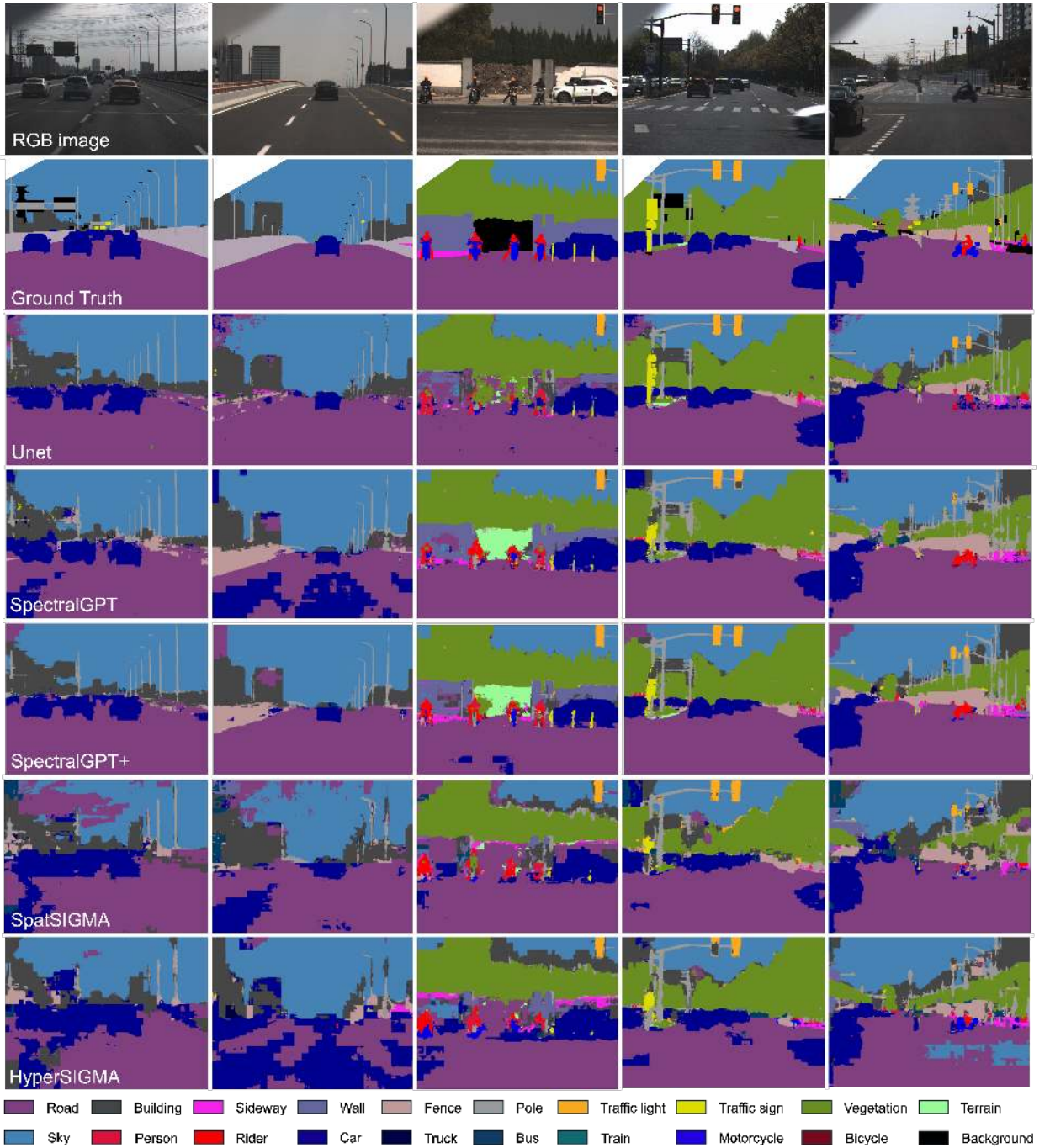


**Figure 39:** Visualization of a sample in FVgNET predicted using different models.



**Figure 40:** Visualization of a sample in FVgNET predicted using different models.

in noise modeling, domain gap, sensor characteristics, and illumination conditions collectively constitute a major bottleneck for transferring remote sensing foundation models to computer vision scenarios.



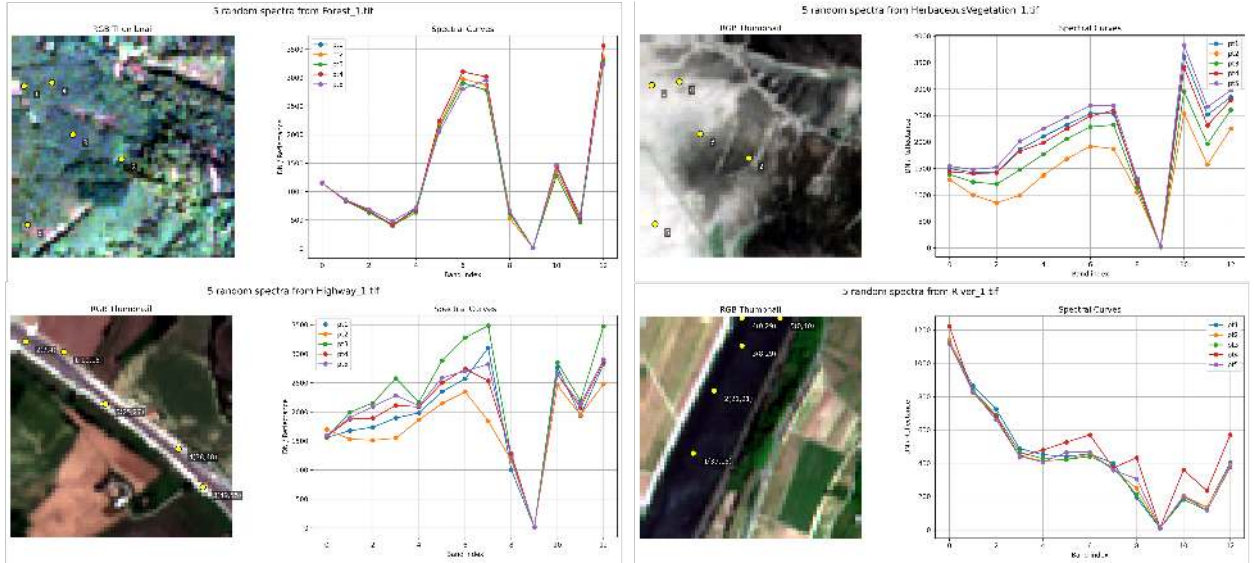
**Figure 41:** Visualization of 5 samples in HSiCityV2 predicted using different models.

## 6. Discussion

Based on the literature review and benchmarking results, several key challenges emerge, including the spectra stability, the domain gap between remote sensing and computer vision HSI, disparities in imaging sensors, outdated HSI network architectures, and the lack of computer vision HSI datasets. These challenges limit the development of HSI in computer vision and highlight important directions for future AI-enabled HSI research in natural scenes.

## 6.1. Noise model and the robustness of HSI

Disregarding atmospheric effects, the signal recorded by a hyperspectral camera at a pixel can be regarded as a mixture of light scattered by substances located in the field of view [28]. Ideally, materials composed of similar substances exhibit similar spectral signatures, while different materials produce distinguishable spectral patterns. In remote sensing (RS), hyperspectral images are commonly annotated with Land-use/Land-cover (LULC) classes [287], such as annual crop, permanent crop, forest, vegetation, water bodies, and industrial areas [120, 283]. These classes can be broadly grouped into four categories: vegetation-dominated land (e.g., forest, permanent crop, pasture), mixed vegetation and soil (e.g., annual crop, herbaceous vegetation), water bodies (e.g., river, sea, lake), and artificial materials (e.g., highways, industrial and residential areas). When sampling pixels from the same category, their spectral curves typically present consistent and robust patterns, as illustrated in Fig. 42.



**Figure 42:** Spectral curves: 4 categories of remote sensing materials in EuroSAT [120], each showcases consistent and robust pattern.

Such robustness largely stems from the relatively simple imaging conditions in remote sensing. In RS scenarios, the Sun acts as the primary illumination source. Due to the extremely large distance between the Sun and the Earth, the illumination can be approximated as uniform and parallel. This assumption greatly simplifies the hyperspectral imaging model. As a result, RS studies usually focus on modeling the interaction between light and materials, such as atmospheric absorption and surface reflectance from vegetation, water, or concrete. Moreover, because of the large imaging distance, scene geometry can often be simplified as a planar surface. The angles of incident and reflected light are therefore largely independent of local surface frames. The spatial resolution of RS images is also relatively low, typically around 10 meters [120, 283, 125] or 30 meters [297] per pixel, which further reduces the influence of fine-scale geometric variations. The simplified imaging configuration is illustrated in Fig. ?? Under such conditions, the captured spectra tend to be stable and consistent across similar materials.

However, hyperspectral imaging in natural scenes, as commonly encountered in computer vision (CV), presents a significantly more complex noise model. Unlike RS scenarios, natural scenes often involve multiple and heterogeneous light sources, leading to complex illumination patterns and stochastic incident angles. In addition, objects in the scene typically exhibit diverse geometries and surface orientations. According to Lambert's reflectance model (see Section 2), the reflected radiance depends on both the incident illumination direction and the surface normal. Consequently, the measured spectral signal is influenced not only by the intrinsic material properties but also by detailed surface geometry and illumination configuration.

As a result, each stage of the hyperspectral imaging pipeline in computer vision—including illumination, transmission, reflection, and sensor acquisition—may introduce additional noise. Compared with remote sensing, the accumulated uncertainties can lead to unstable spectral measurements. Therefore, directly applying raw HSI data without accounting for these noise factors may lead to unreliable spectral representations and sub-optimal performance

$$\begin{aligned}
I(u, v, \lambda) \propto L_r(\omega_r, \lambda) &= \int_{\Omega} f_r(\lambda, \omega_i, \omega_r) L_i(\omega_i, \lambda) \cos \theta_i d\omega_i S(\lambda) \\
&= L_i(\lambda) \int_{\Omega} f_r(\lambda, \omega_i, \omega_r) \cos \theta_i d\omega_i S(\lambda) \\
&= L_i(\lambda) f_r(\lambda) \int_{\Omega} \cos \theta_i d\omega_i S(\lambda) \\
&= L_i(\lambda) R(\lambda) S(\lambda)
\end{aligned}$$


 Uniform/Parallel illumination  
 Surface geometry neglected due to remote distance  
 Hemisphere integral

**Figure 43:** Noise model: A step-by-step simplification from Computer Vision to Remote Sensing.

of downstream algorithms. We build a natural scene HSI noise model as summarized in Fig. 43, involving the main sources of noise.

## 6.2. Domain gap

Beyond differences in imaging physics, hyperspectral datasets used in remote sensing and computer vision also exhibit substantial domain discrepancies. These differences arise from variations in task definitions and dataset characteristics.

**Task-level gap.** Remote sensing hyperspectral datasets are usually annotated with coarse Land-use/Land-cover (LULC) categories such as crops, forests, buildings, and water bodies. In contrast, computer vision tasks often require much finer-grained material discrimination. For instance, urban scene segmentation may need to distinguish vehicles such as cars, buses, and motorcycles [346, 273]. Fruit inspection tasks require separating real fruits from artificial ones while also identifying fruit categories [219]. Building facade analysis involves distinguishing materials such as glass, timber, clay, concrete, and vegetation [110]. Table 21 illustrates the class distribution across different datasets.

Table 21: Class distribution of remote sensing (RS) and computer vision (CV) datasets.

| Materials             | Remote sensing datasets |                    |                | Computer vision datasets |                  |                    |
|-----------------------|-------------------------|--------------------|----------------|--------------------------|------------------|--------------------|
|                       | EuroSAT<br>[120]        | SegMunich<br>[125] | Huston<br>[54] | FVgNET<br>[219]          | LIB-HSI<br>[110] | HSICityV2<br>[273] |
| An. Crop              | ✓                       |                    |                |                          |                  |                    |
| Apple (Real & Fake)   |                         |                    |                | ✓                        |                  |                    |
| Artificial vegetation |                         | ✓                  |                |                          |                  |                    |
| Arable land           |                         | ✓                  |                |                          |                  |                    |
| Avocado (Real & Fake) |                         |                    |                | ✓                        |                  |                    |
| Background            |                         | ✓                  |                | ✓                        |                  | ✓                  |
| Bicycle               |                         |                    |                |                          |                  | ✓                  |
| Blocks                |                         |                    |                |                          | ✓                |                    |
| Bricks                |                         |                    |                |                          | ✓                |                    |
| Buildings             |                         | ✓                  |                |                          |                  | ✓                  |
| Bus                   |                         |                    |                |                          |                  | ✓                  |
| Car                   |                         |                    |                |                          |                  | ✓                  |
| Commercial            |                         |                    | ✓              |                          |                  |                    |
| Concrete              |                         |                    |                |                          | ✓                |                    |
| Door                  |                         |                    |                |                          | ✓                |                    |
| Forests               | ✓                       | ✓                  |                |                          |                  |                    |
| Fence                 |                         |                    |                |                          |                  | ✓                  |

*Continued on next page*

Table 21: Class involvement of RS and CV datasets (continued)

| Materials            | Remote sensing datasets |           |        | Computer vision datasets |         |           |
|----------------------|-------------------------|-----------|--------|--------------------------|---------|-----------|
|                      | EuroSAT                 | SegMunich | Huston | FVgNET                   | LIB-HSI | HSICityV2 |
| Glass window         |                         |           |        |                          | ✓       |           |
| Grape (Real & Fake)  |                         |           |        | ✓                        |         |           |
| Healthy grass        |                         |           | ✓      |                          |         |           |
| Herbaceous           | ✓                       |           |        |                          |         |           |
| Highway              | ✓                       |           | ✓      |                          |         |           |
| Industrial           | ✓                       |           |        |                          |         |           |
| Lemon (Real & Fake)  |                         |           |        | ✓                        |         |           |
| Metal                |                         |           |        |                          | ✓       |           |
| Mine, dump           |                         | ✓         |        |                          |         |           |
| Miscellaneous        |                         |           |        |                          | ✓       |           |
| Motorcycle           |                         |           |        |                          |         | ✓         |
| Open spaces          |                         | ✓         |        |                          |         |           |
| Orange (Real & Fake) |                         |           |        | ✓                        |         |           |
| Parking lot          |                         |           | ✓      |                          |         |           |
| Pastures             | ✓                       | ✓         |        |                          |         |           |
| Pepper (Real & Fake) |                         |           |        | ✓                        |         |           |
| Per. Crop            | ✓                       | ✓         |        |                          |         |           |
| Person               |                         |           |        |                          |         | ✓         |
| Pole                 |                         |           |        |                          |         | ✓         |
| Potato (Real & Fake) |                         |           |        | ✓                        |         |           |
| Railway              |                         |           | ✓      |                          |         |           |
| Residential          | ✓                       |           | ✓      |                          |         |           |
| Rider                |                         |           |        |                          |         | ✓         |
| River                | ✓                       |           |        |                          |         |           |
| Road                 |                         |           | ✓      |                          |         | ✓         |
| Running track        |                         |           | ✓      |                          |         |           |
| Sea & Lake           | ✓                       |           |        |                          |         |           |
| Sidewalk             |                         |           |        |                          |         | ✓         |
| Shrub                |                         | ✓         |        |                          |         |           |
| Soil                 |                         |           | ✓      |                          |         |           |
| Stressed grass       |                         |           | ✓      |                          |         |           |
| Sky                  |                         |           |        |                          |         | ✓         |
| Surface water        |                         | ✓         |        |                          |         |           |
| Synthetic grass      |                         |           | ✓      |                          |         |           |
| Tennis court         |                         |           | ✓      |                          |         |           |
| Terrain              |                         |           |        |                          |         | ✓         |
| Timber               |                         |           |        |                          | ✓       |           |
| Traffic light        |                         |           |        |                          |         | ✓         |
| Traffic sign         |                         |           |        |                          |         | ✓         |
| Train                |                         |           |        |                          |         | ✓         |
| Tree                 |                         |           | ✓      |                          |         |           |
| Truck                |                         |           |        |                          |         | ✓         |

*Continued on next page*

Table 21: Class involvement of RS and CV datasets (continued)

| Materials    | Remote sensing datasets |           |        | Computer vision datasets |         |           |
|--------------|-------------------------|-----------|--------|--------------------------|---------|-----------|
|              | EuroSAT                 | SegMunich | Huston | FVgNET                   | LIB-HSI | HSICityV2 |
| Urban fabric |                         | ✓         |        |                          |         |           |
| Vegetation   |                         |           |        |                          | ✓       | ✓         |
| Wall         |                         |           |        |                          |         | ✓         |
| Water        |                         |           | ✓      |                          |         |           |
| Wetlands     |                         | ✓         |        |                          |         |           |

**Dataset-scale gap.** Another major difference lies in dataset scale. Existing spectral foundation models are typically pretrained on large-scale remote sensing datasets, such as SpectralGPT trained on one million images [125] and HyperSIGMA trained on the HyperGlobal-450K dataset [297]. In contrast, publicly available hyperspectral datasets for computer vision are significantly smaller. For example, CAVE [343] contains 32 images, Harvard [38] contains 50 images, and NUS [240] contains 77 images. Even the relatively larger ICVL dataset [9] contains only 201 images.

Moreover, several hyperspectral datasets lack semantic annotations, including KAUST [180], NTIRE2022 [10], BUSIFusion [168], BJTU-UVA [69], and BeyondRGB [102]. Although annotated datasets such as HSICityV2 [273], HSiDrive [20, 109], LIB-HSI [110], FVgNET [219], HFD100 [379], and HOTC [330] have been introduced recently, they are collected under different conditions and follow different annotation protocols, which further increases the domain discrepancy across datasets.

### 6.3. Sensor and Acquisition Variability

Another important source of discrepancy arises from the diversity of hyperspectral imaging sensors used in different applications. Owing to distinct application scenarios, remote sensing and computer vision employ substantially different hyperspectral acquisition systems. Table 22 summarizes several representative hyperspectral sensors and their configurations.

Remote sensing sensors typically operate over a broad spectral range from 400 to 2500 nm with approximately 200 spectral bands, most of which lie in the infrared region of the electromagnetic spectrum (Fig. 3a). Due to the long imaging distance, each pixel covers a relatively large ground area, and the measured spectrum often represents a mixture of multiple materials. This characteristic contributes to improved spectral stability.

In contrast, hyperspectral cameras used in computer vision are usually positioned close to the observed scene and primarily operate in the visible spectrum or wavelengths below 1000 nm. Currently, three main types of hyperspectral cameras are widely used. The first category consists of Fabry–Perot–based snapshot hyperspectral cameras, which enable real-time acquisition and hyperspectral video recording [330, 186, 43, 234, 259, 187, 185]. However, these systems rely on mosaic sensor chips composed of  $4 \times 4$  or  $5 \times 5$  multispectral filter arrays, limiting the spectral resolution to at most 25 bands. The second category employs spectral scanning using a liquid crystal tunable filter (LCTF) combined with a monochrome camera [50]. This configuration can achieve high spatial resolution but is constrained by the limited number of spectral bands provided by the filter. The third category is represented by line-scan hyperspectral cameras such as Specim IQ [22]. These systems capture rich spectral information while maintaining moderate spatial resolution (typically  $512 \times 512$ ). However, acquiring a complete hyperspectral cube may require tens of seconds, making them unsuitable for dynamic scenes.

The diversity of hyperspectral imaging devices leads to substantial variations in spectral range, spatial resolution, calibration protocols, and acquisition strategies. Consequently, hyperspectral datasets captured by different sensors often exhibit inconsistent characteristics, making it difficult to directly transfer models trained on one dataset to another without domain adaptation or transfer learning.

### 6.4. Network architecture

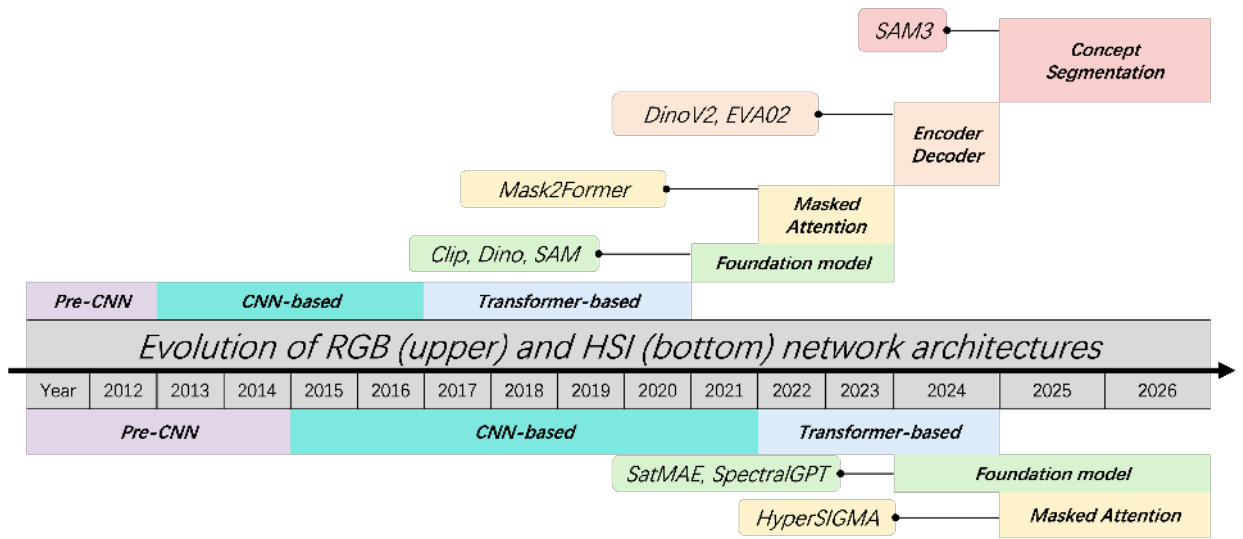
While the previous sections focused on the physical, data, and sensor-level challenges of hyperspectral imaging, another critical issue lies at the algorithmic level—namely, the gap between network architectures developed for traditional RGB vision and those used in hyperspectral computer vision, as shown in Fig. 44.

**First-generation gap: attention mechanisms.** Recent RGB-based segmentation frameworks have introduced sophisticated attention mechanisms to improve feature extraction and region modeling. For instance, Mask2Former [46] proposed a masked-attention mechanism that restricts cross-attention to predicted mask regions, enabling the network

**Table 22**

Comparison of representative hyperspectral imaging systems used in remote sensing and computer vision.

| Category        | Sensor    | Imaging Distance | Spatial Resolution  | Spectral Range (nm) | Data Cube Size               | Imaging Method |
|-----------------|-----------|------------------|---------------------|---------------------|------------------------------|----------------|
| Remote Sensing  | AVIRIS    | 20 km            | 20 m                | 400–2500            | $512 \times 614 \times 224$  | Point scan     |
|                 | HYDICE    | 1.6 km           | 0.75 m              | 400–2500            | $200 \times 320 \times 210$  | Line scan      |
| Computer Vision | 4X4 chip  | 100 mm           | $5.5 \mu\text{m}$   | 470–630             | $512 \times 272 \times 16$   | Snapshot       |
|                 | LCTF      | 100 mm           | $3.69 \mu\text{m}$  | 420–720             | $3376 \times 2704 \times 28$ | Spectral scan  |
|                 | Specim IQ | 150 mm           | $17.58 \mu\text{m}$ | 400–1000            | $512 \times 512 \times 204$  | Line scan      |

**Figure 44:** Evolution of architectures for RGB-based vision and HSI-based vision.

to focus on localized semantic structures. A related idea has been partially adopted in HyperSIGMA [297]. As shown in Fig. 45, the Sparse Sampling Attention (SSA) module operates similarly to deformable attention by learning a set of sampling locations for each query in order to capture inter-region relationships.

However, as demonstrated by our benchmarking experiments in Section 5, the SSA module contributes only marginal improvements when fine-tuning HyperSIGMA or SpatSIGMA on computer-vision hyperspectral datasets. This observation suggests that simply transferring attention designs from remote sensing or RGB vision is insufficient. Dedicated masked-attention mechanisms tailored to HSI in computer vision remain largely unexplored.

**Second-generation gap: encoder–decoder paradigms.** Another major disparity lies in encoder–decoder design. In RGB-based vision, many modern segmentation pipelines rely on powerful foundation-model encoders such as DINOv2 [246], EVA [78], and EVA02 [77]. These encoders are typically coupled with advanced decoders such as Mask2Former [46], forming a highly effective pipeline that fully exploits large-scale pretrained representations.

In contrast, such a paradigm has rarely been explored for hyperspectral computer vision. It remains unclear which types of foundation-model-based encoders and decoders are most suitable for HSI data, or how spectral–spatial information should be integrated within these architectures. Identifying efficient architectural designs for hyperspectral foundation models therefore remains an open research question.

**Third-generation gap: promptable segmentation models.** The generation gap has widened further with the emergence of promptable vision models. The latest version of the Segment Anything Model, SAM3 [?], introduces the concept of Promptable Concept Segmentation (PCS). In this framework, prompts can be expressed as short noun



**Figure 45:** Comparison between (a) Full Attention and (b) Sparse Sampling Attention (SSA) in HyperSIGMA. Stars represent queries, with dots surrounded by corresponding colored lines indicating the attention regions of captured contexts. Green points denote common areas shared by both queries. Courtesy of [297].

phrases, image exemplars, or their combination, and the model returns segmentation masks and identities for all matching object instances.

Such concept-driven segmentation capabilities represent a new paradigm in visual understanding. However, comparable promptable or concept-based segmentation models have not yet been explored in hyperspectral computer vision. Bridging this gap will require developing architectures capable of integrating spectral representations with multimodal prompting mechanisms.

Overall, adapting and extending state-of-the-art network architectures from RGB vision to hyperspectral imaging remains a critical research direction. By systematically exploring modern attention mechanisms, foundation-model encoder–decoder designs, and promptable segmentation frameworks, future studies may significantly advance hyperspectral computer vision.

## 6.5. Future works

Based on the challenges discussed above, several promising research directions deserve further investigation.

**Robust hyperspectral modeling under complex natural scene environment.** Compared with remote sensing, hyperspectral imaging in natural scenes is significantly affected by complex illumination conditions, diverse surface geometries, and dynamic environments. Future work could focus on developing physics-aware or illumination-invariant models that explicitly incorporate reflectance models and scene geometry into the learning process. Such approaches may help improve the robustness of hyperspectral representations under varying lighting conditions and reduce the impact of noise accumulated during the imaging pipeline.

**Bridging the domain gap between remote sensing and computer vision.** The large domain gap between remote sensing hyperspectral datasets and computer vision datasets poses a major challenge for transferring spectral foundation models. Future research may explore domain adaptation, cross-domain pretraining, and self-supervised learning strategies to better leverage large-scale remote sensing hyperspectral data for computer vision tasks. In addition, constructing large-scale hyperspectral datasets with unified annotation protocols would greatly facilitate cross-domain learning.

**Standardization of hyperspectral acquisition and datasets.** The diversity of hyperspectral sensors leads to significant variations in spectral range, spatial resolution, calibration procedures, and acquisition strategies. Establishing standardized acquisition protocols and benchmark datasets could significantly improve reproducibility and model comparability across studies. Furthermore, the development of sensor-invariant learning methods may help mitigate discrepancies caused by different hyperspectral imaging systems.

**Next-generation architectures for hyperspectral vision.** Another promising direction lies in the design of network architectures tailored for hyperspectral data. Current hyperspectral models lag behind the rapid architectural evolution observed in RGB vision, including masked-attention mechanisms, foundation-model encoder–decoder pipelines, and promptable segmentation frameworks. Future research could investigate how modern architectures such as foundation models, promptable segmentation systems, and multimodal learning frameworks can be adapted to hyperspectral data. Bridging this architectural generation gap may significantly enhance the performance and generalization capability of hyperspectral computer vision models.

Overall, advancing hyperspectral computer vision will require joint efforts in imaging physics, dataset construction, sensor design, and learning architectures. Addressing these challenges will help unlock the full potential of hyperspectral imaging for a wide range of real-world natural scene computer vision applications.

## 7. Conclusion

Hyperspectral imaging (HSI) has demonstrated remarkable potential for material understanding and scene analysis due to its rich spectral information. While hyperspectral techniques have been extensively studied in remote sensing, their application to computer vision remains relatively underexplored and presents a number of unique challenges. In this work, we systematically review the progress of hyperspectral imaging in computer vision scenarios.

First, we construct a comprehensive pipeline that spans the fundamental concepts, physical principles, and hardware sensing systems of HSI. Unlike the relatively stable imaging conditions in remote sensing, hyperspectral imaging in computer vision is strongly influenced by complex illumination patterns, diverse object geometries, and multiple sources of noise throughout the imaging pipeline. These factors often lead to unstable spectral measurements and reduce the reliability of raw hyperspectral signals.

Second, we review early explorations of hyperspectral imaging in computer vision applications. According to the twofold objectives of HSI research in computer vision, low-level vision tasks aim to restore high-quality hyperspectral data, including reconstruction, super-resolution, denoising, demosaicing, and illumination spectrum estimation, while high-level applications focus on semantic understanding tasks such as classification, semantic segmentation, face recognition, object detection, and tracking. We place particular emphasis on modern artificial intelligence techniques used in HSI research, with special attention to recent spectral foundation models. Through an in-depth analysis of existing hyperspectral datasets, we also revisit the long-standing problem of data scarcity in this field.

Third, we conduct the first evaluation of two representative spectral foundation models across three hyperspectral computer vision datasets. Based on the benchmarking results, we further analyze the fundamental factors that hinder the effective adoption of hyperspectral imaging in computer vision scenarios.

Finally, we discuss the key challenges that currently limit the development of hyperspectral computer vision and outline several promising future research directions. These challenges span multiple aspects of the HSI pipeline, including physical modeling, dataset characteristics, sensing hardware, and network architectures. In particular, complex noise models compatible with HSI in computer vision, the domain gap between remote sensing and computer vision datasets, the variability of hyperspectral sensing devices, and the architectural generation gap between hyperspectral models and modern RGB-based vision systems all contribute to the fact that HSI research still lags behind mainstream computer vision.

Together, these challenges highlight the need for coordinated progress in imaging physics, dataset construction, sensing hardware, and learning architectures. Addressing these issues will be essential for bridging the gap between hyperspectral imaging and mainstream computer vision. We hope that the analysis and discussions presented in this work will provide useful insights and stimulate further research toward robust, scalable, and versatile hyperspectral computer vision systems.

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